

MWENDO PAMOJA

Continuous Underwriting for Insurtech, Embedded Finance, and Institutional Capital

A Kenya-first architecture for gig-driver economic resilience

Narrative and Technical White Paper

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Part 1

Architecture of the Insurtech Product, Multi-Product Capital Stack, and the Default Cascade

Mwendo Pamoja White Paper

1. Introduction to the Insurtech Product-Issuer Ecosystem

1.1. The Gig Driver as a Unified Asset Class

Core Architectural Question: How can InsureTech and Embedded Finance be employed to continuously understand, predict, and stabilize the cashflow of gig workers while enabling banking partners to embed and finance multiple products satisfying both banking and insurance regulatory frameworks?

The question becomes tangible before it becomes technical. A driver who starts a morning shift with a functioning vehicle has an income-producing asset. If that vehicle is uninsured, under-fuelled, under repair, or deactivated by the platform, its financial value changes within hours. The driver does not experience those failures in the separate product silos used by lenders and insurers. They experience one inability to work.

In the traditional consumer finance and commercial lending paradigms, risk underwriting relies on the structural separation of personal liabilities and corporate assets. Underwriters evaluate individual borrowers based on stable, predictable salary streams verified via tax returns, or corporate borrowers based on balance sheet strength and ring-fenced business cash flows. The gig-economy driver, however, defies this binary classification entirely, representing a highly volatile, hybrid economic entity that occupies neither category cleanly.

In this paper, the driver and vehicle are analysed as a **unified operating system**, not literally as a tradeable asset class. Economic viability is concentrated in a small set of connected conditions: the vehicle must work, insurance must remain valid, fuel and maintenance must be affordable, and the platform account must remain active. This shared dependency is the root of the correlated default risk that follows. The proposed architecture seeks shared value by protecting a viable driver from avoidable cascades while safeguarding lenders, insurers, investors, and platform partners.

For many gig drivers, the vehicle is the primary income-generating operating asset. Auto finance, motor insurance, and personal credit are often assessed by different institutions against different files. In an embedded-finance ecosystem, the vehicle's condition, insurance status, platform access, trip activity, and wallet settlement can nevertheless form one tightly connected cash-flow chain. Net fare income, meaning gross earnings after platform deductions and operating costs, may support several commitments at once. It is not assumed to be the driver's sole source of cash unless the evidence establishes that fact.

Let I_{gross} denote the driver's gross fare volume per day, τ_{platform} the platform commission rate, C_{ops} daily operational costs (fuel, vehicle wear-and-tear, congestion charges), and r the automated split-fare repayment deduction rate on outstanding debt. The driver's net take-home pay I_{net} is then:

$$I_{\text{net},t} = I_{\text{gross},t}(1 - \tau_{\text{platform},t}) - C_{\text{ops},t} - R_t - P_t,$$

where R_t is the actual debt deduction and P_t is the actual premium deduction. If the contractual debt split is r_t , it remains bounded by the amount due and the amount legally and operationally available:

$$R_t = \min \{r_t I_{\text{gross},t}, A_t^{\text{due}}, I_{\text{gross},t}(1 - \tau_{\text{platform},t}) - C_{\text{ops},t} - S_t\}_+.$$

Let S_t be a calibrated sufficiency threshold representing disclosed and supportable minimum household and operating needs. The **cash-flow suffocation boundary** is crossed when:

$$I_{\text{gross},t}(1 - \tau_{\text{platform},t}) - C_{\text{ops},t} - R_t - P_t < S_t.$$

When the inequality holds persistently, a fixed automated deduction can compete with fuel, food, repair, and insurance. That does not make default deterministic: a driver may have savings, another platform, household support, a negotiated hardship arrangement, or an insurer-approved policy option. It does identify a state in which the marginal intervention matters. The platform's task is to estimate that state without claiming private facts it has not observed, then choose a lawful, proportionate response.

1.2. The Triple-Product Capital Stack

To enable and sustain a gig driver's operations, the embedded finance partnership deploys three distinct financial products forming a layered capital stack that addresses different operational needs, cash-flow cycles, and risk time horizons. Understanding the contractual mechanics of each product is prerequisite to understanding how a single external shock can cascade simultaneously across all three.

1.2.1. Product 1: Insurance Premium Financing (IPF)

Appropriate motor cover may be required by law, policy terms, vehicle use, or platform rules. The exact class and limit must be confirmed for each use case. High-utilisation ride-hailing vehicles can attract substantial annual premiums. A working range of KES 150,000 to KES 300,000 may be tested in the financial model, but it remains an illustrative assumption until quotations and portfolio data establish product, geography, vehicle, and driver effects. The financing opportunity arises when an otherwise viable driver cannot pay the annual amount upfront.

Under the proposed IPF structure, the finance provider pays an agreed premium amount to the insurer at inception. The driver repays under a weekly or monthly schedule. A cancellation refund may provide credit support, subject to policy wording, law, elapsed cover, taxes, commissions, fees, claims, notice, cancellation effectiveness, and insurer set-off. A simplified gross unearned-premium starting point is:

$$UP(t) = P_{\text{total}} \left(1 - \frac{t}{T}\right)$$

The bank structures the driver's repayment schedule such that the outstanding loan balance $L_{\text{outstanding}}(t)$ remains below the unearned premium value at all times:

$$L_{\text{outstanding}}(t) \leq UP_{\text{eligible}}(t)(1 - \delta)$$

where δ is an approved haircut for timing, expenses, deductions, uncertainty, and other ineligible amounts. A 10 to 15 percent range is illustrative. Cancellation authority, notice, refund direction, coverage status, and application of proceeds must be established by the finance and insurance contracts. A missed instalment does not itself grant the platform a universal power to cancel cover.

The operational vulnerability lies in the interval between missed payment, notice, cure, cancellation, refund, and platform action. Its length and coverage effect are contractual facts, not a universal 10 to 15 day rule. In some arrangements cover may remain effective during a cure period; in others suspension or cancellation may occur after specified steps. The system must therefore ingest authoritative policy status rather than infer that a late payment means the vehicle is uninsured. A collision during genuine non-cover can impose a severe personal loss and interrupt earning capacity. It can then propagate into other products, but savings, support, another vehicle, claim outcome, restructuring, and recovery may interrupt the cascade.

1.2.2. Product 2: Microloans

Microloans are high-frequency, short-duration advances, with an illustrative range of KES 5,000 to KES 25,000 and tenors of 7 to 30 days, designed to cover immediate cash-flow gaps. The licensed lender originates the credit. Funding may later be provided or refinanced through the SPV, and authorised daily or weekly split-fare deductions may service the balance.

The potentially pro-cyclical repayment design is the key risk feature. If the split-fare deduction rate r remains fixed while gross fare volume I_{gross} falls because of fuel costs, demand, weather, or platform changes, the same contractual rule can consume a larger share of cash available after essential operating costs. The suffocation boundary tightens when I_{gross} declines while C_{ops} rises. A sufficiency floor, variable deduction, hardship pause, or payment cap can interrupt this mechanism; the product should test those controls rather than assume extraction has priority over the driver's ability to keep working.

1.2.3. Product 3: Revolving Credit Lines

Revolving credit lines, with an illustrative range of KES 25,000 to KES 100,000, are working-capital buffers for lumpy operating shocks such as servicing, tyre replacement, medical expense, or vehicle licensing. A viable driver may draw for a repair and repay as earnings recover. The line supports uptime only when the repair, limit, price, repayment design, and later income remain workable.

Let $U_i(t) = B_i(t)/L_i$ be the credit utilisation rate for driver i at time t , where $B_i(t)$ is the outstanding balance and L_i is the total credit limit. Under normal conditions, $U_i(t)$ can rise and fall as the driver cycles draws and repayments. Under a sustained structural income shock, the line may stop smoothing temporary operating costs and begin financing a persistent household or business deficit. This is the **adverse utilisation trap**. In the deliberately narrow stress model where the deficit persists, undrawn capacity remains available, no external support arrives, and draws mechanically fund the shortfall, utilisation approaches the limit:

$$\lim_{h \rightarrow \infty} P(U_i(t+h) \geq u^* \mid D_{i,t:t+h} > 0, \text{draw access, no cure}) = 1.$$

Here, $D_{i,t:t+h} > 0$ denotes a persistent net deficit and u^* is the operational full-utilisation threshold. This is a limiting mechanism under stated assumptions, not a claim that observed drivers inevitably max out. When many drivers approach u^* together, revolving assets can extend in behavioural maturity and EAD can rise. LGD may also rise if the same shock weakens recoveries, but that relationship must be estimated rather than inferred from utilisation alone.

1.3. Partnership and Counterparty Structure

Operationalising the triple-product stack requires several counterparties, not a single synthetic “issuer.” The **Mwendo Pamoja platform** occupies the central technology and servicing node: it maintains contracted integrations, hosts the driver experience, calculates features and decision-support outputs, and orchestrates authorised payment instructions. It does not issue insurance. It does not originate regulated credit unless the relevant entity has authority to do so. It may retain first-loss exposure or other contingent obligations, so the HoldCo should not be described as risk-free.

The **licensed insurer** writes the commercial motor policy, owns the actuarial and policy decision, holds insurance risk, and administers cancellation and any refund under the policy. The **licensed lender or finance provider** originates each microloan, revolving facility, or IPF exposure under its own authority and owns the customer credit terms before any eligible transfer. The three instruments remain legally and operationally distinct even when they protect the same productive vehicle.

The **SPV** is a ring-fenced financing and receivables-holding vehicle. It is not the insurer, the originating lender, the technology platform, or the customer-facing product issuer. Under a documented eligibility schedule, it may purchase only receivables and associated rights that the seller owns and can legally transfer. Eligibility should test executed documentation, payment and arrears status, currency, product and policy status where relevant, platform and geographic concentration, data completeness, prior security, and any other condition negotiated with investors. The trustee, account bank, servicer, backup servicer, calculation agent, and security trustee then perform the separate duties allocated in the transaction documents. Controlled accounts direct eligible collections through the agreed priority of payments.

The legal separation is achieved by substance, not by the SPV label. True sale, enforceability, perfection, account control, servicing continuity, commingling protection, set-off, tax, data transfer, and insolvency treatment require Kenyan legal and transaction analysis. Part 5 develops the capital stack, asset-liability management, matched-funding attribution, overcollateralisation, Cash Reserve Account, note pricing, waterfall, and investor cash-flow mechanics in detail.

IPF requires an additional distinction among **premium**, **premium funding**, and the resulting **finance receivable**. Section 156 of the Kenyan Insurance Act requires premium receipt by the insurer before risk is assumed and restricts an intermediary from receiving premium on the insurer’s behalf [11]. The base architecture therefore routes the financed premium directly,

or through another legally approved controlled mechanism, to the licensed insurer. Once that funding has occurred, the lender may hold a separate contractual receivable against the driver. The SPV may purchase that IPF finance receivable and validly assigned related rights if it generates an identifiable cash flow, is legally transferable, and satisfies the transaction's true-sale and eligibility requirements [12], [13]. It should not purchase or account for insurer-owned or unremitted premium money as an ordinary loan asset.

One financing SPV can therefore be designed to hold eligible microloan, revolving-credit, and IPF finance receivables without pretending that the three assets are homogeneous. Each product requires its own subledger, eligibility rules, collection code, loss and recovery curve, concentration test, and reporting line. IPF also requires a distinct premium-funding rail and cancellation-refund process. Any assigned net refund follows the policy, finance agreement, account-control arrangements, and waterfall. A separate IPF series, compartment, warehouse, or legal SPV remains available if counsel, a lender mandate, carrier exposure, asset-homogeneity analysis, or insolvency treatment makes stronger separation appropriate.

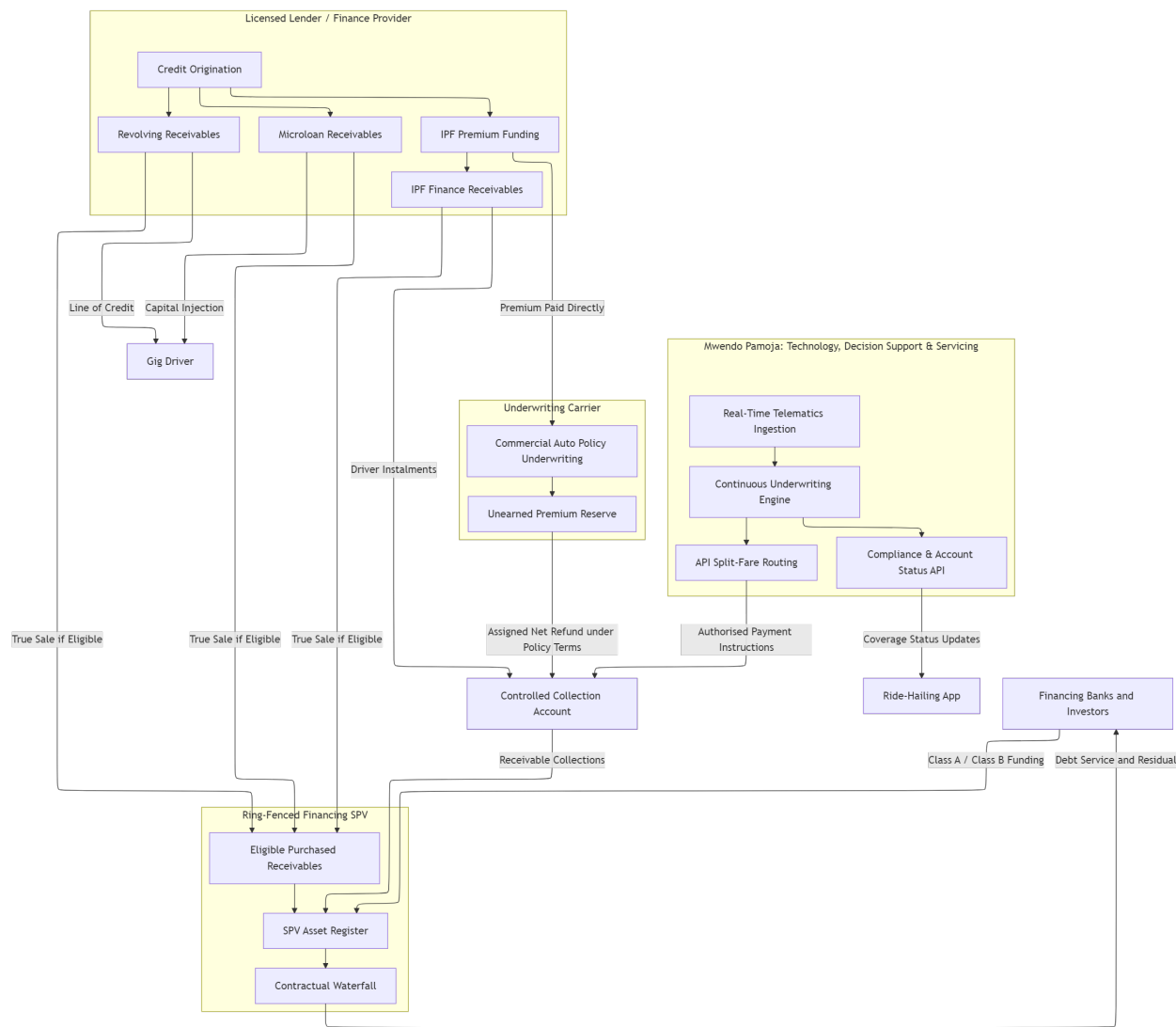
2. Siloed Product Mechanics and Structural Interdependencies

2.1. Microloan Repayment Mechanics

Microloans in this context can reduce payment friction through authorised, high-frequency repayment. Instead of relying only on a month-end transfer, a contracted **split-fare API** can allocate a defined share r of eligible settlement cash. The repayment amount routes to a controlled collection account or product ledger under the actual account structure, while the residual reaches the driver's wallet. The mechanism requires clear authority, reversals, refunds, disputes, downtime handling, and a sufficiency control.

The mechanism can reduce payment friction during high-earnings periods, but it creates a structural asymmetry. The deduction rI_{gross} follows eligible gross settlement, while the suffocation threshold depends on cash remaining after operating and essential commitments. During a fuel shock, a driver may need more gross fare merely to preserve the same net income. Without a floor or cap, deductions can then compete with the fuel required to earn tomorrow's fare. Priority is a contractual and policy choice, not a constitutional fact, and the platform should preserve enough working cash for a viable driver to continue operating.

Figure 1: Partnership and Counterparty Structure



2.2. Revolving Credit Lines as Working Capital Buffers

Revolving credit lines are intended to act as operational shock absorbers. A transmission failure requiring an illustrative KES 60,000 repair can sideline a driver and interrupt income. Where repair capacity, parts, and platform reactivation permit, a revolving line may finance the repair and support return within 24 to 48 hours, followed by repayment over later earnings cycles. Those timings and outcomes remain pilot assumptions.

The revolving product's risk profile can change during a persistent market-wide income downturn. A useful diagnostic is sustained rising utilisation combined with declining Repayment Velocity. Monitoring should distinguish a temporary operating shock, where utilisation later cures, from a persistent deficit, where draws remain high and repayment weakens. Neither state is inferred from one observation, and the policy response should account for uncertainty and the reason for the draw.

2.3. Insurance Premium Financing Operational Asset Structure

The cancellation-refund mechanism that supports IPF under ordinary conditions can still amplify a cascade [3]. After a missed payment, the policy may pass through notice, cure, cancellation, and refund states defined by its terms. Cover cannot be inferred from the loan status. If cover genuinely ends and the platform requires active proof, earning access may be suspended. A severe collision, cancellation, or platform block can then damage several products through their shared dependence on the vehicle and settlement stream. The products do not legally default at one instant; their arrears, cure, default, recovery, and accounting clocks remain distinct.

3. The Mechanics of the Correlated Default Cascade

3.1. Anatomy of an External Shock

The central vulnerability of the hybrid insurtech and embedded-finance portfolio is exposure to common shocks [4]. Thousands of drivers do not create full diversification when many depend on the same platforms, fuel markets, urban demand, payment rails, vehicle types, or regulatory setting. Exposure is still heterogeneous, and no shock affects every borrower identically. The portfolio model must estimate concentration and conditional dependence instead of treating borrower count as diversification by itself.

Three primary shock categories are relevant:

- **1. Macroeconomic Cost Shocks:** A sustained spike in global crude oil prices translates into localized fuel price increases. Fuel is the primary variable operating cost C_{ops} for ride-hailing drivers. This shock immediately compresses net margins without any change in gross revenue, bringing drivers below the suffocation threshold $I_{net} < S$ en masse.
- **2. Platform Algorithmic Shocks:** Ride-hailing platforms frequently adjust algorithms, changing base fare rates, surge pricing multipliers, or their own take-rates $\tau_{platform}$. A 5% increase in the platform's commission fee directly reduces every driver's net revenue without any corresponding change in passenger demand. The financial impact is equivalent to a proportional cut in gross fares, with no recourse for the driver.
- **3. Consumer Demand Contractions:** A macroeconomic recession or drop in local economic activity reduces consumer discretionary spending on ride-hailing services. Longer inter-trip wait times reduce the driver's effective hourly earning rate even without any change in per-trip economics, causing I_{gross} to decline as a function of platform utilization.
- **4. KESONIA and funding-cost shocks:** KESONIA reflects overnight unsecured KES interbank transactions and can affect floating-rate funding and, where contracts permit, customer rates [7]. Transmission is not instantaneous or universal. It depends on the product contract, reset date, compounding convention, margin, fixed-rate period, caps, customer notice, and applicable pricing rules. The cascade model therefore treats repricing as a timed scenario, not as an automatic jump.

3.1.1. Three credit clocks between strain and realised loss

The earliest sign of strain, the contractual credit event, and the eventual cash loss do not occur on one clock. The architecture therefore follows each driver-product-risk episode as an observation unit (i, p, e, t) : driver i , product p , risk episode e , and decision time t . This keeps a morning liquidity warning from being confused with a later missed payment, a still later default definition, or the recovery cash that may arrive months afterwards.

The first clock is the **operating and decision clock**. Trips, wallet movements, vehicle availability, policy status, fuel costs, and repayments change the evidence available at a particular decision time. The second is the **contractual performance clock**. Amounts become due, cure periods open and close, revolving limits change, policies move through their own states,

and each product reaches arrears or default only under its approved definition. The third is the **loss and recovery clock**. EAD can rise before default, cure can restore performance, IPF can generate an eligible cancellation refund, recoveries arrive with costs and delay, and the SPV experiences the resulting monthly cash shortfall through its waterfall.

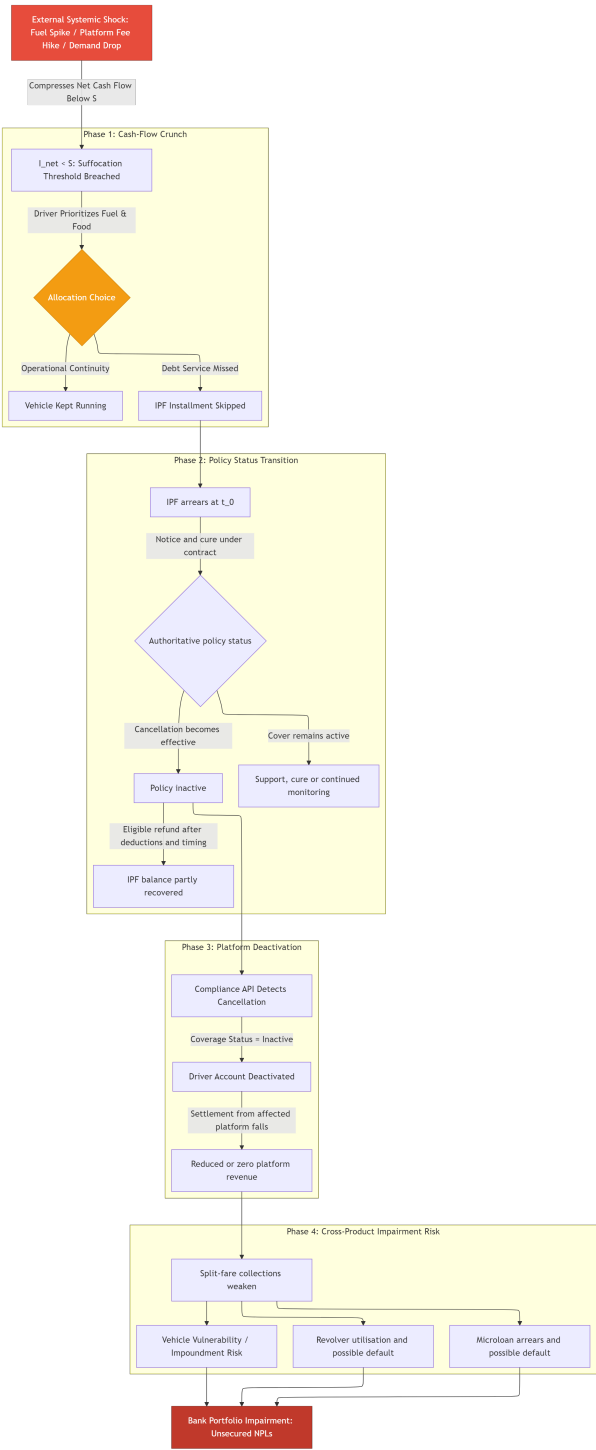
This separation matters economically. An intervention may improve the operating clock without erasing an accounting fact; a cure may improve the contractual clock without returning every collection on its original date; and a low fixed-horizon PD may coexist with a material liquidity effect if the expected collection profile becomes slower. Parts 2a and 2b turn the three clocks into point-in-time features, default horizons, survival paths, and calibrated loss components. Part 5 carries them into cohort cash flows, reserves, tranche loss, and investor timing.

The apparent contagion also has three distinguishable sources. **Shared-factor dependence** arises when fuel, demand, platform rules, payment rails, or geography affect many drivers together. **Within-driver dependence** arises because one vehicle and settlement stream support several products. **Transmission dependence** arises when a product or policy event changes earning capacity, collection routing, or available liquidity and thereby changes another exposure. The hierarchy absorbs observed and partially pooled common structure first; residual joint-tail dependence is modelled only after those channels have been represented.

3.2. The Multi-Product Default Domino Effect

An external shock can trigger a sequential, multi-product default cascade. The mechanism is path-dependent, not deterministic: reserves, another income source, insurer and lender hardship options, repair support, platform portability, and the timing of cancellation can interrupt it. Figure 2 maps the vulnerable path so the architecture can intervene at each edge.

Figure 2: The Multi-Product Default Domino Effect



- **Phase 1, cash-flow crunch:** An external shock can compress I_{net} below S . The driver may allocate scarce cash to fuel, food, repair, insurance, or debt according to urgency and available support. An IPF payment may become overdue at t_0 .
- **Phase 2, policy-status transition:** The finance provider and insurer follow the notice, cure, cancellation, and refund process in the executed contracts. Cover may continue during a cure period. If cancellation becomes effective, an eligible net refund may reduce the IPF balance after timing, deductions, and other contract effects. The driver must receive accurate status and an opportunity to act where the terms provide it.
- **Phase 3, platform eligibility:** Where the mobility platform requires the affected cover, an authenticated cancellation event can enter its compliance process. The platform owns the eligibility decision, timing, notice, correction, and appeal. Deactivation can reduce settlement income from that platform, but the model must still observe other platforms and lawful income sources.
- **Phase 4 - Credit impairment risk:** With the platform account deactivated, that platform produces no settlement events. Microloan and revolving collections can fall together, especially where the driver has no substitute platform or income source. Arrears, default, and cure still follow their own contractual definitions and timing. The model must estimate those transitions rather than assume simultaneous legal default.

3.3. The Diversification Illusion: Endogenous Default Correlation

Traditional credit risk portfolio management relies on the assumption that a diverse portfolio of financial products is structurally hedged against individual obligor failures. The mathematical representation of portfolio variance σ_p^2 for a multi-asset portfolio is:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

where w_i are portfolio weights, σ_i^2 are product loss variances, and ρ_{12} is pairwise dependence under a stated horizon and scenario. No universal 0.15 to 0.30 correlation is assumed for these products. Dependence varies with borrower overlap, platform, geography, vehicle, income source, policy status, fuel exposure, and the severity of the state being measured.

In the gig-economy insurtech-banking partnership, diversification can be overstated because products share the driver's earning system and portfolios share platforms, places, vehicles, and operating costs. KESONIA is a common factor only for exposures whose contractual pric-

ing or funding resets to it. Under ordinary conditions, idiosyncratic events can dominate. Under a systemic shock, dependence can rise materially, but unity is a limiting stress idea rather than an empirical forecast:

$$\rho_{ij}(x) \uparrow \rho_{ij}^{\text{stress}} \quad \text{as shock severity } x \text{ increases,} \quad 0 \leq \rho_{ij}^{\text{stress}} \leq 1.$$

Under these conditions, the joint probability of simultaneous default across all three products:

$$P(D_{\text{micro}} \cap D_{\text{revol}} \cap D_{\text{IPF}})$$

can increase non-linearly. Simple constant-correlation specifications may understate joint downside risk when dependence changes by regime. A Gaussian copula has zero asymptotic tail dependence for correlation below one, although it can still exhibit finite-threshold association. The observed shape of gig-economy dependence must be estimated rather than declared. This motivates testing the **Clayton copula** in Part 2b alongside independence, Gaussian, Student-*t*, rotated, mixture, and stress-overlay alternatives. Clayton supplies lower-tail dependence with zero upper-tail dependence:

$$\lambda_L = 2^{-1/\alpha_c} > 0 \quad \lambda_U = 0$$

If fitted α_c rises across stress regimes, estimated lower-tail dependence rises. That relation is a model output subject to marginal calibration, sample size, regime definition, and validation, not proof of absolute lockstep default.

4. Portfolio-Level Strategic Risk Mitigation

To reduce the probability and severity of a correlated cascade, the parties need controls at the points where one product transmits stress to another. The four mechanisms below are design proposals. Their effect on dependence must be measured through intervention experiments and portfolio outcomes; a control does not directly “set” a copula parameter.

4.1. Cross-Product Exposure Caps

Banks must abandon independent credit limits for individual products and instead calculate a **dynamic aggregate exposure cap** $EC_i(t)$ for each driver, continuously updated from real-time telematics and transaction data:

$$EC_i(t) = f(EV_i(t), DLR_i(t), U_i(t), R_i(t), PD_i^{\text{cal}}(t), \mathcal{U}_i(t), X_t),$$

where EV is Earnings Velocity, DLR is the Dynamic Debt-to-Liquidity Ratio, U is utilisation, R is the driver reserve balance, PD^{cal} is calibrated posterior default probability, $\mathcal{U}$ is model uncertainty, and X_t is external stress. The function is bounded by licensed-lender product policy, affordability, consent, and contractual rules. Thresholds such as $U_i(t) > 0.80$ remain calibration candidates. The final Credit Policy and Compliance Gate, not the feature service or HLR alone, freezes draws or changes limits.

4.2. Dynamic Premium Models (Usage-Based Insurance)

The fixed monthly IPF instalment may be poorly aligned with variable weekly earnings. Subject to insurer ownership, approved rating, policy wording, floors, caps, and data quality, the partnership can test a usage-sensitive premium model:

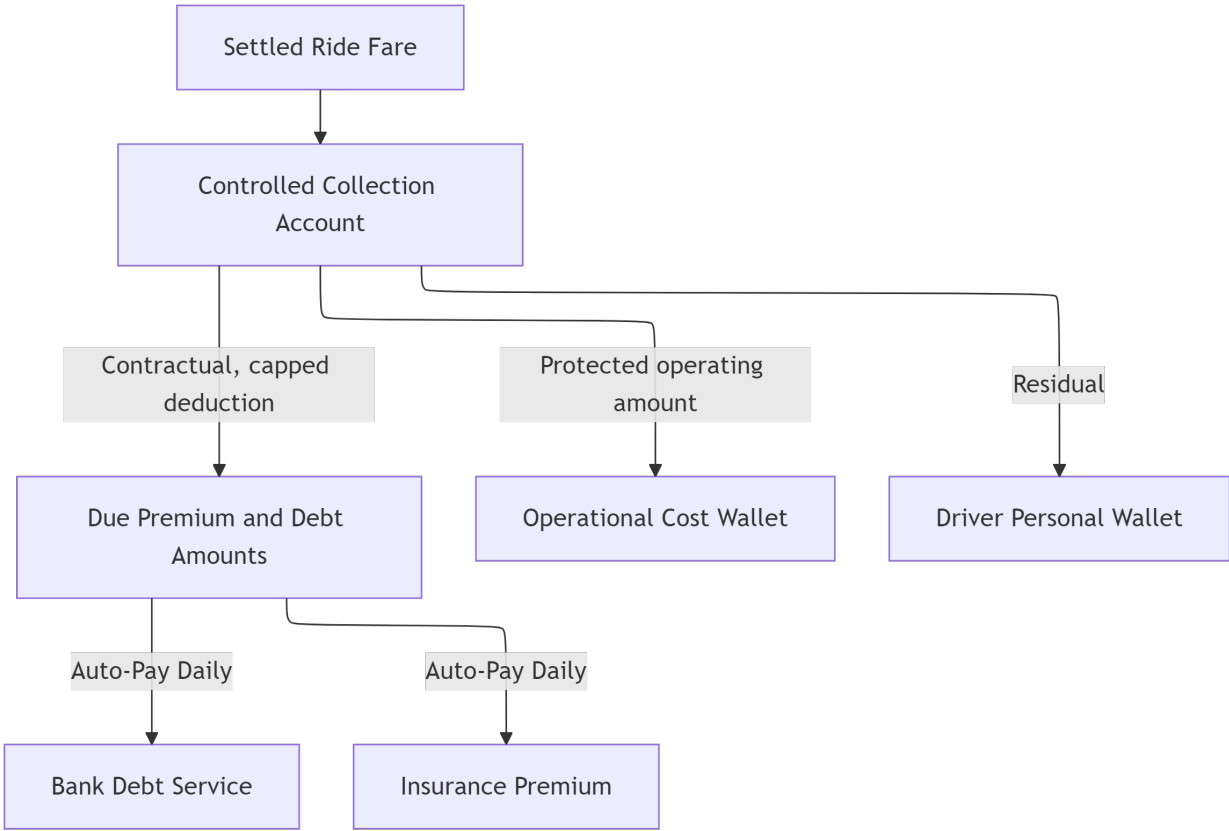
$$\text{Premium}_{\text{daily}} = \text{Base}_{\text{static}} + \gamma \cdot \text{MilesDriven}_{\text{daily}}$$

where $\text{Base}_{\text{static}}$ covers defined fixed exposure, γ is an actuarially approved variable rate, and $\text{MilesDriven}_{\text{daily}}$ is a validated exposure measure. During a low-mileage period the variable component falls, but the fixed component, taxes, fees, and minimum premium may remain. The design can reduce payment mismatch; it cannot eliminate lapse risk.

4.3. Controlled Platform Collection Accounts

Where the driver, platform, originator, servicer, account bank, and other required parties have executed the necessary terms, the platform can send an authorised portion of settlement to a controlled collection account. The account should not be held out as “Insurtech-managed escrow” unless its legal status, owner, bank, trust or security arrangement, permitted withdrawals, reconciliation, and insolvency treatment support that description.

Figure 3: Contractual split settlement through a controlled account



The split reduces collection friction but does not guarantee policy continuity or debt service. Amounts due, sufficiency protection, reversals, refunds, hardship, disputes, platform downtime, cash payments, and other income sources must be handled. A collection mechanism should stabilise the productive system rather than remove so much working cash that it causes the failure it was meant to prevent.

4.4. Collateralized Reserve Pockets

One pilot design funds a driver reserve pocket with 5% of principal and a 1% fare sweep. These are illustrative parameters. Ownership, disclosure, withdrawal, yield, security, affordability, tax, exit, and treatment on default require written terms. The driver reserve is distinct from the SPV Cash Reserve Account. Let $R_i(t)$ be the driver reserve balance. A permitted cure is:

$$R_i(t + 1) = \max\{0, R_i(t) - \Delta P_i(t)\},$$

where $\Delta P_i(t)$ is the authorised amount applied to the shortfall. A two-week target remains a pilot input. The reserve balance is an Explicit Liquidity Feature that enters the HLR directly, alongside a separately defined DLR. It should not be buried inside DLR in one paper and treated as an independent feature in another without an explicit decomposition.

Part 2a follows the information journey from edge capture through event-time processing and explicit feature ownership. Part 2b asks how neural representations and interpretable liquidity measures can enter a redundancy-controlled HLR, then tests candidate copulas for joint tail risk. Part 3 translates model outputs into interventions, accounting evidence, policy controls, and a Kenya-first governance map, using Basel, Solvency II, and foreign fair-lending frameworks as comparative references rather than automatically applicable Kenyan law.

The narrative has moved from a driver's working day to a portfolio dependence problem. The next part begins one level deeper: what exactly must be recorded, at what timestamp, and by which feature path, so that the mathematics does not learn from the future or count the same liquidity signal twice?

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Part 2a

Advanced Predictive Modeling, Deep Temporal Representation, and Feature Engineering

Mwendo Pamoja White Paper

1. Introduction to Continuous Underwriting Data Architectures

Part 1 ended at the moment a driver, a vehicle, and three financial products begin to fail together. Part 2a moves inside that moment. It asks what the platform could actually have known before the cascade, which timestamp makes that knowledge legitimate, how a physical signal becomes a feature, and how a feature reaches a model without being counted twice.

The ambition is to predict and interrupt avoidable failure before it reaches the loss ledger. A bureau score remains valuable evidence of prior credit behaviour; it does not have “zero” predictive weight. Its limitation is temporal resolution. A driver may cross the sufficiency boundary $I_{\text{net}} < S$ between reporting cycles after a repair, fuel, platform, or demand shock. Higher-frequency information may improve the timing of a response, provided it is consented, point-in-time correct, relevant, stable, and tested against simpler baselines.

Two structural failure modes motivate the architecture. First, **temporal blindness**: a feature can reflect the last bureau pull rather than the decision moment. Second, **population and label mismatch**: a model trained on a different employment and product population may not transfer cleanly to variable platform income. The empirical distribution may be skewed, heavy-tailed, seasonal, zero-inflated, or regime-dependent; it should not be declared Gaussian or power-law without testing. The design therefore keeps distributional assumptions visible and validates them by product, platform, geography, and time.

The system complements conventional evidence with a daily or event-triggered posterior distribution. A GRU branch learns short-term operational and sequence state [1]. A Transformer branch represents slower macroeconomic, platform, and regulatory context [2]. A separate Explicit Liquidity Feature Path calculates CFA, DLR, Earnings Velocity, Repayment Velocity, wallet volatility, reserve balance, time since depletion, utilisation, and defined interactions. These engineered financial variables do not enter either neural encoder. The branches meet only at the redundancy-controlled HLR in Part 2b.

Point-in-time controls can materially reduce look-ahead leakage, but no library call can guarantee it. Source corrections, late events, label construction, clock skew, backfills, and changing definitions still require tests and lineage.

2. Edge Capture: Hardware Layer and Raw Data Format

Before any streaming architecture can be discussed, the physical origin of the data must be understood precisely. The gig driver's vehicle is a rolling sensor array. At the telematics hardware level, three distinct sensor families generate the raw input streams:

IMU / gyrometer layer: The Inertial Measurement Unit records tri-axial linear acceleration (a_x, a_y, a_z) and angular velocity ($\omega_x, \omega_y, \omega_z$). At 10 Hz, six channels yield 60 scalar readings per second before other sensor fields. Thresholds such as $|a_{\text{brake}}| > 0.6g$ for 500 milliseconds are candidate event definitions, not universal facts. Road geometry, device mounting, vehicle type, speed, calibration, and sensor noise can change the observed pattern. Fatigue is not directly observed by yaw oscillation; it is a risk hypothesis requiring safe validation.

GNSS Layer: The Global Navigation Satellite System receiver generates latitude, longitude, altitude, instantaneous speed (km/h), and heading at 1 Hz (1 sample per second). Speed data cross-validates the OBD-II reading, enables geofence computation, and provides the ground-truth mileage feed for the UBI per-kilometer premium calculation $\text{Premium}_{\text{daily}} = \text{Base}_{\text{static}} + \gamma \cdot \text{MilesDriven}_{\text{daily}}$.

OBD-II / CAN-bus layer: Subject to vehicle compatibility and permission, the interface can expose engine RPM, throttle position, fuel level, manifold pressure, coolant temperature, and Diagnostic Trouble Codes. A code such as P0301 indicates a cylinder-one misfire condition; it does not create a universal 72-hour forecast of engine failure. The useful feature is the code, recurrence, severity, associated measurements, service history, and verified outcome.

On arrival at the cloud infrastructure, all sensor streams converge into tabular records in PostgreSQL or MySQL operational databases with the schema:

```
timestamp (BIGINT, unix microseconds)
vehicle_id (UUID)
lat (FLOAT8), lon (FLOAT8), altitude_m (FLOAT4), speed_kmh (FLOAT4)
accel_x, accel_y, accel_z (FLOAT4)    -- m/s^2
g_force_lateral, g_force_longitudinal (FLOAT4)
angular_vel_x, angular_vel_y, angular_vel_z (FLOAT4)  -- rad/s
rpm (INT), fuel_pct (FLOAT4)
dtc_codes (TEXT[])
```

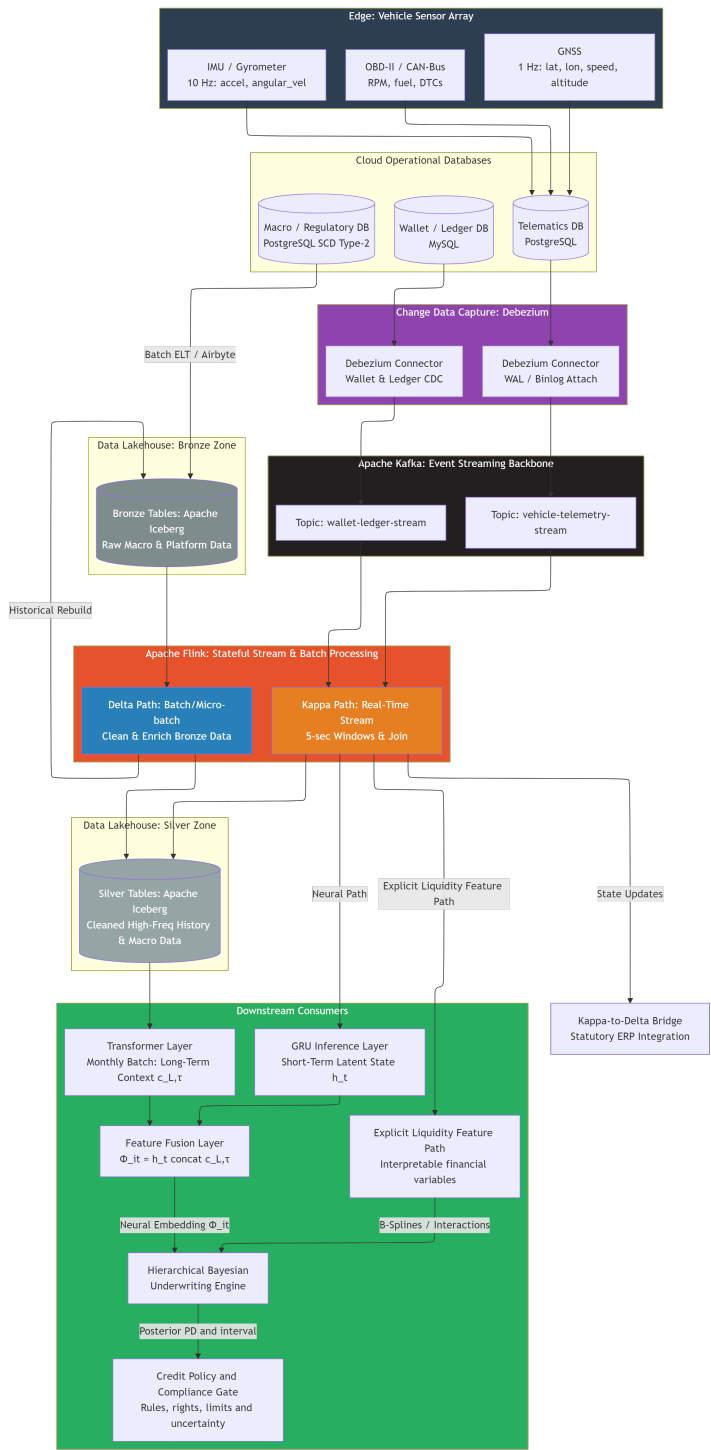
A fleet of 50,000 vehicles sampled continuously at 10 Hz would generate 43.2 billion device-time observations per day before aggregation, or far more scalar readings if every sensor chan-

nel is stored separately. Real duty cycles, edge aggregation, compression, sampling policies, connectivity, and retention will change the volume. The calculation is therefore a capacity envelope, not a production forecast. It nonetheless explains why raw sensor ingestion should be decoupled from transactional ledgers and why event streaming, edge summarisation, and tiered storage are required.

3. Architectural Implementation: Refined Data Engineering Pipeline

The Insurtech's data engineering backbone implements a modern **Lakehouse streaming architecture** organized around four sequential layers: Change Data Capture, Event Streaming, Real-Time Stream Processing, and Open-Format Lakehouse storage. This pipeline ensures that high-frequency telematics events are scored in near-real-time by the GRU inference layer, while simultaneously building the immutable long-term historical record that the Transformer queries for macro context encoding, as detailed in **Figure 1**.

Figure 1: Multimodal Data Engineering Pipeline



3.1. Change Data Capture via Debezium

Debezium can read the PostgreSQL Write-Ahead Log or MySQL binary log and publish committed row-level changes without periodic table scans. A connector envelope can include before, after, operation, source metadata, and processing timestamps in Avro or JSON. Field availability and timestamp meaning depend on connector and configuration [3]. End-to-end latency such as 50 milliseconds is a benchmark target, not an architectural fact. More importantly, source commit time, connector time, ingestion time, and event time must not be collapsed into a misleading two-clock story.

Log-based CDC can reduce polling load, but snapshotting, log retention, connector lag, back-pressure, schema changes, and recovery still require controls. Telematics does not normally originate in a relational WAL; its device or gateway path is separate. A 10 Hz stream and sub-second downstream latency are capacity and service-level assumptions to test on representative devices, networks, partitions, and failure modes.

3.2. Apache Kafka: Fault-Tolerant Event Streaming

Connectors publish changes to Kafka topics organised by data domain. A `vehicle-telemetry-stream` topic may be keyed by `vehicle_id` so records with the same effective key are assigned to the same partition under a stable partitioning strategy. Kafka preserves order within a partition, not across partitions [4], and producer retries, key changes, duplicate delivery, late device events, and replay still require sequence numbers, idempotency, event-time buffering, and tests. The GRU input builder, rather than Kafka alone, is responsible for reconstructing the ordered sequence $\mathbf{x}_{t-1}, \mathbf{x}_t$.

Kafka's retention policy implements a tiered storage strategy: a 7-day hot window on NVMe SSD for low-latency GRU stream consumption, with automated tiered offload to object storage (S3 or GCS) for long-term archival. The archived topics serve as the historical data source for Transformer training and periodic full-model retraining.

3.3. Apache Flink: Stateful Real-Time Stream Processing

Apache Flink consumes the Kafka topics via the Flink SQL or DataStream API, executing four classes of real-time operations in parallel [5]:

- **1. Tumbling Window Aggregations:** 5-second tumbling windows aggregate raw telematics into the per-window statistics required for feature engineering, mean speed,

peak G-force, count of hard-braking events, fuel consumption delta. These window outputs serve as the input vectors $\mathbf{x}_t^{\text{tele}}$ for the GRU layer.

- **2. Hard-Braking and Operational Stress Rules:** Stateless rule evaluators flag individual events exceeding configurable thresholds. A hard-braking event is flagged when $|a_{\text{brake}}| > 0.6g$ sustained over 500ms. Fatigue detection fires when cumulative consecutive driving hours within a single session exceed a configurable threshold (typically 10 hours without a 30-minute break). DTC code detection flags any non-empty `dtc_codes` array, immediately triggering an alert to the underwriting engine.
- **3. Temporal Table Joins for Point-in-Time Correctness:** Flink's `FOR SYSTEM_TIME AS OF` syntax implements zero-look-ahead joins between the high-frequency telematics stream and low-frequency macro state tables:

```
SELECT
    t.vehicle_id,
    t.window_end_ts,
    t.mean_speed_kmh,
    t.peak_g_force,
    m.gas_price_index,
    m.platform_commission_pct,
    r.zev_mandate_tier
FROM telematics_windows AS t
JOIN macro_table FOR SYSTEM_TIME AS OF t.window_end_ts AS m
    ON t.jurisdiction_code = m.jurisdiction_code
JOIN regulatory_table FOR SYSTEM_TIME AS OF t.window_end_ts AS r
    ON t.jurisdiction_code = r.jurisdiction_code
```

The intended temporal join selects the version valid and available at the historical decision time. The exact Flink syntax depends on whether the table is versioned by event time or processing time and on connector support. The test suite must prove the behaviour under late events, corrections, backfills, and out-of-order arrival.

- **4. ACID Lakehouse Sink:** In parallel with GRU inference feeding, Flink sinks all processed records to the Lakehouse via Apache Iceberg's transactional write protocol. Iceberg snapshot isolation supports consistent reads of committed historical state for Transformer training [6].

3.4. Data Lakehouse: Unified Tiered Storage (The Delta Architecture)

A Delta Architecture (often called a Data Lakehouse) is designed to store massive amounts of historical data reliably so that data scientists can train AI models on it offline. It uses a concept called the “Medallion Architecture” to represent data as it gets cleaner and more refined. The Lakehouse implements a three-tier table architecture:

- **Bronze tables (retained source events):** This layer preserves received payloads, source metadata, schema version, and arrival lineage. It is evidence of what the platform received, not proof that the upstream fact was correct. Versioned macro, regulatory, and platform configuration records should distinguish effective time from knowledge and ingestion time rather than relying on `valid_from` and `valid_to` alone.
- **Silver Tables (Cleaned & Filtered):** The data engineering pipeline (Apache Flink) cleans the Bronze data. It handles missing values, converts timestamps, and partitions the data by `vehicle_id` and `date`. Each partition contains the chronologically ordered sequence of Flink-windowed feature vectors for a specific driver. These are the input sequences consumed by the Transformer’s monthly batch processing to construct the 12, 24 month historical token window $\mathbf{X}^{\text{macro}}$.
- **Gold tables (approved analytical and integration views):** Gold views can hold reconciled KESONIA calculations, approved risk and accounting inputs, SPV waterfall results, and reporting aggregates. They are not statutory ledgers and should not be called immutable merely because they are curated. Product subledgers, the SPV register, and D365 remain authoritative for their allocated records, while Gold views preserve lineage to the approved source events.

The Problem with Delta for Real-Time Underwriting: Delta architecture relies on writing files to disk (like AWS S3 or GCS). Writing and reading from disk takes seconds or minutes. If a driver requests a microloan, the system cannot wait 30 seconds for the Delta Lake to retrieve their wallet balance. Therefore, while the Gold layer is the ultimate destination for statutory accounting (resolved via the **Asynchronous Kappa-to-Delta Bridge** detailed in **Part 4**), the actual real-time credit decision must bypass it.

3.5. The Kappa Architecture (The “Stream”)

The Kappa Architecture was invented to solve the speed problem. In a Kappa architecture, everything is treated as a continuous stream of events. There is no “batch” database to query for real-time decisions. When data streams in (like a Debezium CDC event showing a driver

just spent \$10), a stream processor like Apache Flink updates the driver's total wallet balance in its internal RocksDB memory (or an external high-speed memory cache like Redis) instantly. Because the data lives in active memory (not written to a slow disk), the inference engine can fetch the driver's exact wallet balance in under 2 milliseconds.

3.5.1. The Mwendo Pamoja Case Study: Why Both Are Required

Consider a real-time scenario where a driver requests a microloan at 2:00 PM on a Tuesday. The system has 100 milliseconds to approve or deny it. This requires orchestrating data across both offline batch and real-time streaming architectures.

- **1. The Transformer (Using Delta / Low-Frequency):** The Transformer model needs to know the macroeconomic context (fuel prices, inflation, platform commission rates over the last 24 months). This data changes very slowly. Once a month, the system trains the Transformer on the massive historical Silver Tables in the Delta Lake. The Transformer outputs a single dense “context vector” ($\mathbf{c}_{L,\tau}$) that represents the current economy. That vector just sits quietly waiting to be used.
- **2. The GRU & Cross-Attention (Kappa / High-Frequency):** Simultaneously, the driver's real-time kinematic telemetry is processed by the GRU ($\mathbf{h}_t^{\text{GRU}}$). The system uses a Cross-Attention Fusion layer to fuse the GRU's current behavioral state with the Transformer's macro context vector, generating a single, unified neural representation of the driver's risk state (Φ_{it}).
- **3. The Explicit Liquidity Feature Path (Using Kappa / Liquidity):** The Bayesian risk model *also* needs to know the driver's exact Wallet Cash-Flow Asymmetry (CFA) right at this second. The system cannot query the Delta Lake for this; it's too slow. Instead, every time the driver earns or spends a cent, the Debezium stream updates the driver's CFA in a lightning-fast Redis cache (Kappa Architecture).
- **The moment of decision:** At 2:00 PM, the system retrieves the versioned neural representation and the latest valid Explicit Liquidity Features. The HLR produces a posterior PD and uncertainty summary; the Credit Policy and Compliance Gate then determines the permitted action. Forty-five milliseconds remains an end-to-end latency target with a defined percentile and load profile. The lakehouse supports reproducible history; the online path supports the present.

3.6. Low-Frequency Data Ingestion: Macro, Regulatory, and Platform Context

The Transformer's effectiveness depends on ingesting all structural factors that affect gig-driver earning capacity and credit risk on timescales longer than the GRU's daily window. Four categories are relevant:

- **Macroeconomic conditions** (fuel-price measures, CPI, exchange rates, and central-bank policy or benchmark rates): Monthly or release-driven ingestion uses Central Bank of Kenya publications and benchmark series, World Bank indicator APIs, and IMF SDMX or Data APIs [7]-[9]. Where a Kenyan source publishes files rather than an API, a scheduled ingestion job retrieves the authoritative release. Python Airflow DAGs commit results to Bronze SCD Type 2 tables with the publication date, effective date, source version, and revision lineage.
- **Platform algorithmic regimes** (base-fare floors, surge policies, and commission rates): Contracted webhooks, published notices, or other authorised interfaces can provide versioned configuration evidence. An illustrative one to four hour refresh target must be tested against what the partner actually exposes. The system must not assume access to a platform's internal configuration database.
- **Regulatory and ZEV mandates** (gig-worker reclassification laws, metropolitan vehicle permit caps, zero-emission vehicle transition deadlines): Web-scraping pipelines and legal API feeds, committed with the regulatory `effective_date` and `jurisdiction_code`. Mapped to geohash zones for regional driver impact computation.
- **Urban infrastructure events** (prolonged highway reconstructions, corridor capacity changes): Open Street Map and city open-data feeds. Relevant because permanent road closures or capacity degradation durably depress hourly earning capacity for all drivers operating specific geohash corridors.

4. Point-in-Time Correctness: The Asynchronous As-Of Join Matrix

One of the greatest engineering hazards in retrospective underwriting is **look-ahead bias**: allowing a historical feature to contain information that was not available at the decision time. If a training record at 2024-03-15 11:30 AM contains a gas-price observation first knowable to the platform at 2024-03-15 12:00 PM, perhaps because event time was confused with arrival

time, the back-test can learn a relationship unavailable in production. Apparent accuracy then disappears when the model meets the live clock.

The event envelope and source metadata should preserve at least business-effective time T_e , source commit time T_c , ingestion time T_i , and model decision time T_d . Field names depend on the connector and database and should not be inferred from precision suffixes alone. A necessary availability constraint is:

$$T_i \leq T_d, \quad T_e \leq T_d, \quad \text{version}(x, T_d) = \max\{v : T_i(v) \leq T_d\}.$$

A decision may only use a version that had arrived by the decision timestamp and whose effective-time treatment is valid for the feature definition. This prevents a known future version from entering the row when the source and pipeline honour the contract. It does not eliminate label leakage, incorrect effective dates, backfilled facts, or accidental reuse of present-day dimensions.

When no new low-frequency record has been committed since the last join, the pipeline applies a **Zero-Order Hold**: the last committed macro state is broadcast unchanged across all subsequent high-frequency events until a new commitment arrives. This is equivalent to the asynchronous decoupled clock cycle of the GRU/Transformer dual-regime architecture: the Transformer's $\mathbf{c}_{L,\tau}$ vector is held constant across all daily GRU cycles until a new monthly Transformer run produces $\mathbf{c}_{L,\tau+1}$.

The feature-store pattern formalises the retrieval contract. A spine containing decision time and entity identifiers is joined to versioned feature views using documented availability and validity rules. The returned training frame is accepted only after automated leakage tests, boundary cases, replay comparison, source-to-feature lineage, and sign-off on label timing.

4.1. The Credit-State Ledger and Five Clocks

Point-in-time correctness becomes operational when every training row can be reconstructed from five connected ledgers. The **identity and exposure ledger** links the driver, product, facility, contract, vehicle, platform, policy, geography, and risk episode. The **obligation ledger** records schedules, due amounts, limits, draws, repayments, policy finance, arrears, and modifications. The **decision and action ledger** records the model artifact, feature snapshot, calibrated estimate, policy version, human approval, intervention, notice, and override. The **outcome and maturity ledger** records product-specific default, cure, closure,

censoring, and the date on which a label became observable. The **recovery and cash-flow ledger** records collections, IPF refunds, write-offs, recoveries, costs, delays, and SPV allocation.

These ledgers preserve five times whose names should not be collapsed:

Clock	Meaning	Principal control
Event time T_e	When the underlying activity occurred	Source sequence and correction lineage
Availability time T_a	When the platform could lawfully and technically use it	Point-in-time feature join
Decision time T_d	When a score or action was produced	Immutable decision snapshot
Label-maturity time T_m	When the full product outcome window became observable	Training eligibility and censoring
Cash-realisation time T_c	When collection, refund, cost, or recovery affected cash	ECL, loss, and SPV cohort timing

A feature is eligible only when

$$T_a \leq T_d,$$

and a fully matured supervised label can enter a training cutoff T_{train} only when

$$T_m \leq T_{\text{train}}.$$

Rows that have entered the risk set but have not completed the outcome window are not silently labelled as non-defaults. The fixed-horizon HLR uses matured binary outcomes or a formally approved censoring treatment. The timing challenger in Part 2b uses the same ledgers to construct at-risk exposure intervals and right-censored event histories. This common source prevents the logistic and survival views from becoming two inconsistent credit histories.

5. Feature Engineering for the Gig Economy

Before feeding data into the neural architectures, raw telemetry and transactional logs must be engineered into financially meaningful feature vectors. The gig driver's risk profile demands features that capture liquidity constraints, operational stress, behavioral degradation, and structural macro exposure simultaneously.

5.1. High-Frequency Behavioral Stream (GRU Input Path)

Kinematic features: Speed EMA computed over 1-minute, 5-minute, and 15-minute trailing windows. Braking delta $\Delta|a_{\text{brake}}|$ (change in peak braking G-force between consecutive windows). Cornering lateral G-force. Angular velocity variance (yaw oscillation, indicative of fatigue-driven lane instability). These features collectively constitute the driver's real-time **operational stress signature**.

Temporal session markers: Consecutive driving hours within the current session (reset on breaks > 30 min). Shift-start time encoded as a cyclical feature: $\sin(2\pi h/24)$, $\cos(2\pi h/24)$ where h is the hour of shift start. Hours since last 30-minute break. Circadian fatigue accumulation: integration of trailing 7-day hours driven during the driver's biological sleep window (the highest-stress but highest-surge-opportunity period).

Permitted raw wallet-event sequence: The GRU may observe a minimised sequence of event type, direction, amount bucket, and time gap when the data basis permits. It does not receive CFA, DLR, Earnings Velocity, Repayment Velocity, wallet volatility, cash-out ratio, reserve balance, time since depletion, or deterministic transformations of them. Those engineered concepts belong to the Explicit Liquidity Feature Path. This boundary lets the neural branch learn timing and rhythm without giving the downstream HLR two versions of the same named financial variable.

Engineering transformations: EMA decay, window size, thresholds, and clipping are versioned hyperparameters estimated from training evidence. Scaling parameters are fitted without validation or future leakage, then frozen for the model version or updated through a governed recalibration. A rolling global scaler does not prevent drift and can conceal it. Irregular sampling receives a mask, age-of-data, coverage, and reason code where available.

Input tensor: $\mathbf{X}^{\text{tele}} \in \mathbb{R}^{B \times T_{hf} \times d_{hf}}$ where B is batch size, $T_{hf} = 14$ days (daily resolution lookback), and d_{hf} is the telematics feature dimension.

5.2. Low-Frequency Structural Context (Transformer Input Path)

Macroeconomic indicators: Gas price index (7-day percentage change from 30-day trailing mean, removes level effects; captures rate-of-change signal). CPI-adjusted real wage index for the driver's metropolitan area (normalizes nominal earnings against purchasing power drift). Central bank policy rate (affects driver consumer demand through discretionary income compression).

Platform configuration: Surge dampening coefficient (weekly update frequency). Base fare floor percentage change from the previous month. New competitor market entry indicator (binary flag; entry of a new ride-hailing competitor typically compresses surge pricing across the market). Algorithmic matching elasticity (internal metric proxying the localized dispatch density).

Regulatory environment: Emissions zone restriction intensity for the driver's primary jurisdiction (ordinal encoding: 0 = no restriction, 1 = announced pricing, 2 = active tolling, 3 = complete combustion ban). Gig-worker classification statute status (binary: reclassified to employee status / independent contractor). Metropolitan vehicle permit utilization rate (fraction of total licensed permits currently active, a proxy for market saturation).

Urban infrastructure: Major corridor disruption indicator for geohashes where the driver operates more than 30% of their trips. Highway reconstruction duration in weeks (longer duration = permanent earning capacity depression for affected geohash corridors).

Engineering: Target encoding for categorical regulatory zone identifiers (smoothed target encoding using a Bayesian hierarchical mean to prevent overfitting to small jurisdiction groups). Percentage change from trailing 30-day average for all numerical macro indicators.

Input tensor: $\mathbf{X}^{\text{macro}} \in \mathbb{R}^{B \times T_{lf} \times d_{lf}}$ where $T_{lf} = 24$ monthly tokens and d_{lf} is the macro feature dimension.

5.3. Driver Liquidity and Solvency Features (Explicit Liquidity Feature Path)

Architectural routing note: controlling redundancy The central vulnerability in a hybrid neural-Bayesian model is representation redundancy. The same raw event can legitimately be available to multiple services, but a named engineered liquidity metric must have one owner. Explicit metrics strictly bypass the neural encoders. Part 2b then residualises the neural representation against the explicit feature block out of fold, orthogonalises spline bases, and uses block shrinkage. These controls reduce instability and multicollinearity; identifiability still has to be tested.

Unlike slower contextual variables read from versioned lakehouse tables, the high-velocity Explicit Liquidity Features use a Kappa-style online path. Flink processes source events and materialises live state into a low-latency key-value store such as Redis. At decision time, the service fetches the latest valid versions and quality flags, routes them directly into the HLR, and records the feature timestamp and definition version. A sub-millisecond cache read is a performance target to be benchmarked, not a property guaranteed by selecting Redis.

Wallet Cash-Flow Asymmetry (CFA):

$$\text{CFA}_i(t) = \frac{\text{mean}(E_{i,d} : E_{i,d} \geq Q_{0.95,i}^{(w)})}{\text{median}(E_{i,d} : E_{i,d} > 0) + \varepsilon}.$$

Here $E_{i,d}$ is net settled daily earnings over window w , and $Q_{0.95,i}^{(w)}$ is the driver's 95th percentile. CFA is dimensionless. A large value means upper-tail earning days are much larger than the typical positive day; it does not by itself prove dependency or insolvency. Minimum observations, clipping, seasonality, and the relationship to downside sufficiency must be calibrated.

Dynamic Debt-to-Liquidity Ratio (DLR):

$$\text{DLR}_i^{(h)}(t) = \frac{D_i^{\text{due}}(t, t+h)}{W_i^{\text{available}}(t) + R_i^{\text{available}}(t) + S_i^{\text{earned but pending}}(t) + \varepsilon}.$$

The numerator is debt contractually due over horizon h , initially seven days. The denominator contains cash currently available for those obligations, any reserve amount legally and operationally available, and settlement already earned but pending when that feed is reliable. Assets with withdrawal restrictions are excluded or haircutted. DLR is dimensionless and horizon-matched. It does not divide total debt by a daily income average, which would mix maturity and liquidity concepts and duplicate more of Earnings Velocity.

Repayment Velocity:

$$v_{\text{repay},i}^{(h)}(t) = \frac{P_i^{\text{principal paid}}(t-h, t)}{P_i^{\text{principal scheduled}}(t-h, t) + \varepsilon}.$$

Fees, interest, reversals, prepayments, restructurings, and products with no scheduled principal receive explicit treatment. A zero scheduled denominator is “not applicable,” not evidence of failed repayment. The feature can be clipped for model stability while the raw ratio remains in the audit record.

Wallet Cash Flow Volatility (σ_{wallet}): A robust rolling dispersion measure of daily net settled earnings, reported with its window, scale, and missing-day rule. Two drivers with the same mean can have different liquidity risk. The direction and magnitude of the effect remain empirical because high variance can also accompany seasonal opportunity and strong buffers.

Time Since Last Zero-Balance Event, Exponential Recency Decay: A complete wallet depletion yesterday carries exponentially more predictive weight regarding an impending IPF lapse than a depletion from 24 months prior. This recency structure is captured via dual encoding:

- **Decay transform:** $e^{-\lambda \Delta T}$, where $\Delta T = T_{\text{current}} - T_{\text{last depletion}}$ in days. The initial $\lambda = 0.01$ is a candidate, not a calibrated fact until cohort evidence is available.
- **Binary indicator:** $1[\text{has_prior_observed_depletion}] \in \{0, 1\}$. The decay captures recency and the indicator distinguishes no observed depletion from a remote one. Neither proves permanent fragility, especially when wallet history is short or incomplete.

Earnings Velocity (v_{earn}): The ratio of average net settled earnings over the trailing seven days to the average over the preceding 90-day baseline:

$$v_{\text{earn},i,t} = \frac{\frac{1}{7} \sum_{d=t-6}^t E_{i,d}}{\frac{1}{N_B} \sum_{d=t-96}^{t-7} E_{i,d} + \varepsilon}.$$

Here N_B is the number of valid days in the preceding 90-day baseline, and missing or non-working days follow a documented rule. A value below one indicates recent earnings below the historical daily baseline, not necessarily suffocation; operating costs, obligations, seasonality, and buffers still matter. Its spline effect is learned rather than forced to be linear or monotonic.

Reserve Pocket Balance ($R_i(t)$): The driver-owned or contractually controlled reserve balance and a separately calculated ratio to due obligations. The feature registry must state ownership and availability. It is not the SPV Cash Reserve Account.

Processing: These features bypass the neural encoders. The governed main specification evaluates a moderately rich B-spline basis and applies a difference penalty, producing a P-spline effect whose roughness is controlled separately from knot count. The basis degree, boundary knots, interior knots, difference order, penalty matrix, centring transform, support range, clipping rule, and back-transformation are versioned in the feature registry. A QR or penalty-compatible reparameterisation may be used for numerical conditioning, but it must preserve the penalty's null space and interpretation. B-spline with RW1 and AR(1) coefficient

priors remain documented challengers in Part 2b's appendix rather than disappearing from the research programme. Half-Student- t priors may be used for positive group scales, while coefficient blocks receive regularising shrinkage [10]-[12].

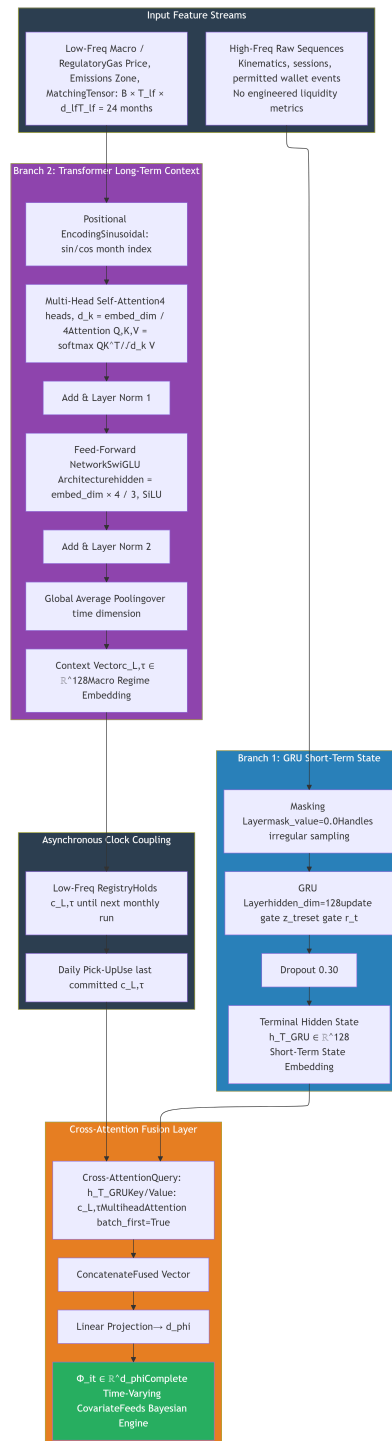
5.4. Gig-Economy Specific Interaction Effects

Standard credit models include generic interaction terms. In the gig-economy context, four specific pairwise interactions carry structural economic meaning. To prevent structural multicollinearity with the neural embeddings (which dynamically model implicit interactions), these four specific explicit interaction pairs (forming the set $\mathcal{G}$) **strictly bypass the neural encoder**. They are routed directly to the Hierarchical Bayesian Logistic Regression layer as orthogonal linear terms:

1. **High revolving utilisation $\times$ prior delinquency:** A strong-heredity interaction testing whether utilisation is more informative after delinquency. The 0.80 threshold is an illustrative policy knot, not a universal boundary or a diagnosis of income supplementation.
2. **Platform commission increase $\times$ high DLR:** Tests whether a commission change has a larger effect where due obligations already consume liquidity. DLR values such as 0.90 and 1.00 are candidate spline knots pending product and outcome calibration.
3. **Late-night driving concentration $\times$ microloan arrears:** Tests whether arrears and a change in working hours jointly predict risk. It is not labelled “desperation” because night work may be an ordinary preference or market response. The feature requires a within-driver baseline and careful fairness review.
4. **ZEV mandate proximity $\times$ ICE vehicle ownership:** Represents a scenario interaction only where a verified rule, date, jurisdiction, and affected vehicle class exist. It should not manufacture a mandate or deactivation event from an announcement.

6. The Dual-Regime Neural Architecture

The multi-scale feature extraction architecture processes the two neural feature streams - telematics (GRU) and macro (Transformer) - in parallel and fuses their outputs into a single high-dimensional covariate vector Φ_{it} . The financial liquidity variables completely bypass this neural network and are fed directly into the downstream Hierarchical Bayesian Logistic Regression engine alongside Φ_{it} , as illustrated in **Figure 2**.

Figure 2: The Dual-Regime Neural Architecture

6.1. Branch 1: Gated Recurrent Units for Short-Term Latent State

The GRU processes the high-frequency telematics tensor $\mathbf{X}^{\text{tele}} = [\mathbf{x}_1^{\text{tele}}, \dots, \mathbf{x}_{T_{hf}}^{\text{tele}}]$ sequentially, maintaining a hidden state $\mathbf{h}_t^{\text{GRU}} \in \mathbb{R}^{d_1}$ that accumulates the driver's behavioral trajectory. At each time step t , the GRU executes:

Update gate: determines how much of the previous hidden state to carry forward:

$$\mathbf{z}_t = \sigma(\mathbf{W}_z \mathbf{x}_t^{\text{tele}} + \mathbf{U}_z \mathbf{h}_{t-1}^{\text{GRU}} + \mathbf{b}_z)$$

Reset gate: controls how much of the previous hidden state is relevant to the candidate:

$$\mathbf{r}_t = \sigma(\mathbf{W}_r \mathbf{x}_t^{\text{tele}} + \mathbf{U}_r \mathbf{h}_{t-1}^{\text{GRU}} + \mathbf{b}_r)$$

Candidate hidden state: the proposed new content for the hidden state, modulated by the reset gate:

$$\tilde{\mathbf{h}}_t = \tanh(\mathbf{W}_h \mathbf{x}_t^{\text{tele}} + \mathbf{U}_h (\mathbf{r}_t \odot \mathbf{h}_{t-1}^{\text{GRU}}) + \mathbf{b}_h)$$

Final hidden state: interpolation between old and candidate states, controlled by the update gate:

$$\mathbf{h}_t^{\text{GRU}} = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1}^{\text{GRU}} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$$

where $\sigma(\cdot)$ denotes the sigmoid function and $\odot$ the Hadamard (element-wise) product. The weight matrices $\mathbf{W}_z, \mathbf{W}_r, \mathbf{W}_h \in \mathbb{R}^{d_1 \times d_{hf}}$ operate on the input, while $\mathbf{U}_z, \mathbf{U}_r, \mathbf{U}_h \in \mathbb{R}^{d_1 \times d_1}$ operate on the recurrent state.

At the end of the evaluation window T_{hf} , the terminal hidden state $\mathbf{h}_T^{\text{GRU}}$ is a compressed short-term representation of permitted sequence information. It may encode combinations of session duration, kinematic change, trip rhythm, and event timing. It is not interpreted as a “desperation profile,” and it does not own declining Earnings Velocity. Its economic content is investigated through controlled probes and ablation, while the downstream HLR receives only its redundancy-controlled residual representation.

The GRU is the initial candidate because it is parameter-efficient and suitable for bounded sequential windows. LSTM, temporal convolution, simpler state-space, and non-neural baselines remain challengers. Model choice follows calibrated out-of-time decision value, stability, fairness, latency, and cost rather than an assumed universal performance advantage.

6.2. Branch 2: Multi-Head Self-Attention Transformers for Long-Term Context

The Transformer branch processes the low-frequency structural context sequence $\mathbf{X}^{\text{macro}}$, such as a 24-month history of macro indicators, regulatory changes, and authorised platform configuration evidence. Unlike the GRU, which processes tokens recurrently, self-attention can create direct paths among positions in the context window. This can help represent long-range relationships, but it does not eliminate all optimisation problems or prove that an event 18 months ago caused a current risk state.

Positional encoding is applied before the attention layers to inject temporal ordering information that the position-agnostic attention mechanism cannot infer from content alone:

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right), \quad \text{PE}(\text{pos}, 2i + 1) = \cos\left(\frac{\text{pos}}{10000^{2i/d_{\text{model}}}}\right)$$

The core computation is scaled dot-product attention. The input sequence is linearly projected into Query (**Q**), Key (**K**), and Value (**V**) matrices, each of dimension d_k :

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}$$

The $\sqrt{d_k}$ scaling term prevents the dot products from growing in magnitude with embedding dimension, which would push the softmax into saturation regions with near-zero gradients. With $H = 4$ parallel attention heads, the multi-head variant runs 4 independent attention computations on projected subspaces of dimension $d_k = d_{\text{model}}/H$ and concatenates the results, allowing different heads to specialize on different structural dependencies, one head may learn to track platform commission regime changes, another to track ZEV mandate escalation trajectories, another to track macro fuel price cycles.

Each Transformer block follows a pre-norm architecture: Layer Normalization → Attention → Residual Add → Layer Normalization → **SwiGLU Feed-Forward Network (SiLU activation)** → Residual Add [13], [14]. The legacy ReLU activation is replaced by the smooth, non-monotonic **SiLU (Swish)** function ($f(x) = x \cdot \sigma(x)$). By dynamically gating the linear projections, SwiGLU preserves subtle negative macroeconomic signals and provides superior representational capacity and smooth gradient flow compared to standard two-layer FFNs. Global average pooling over the time dimension aggregates the T_{lf} token representations into a single dense context vector $\mathbf{c}_L \in \mathbb{R}^{d_2}$.

6.3. Asynchronous Dual-Rate Clock Coupling

The GRU operates on a daily cycle: at midnight, the system processes the day's accumulated telematics and wallet vectors to produce $\mathbf{h}_t^{\text{GRU}}$. The Transformer operates on a monthly cycle (or event-triggered cycle upon receipt of a new macro/regulatory update): it ingests the updated 24-month token sequence and generates a new context vector $\mathbf{c}_{L,\tau+1}$.

The **low-frequency registry** holds the last completed Transformer output $\mathbf{c}_{L,\tau}$ as a static vector. At any given day t , the daily GRU pipeline queries the registry, retrieves $\mathbf{c}_{L,\tau}$ (the context from the last monthly run), and passes it to the fusion layer alongside $\mathbf{h}_t^{\text{GRU}}$:

$$\mathbf{e}_{it} = [\mathbf{h}_t^{\text{GRU}} \parallel \mathbf{c}_{L,\tau}]$$

The zero-order hold avoids recomputing the Transformer every day and lets the daily path use the most recent approved context vector. Point-in-time correctness still depends on the input versions and publication availability used to create $\mathbf{c}_{L,\tau}$. A stale-context flag and maximum age are therefore stored with the vector.

6.4. Cross-Attention Feature Fusion

Naive concatenation $\Phi_{it} = [\mathbf{h}_t^{\text{GRU}} \parallel \mathbf{c}_{L,\tau}]$ does not explicitly model relationships between the two neural streams. If short-term operation changes during a fuel-price regime, a learned fusion may add value. Explicit financial indicators are intentionally absent from this fusion and enter the HLR through their own path.

Cross-attention fusion resolves this: the GRU output $\mathbf{h}_t^{\text{GRU}}$ serves as the **Query** (the question: “given my current behavioral state, which structural factors are most relevant?”). The Transformer vector $\mathbf{c}_{L,\tau}$ serves as the **Keys and Values** (the structural factors available to condition on). The cross-attention mechanism assigns dynamic attention weights that amplify macro-structural context precisely when the behavioral state signals imminent cascade onset. The result is projected through a linear layer to produce the final fused covariate vector $\Phi_{it} \in \mathbb{R}^{d_\phi}$.

7. Handling Asynchronous Clock Cycles and Missing Data

7.1. Missing-data strategies and missingness mechanisms

Underwriting streams can be missing because a source is unavailable, a driver is new, consent is absent, a device is incompatible, a platform integration failed, the event did not occur, or the value depends on an unobserved state. Missingness may be MCAR, MAR, or MNAR. It must not be interpreted as intentional concealment without evidence. A missingness indicator can capture operational pattern, but may also proxy device, platform, geography, or income and therefore requires fairness, lawful-processing, and data-quality review [15].

- **Strategy A, Learned Masking Embedding Tokens:** For every low-frequency input feature x_k , a binary indicator $m_k \in \{0, 1\}$ is appended, where $m_k = 1$ signals a missing value. Both are passed through the Transformer's input projection layer:

$$z_k = W_x x_k + W_m m_k$$

The Transformer can learn whether a missingness pattern adds stable signal. The indicator is not labelled as concealment. Source, consent, outage, applicability, and age-of-data reason codes should be supplied where possible so operational failure is not mistaken for borrower risk.

- **Strategy B, Bayesian generative imputation:** For approved variables and a defensible missingness model, missing values can be treated as unknowns rather than fixed means. A hierarchical prior may use broader cohort information:

$$X_{\text{missing},j} \sim \mathcal{N}(\mu_{\gamma_j}, \sigma_{\gamma_j}^2)$$

Joint inference can propagate imputation uncertainty into PD, but a wider interval is an expected tendency rather than a guaranteed ordering for every record. The Credit Policy and Compliance Gate applies an uncertainty rule whose action depends on product, reason for missingness, available alternatives, and consumer impact.

- **Strategy C - validated fallback engines:** Maintain an explicit-only baseline, full hybrid model, GRU-plus-explicit challenger, and Transformer-plus-explicit challenger. The router selects only a version validated for the observed availability pattern. Wider uncertainty is passed to the policy gate; an outage does not automatically penalise a driver if a safe explicit-only decision is available.

7.2. The Feature Fusion Vector as the Interface to Bayesian Inference

At each defined evaluation interval, the fusion layer outputs Φ_{it} , a neural representation of short-term sequence and longer structural context. Before it reaches the HLR, Part 2b residualises it against the explicit and metadata blocks out of fold. The Explicit Liquidity Features enter separately through centered and orthogonalised spline and interaction bases.

Part 2b begins at that interface. It preserves the narrative ambition of a full posterior distribution, but calibrates the HLR so that dimensionality, redundancy, multicollinearity, hierarchy, and out-of-time uncertainty are controlled rather than hidden.

The boundary is implemented as a signed **model-input artifact**, not an informal tensor handoff. For every score it carries the observation-unit key, product and horizon, decision timestamp, explicit-feature values and versions, P-spline basis version, raw and residual neural artifact versions, metadata and quality flags, feature support indicators, missingness reasons, source ages, and a snapshot manifest. Training also stores the fold-specific residualisation and scaling transformations. Production stores the approved full-development transformations and a numerical parity hash.

The online service never estimates a posterior or refits a residualisation model during the request. It applies the approved feature definitions, basis, projection, calibration, and posterior scoring artifact. A parity suite replays representative observations through offline and online implementations and compares explicit features, basis rows, residual embeddings, linear predictors, calibrated PDs, uncertainty summaries, and reason contributions within approved tolerances. A mismatch routes to the explicit-only fallback or manual process rather than becoming an invisible model change.

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Part 2b

Continuous Underwriting via Hierarchical Bayesian Logistic Regression and Asymmetric Copulas

Mwendo Pamoja White Paper

1. Continuous Underwriting via Hierarchical Bayesian Logistic Regression

1.1. From Deterministic Embeddings to Posterior Distributions

Part 2a ended with two objects arriving at the same inferential boundary. The first is a neural representation Φ_{it} of short-term sequence and slower context. The second is an explicit vector $\mathbf{z}_{it}$ of named liquidity variables. Neither is a sufficient statistic, neither is a probability, and neither is allowed to decide credit on its own. Part 2b asks how to combine them without counting the same economic signal twice, how to represent uncertainty in sparse cohorts, and how to carry marginal risk into a portfolio dependence model.

The Hierarchical Bayesian Logistic Regression translates those inputs into a posterior predictive distribution [1], [2]. Partial pooling can stabilise sparse geography, platform, product, and time effects. Posterior intervals expose parameter and predictive uncertainty under the specified model. Explicit spline terms remain economically interpretable, while latent neural coefficients do not become causal simply because they sit inside a regression. The model is therefore a governed predictive instrument, not a causal proof or an accounting engine.

The human use of that uncertainty is central. A thin-file driver should not be described as unsafe merely because the model knows less. The posterior and its data-quality context move to a separate Credit Policy and Compliance Gate, which can select a smaller limit, request another lawful signal, use an explicit-only fallback, refer the case, or decline where the product rule requires it.

1.1.1. Observation unit, risk episode, and horizon

The model does not score an abstract driver in isolation. Its atomic record is a driver-product-risk episode at a decision time, denoted (i, p, e, t) . The episode identifies the contract or facility, product, observation date, decision purpose, performance horizon, exposure state, intervention history, and label-maturity date. A seven-day operating microloan, a revolving line, and an IPF receivable can therefore share a driver while retaining different horizons, EAD dynamics, cure rules, recovery paths, and default definitions.

For a fixed horizon H_p , the champion HLR outcome is

$$Y_{i,p,e,t}^{(H_p)} = \mathbf{1}\{\tau_{i,p,e} - t \leq H_p\},$$

where $\tau_{i,p,e}$ is the first event time satisfying the approved product default definition. Only labels mature by the training cutoff are treated as observed binary outcomes. Open or right-censored episodes feed the timing challenger described later; they are not recoded as successful repayments. Repeated decisions from one driver are kept in the same validation unit so that a later record cannot leak the driver's outcome into an earlier fold.

This definition makes two outputs complementary rather than competitive. The HLR estimates calibrated default probability at a decision-relevant horizon. A survival challenger estimates the path and timing of default. Both use the same product definitions and event ledgers, and their overlapping horizon probabilities must reconcile within approved tolerances.

1.2. The Partial Pooling Framework and Model Taxonomy

The choice of partial pooling, hierarchical Bayesian modeling, over alternative approaches requires precise justification, since it is a non-trivial modeling decision with substantial implications for both predictive performance and regulatory acceptability.

- **Complete Pooling (standard logistic regression):** A single global coefficient vector β is estimated across all drivers, treating the portfolio as a homogeneous population. This ignores the structural heterogeneity of the gig-economy portfolio: a driver in a dense Nairobi CBD corridor faces fundamentally different surge patterns, traffic kinematics, and macro exposure than a driver in a peri-urban Kisumu route with degraded infrastructure and lower baseline demand. A single global β systematically mis-estimates PD for every specific cluster.
- **No Pooling (independent models per cluster):** An independent logistic regression is fit for each cluster $j = [\text{geohash}, \text{platform}]$. For data-rich clusters (hundreds of observed defaults), this is unproblematic. For newly launched clusters (zero observed defaults in the first 60 days), the maximum likelihood estimate of the default probability is identically zero, the model learns that default is impossible in a new corridor simply because it hasn't happened yet. This is not a feature; it is catastrophic overfitting in sparse data.
- **Partial Pooling (hierarchical Bayesian):** Cluster-specific parameters $[\alpha_j, \beta_j]$ are treated as random variables drawn from a global portfolio-level distribution. Data-sparse clusters borrow statistical strength from the global posterior, their PD estimates are pulled toward the portfolio mean, preventing the zero-default pathology. Data-rich clusters have sufficient evidence to pull their estimates away from the mean, capturing genuine local heterogeneity. This is the Bayesian analogue of shrinkage estimation:

sparse clusters are shrunk toward the global mean; dense clusters are shrunk less. The degree of shrinkage is not set by the analyst but is inferred from the data through the hyperprior on the between-cluster variance.

The three-level hierarchical structure is:

- **Level 1, Individual Driver Likelihood:**

$$Y_{ijt}^{(p)} \sim \text{Bernoulli}(\theta_{ijt}^{(p)})$$

for driver i in cluster j , concerning product $p \in \{\text{micro, revol, IPF}\}$, at time t .

- **Level 2, Cluster Prior (Non-Centered Parameterization):** To avoid **Neal's Funnel** geometry, a pathology in hierarchical models where gradient-based samplers (like HMC/NUTS) struggle as the scale parameter approaches zero, creating a narrow bottleneck, we utilize the “Matt trick” (non-centered parameterization). This mathematically detaches the dependence between the group-level effects and their overarching scale parameter, transforming a highly curved probability space into a smooth, isotropic Gaussian distribution.

The raw, independent standard normal noise components are defined as:

$$\tilde{\alpha}_j^{(p)} \sim \mathcal{N}(0, 1), \quad \tilde{\beta}_j^{(p)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

The actual cluster-level effects are then deterministically shifted and scaled:

$$\alpha_j^{(p)} = \mu_\alpha^{(p)} + \sigma_\alpha^{(p)} \cdot \tilde{\alpha}_j^{(p)}$$

$$\beta_j^{(p)} = \mu_\beta^{(p)} + \sigma_\beta^{(p)} \odot \tilde{\beta}_j^{(p)}$$

- **Level 3, Portfolio Hyperpriors:**

$$\mu_\alpha^{(p)} \sim \mathcal{N}(0, 2), \quad \mu_\beta^{(p)} \sim \mathcal{N}(\mathbf{0}, 2\mathbf{I})$$

$$\sigma_\alpha^{(p)} \sim \text{Half-StudentT}(\nu = 3, \sigma_0 = 1), \quad \sigma_\beta^{(p)} \sim \text{Half-StudentT}(\nu = 3, \sigma_0 = 1)$$

The scale parameters (σ) are regularized using a **Half-Student T** prior. Its generalized probability density function is:

$$f(\sigma) = \frac{2}{\sigma_0} \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{\nu\pi}} \left(1 + \frac{1}{\nu} \left(\frac{\sigma}{\sigma_0}\right)^2\right)^{-\frac{\nu+1}{2}}, \quad \sigma > 0$$

Structural Note on Tail Geometry: While the non-centered parameterization fixes the underlying coordinate system, we must still carefully govern the scale prior. Setting the degrees of freedom to $\nu = 1$ collapses the Half-Student T into a **Half-Cauchy** distribution. While a Half-Cauchy(1) maximizes sparsity, its extreme heavy tails allow the scale parameter to drift into unproportionally large values when data is sparse, forcing the NUTS sampler to diverge across the wide body of the funnel. To gently curb this extreme tail behavior while preserving robustness to genuine structural breaks, we explicitly regularize the prior by setting $\nu = 3$ (or alternatively, utilizing a Half-Normal). This achieves the optimal mathematical compromise between heavy-tailed adaptability and HMC sampling efficiency.

Unlike the Normal distribution (which would assign negative mass to variance parameters and requires truncation), and unlike the Inverse-Gamma distribution (which concentrates too much mass away from zero and over-regularizes in the presence of genuine between-cluster heterogeneity, the Gelman critique), the Half-Student T ($\nu = 3$) allows the model to discover genuinely large inter-cluster variance when the data supports it without destabilizing the leapfrog integrator.

1.3. Full Mathematical Model Specification

The production specification is intentionally block-structured:

$$Y_i^{(p)} \sim \text{Bernoulli}(p_i^{(p)}), \quad \text{logit}(p_i^{(p)}) = \eta_i^{(p)},$$

$$\eta_i^{(p)} = \alpha^{(p)} + a_{g[i]}^{(p)} + b_{\ell[i]}^{(p)} + c_{t[i]}^{(p)} + \mathbf{q}_i^\top \beta_z^{(p)} + \tilde{\mathbf{h}}_i^\top \beta_h^{(p)} + \psi_i^\top \gamma^{(p)} + \mathbf{m}_i^\top \delta^{(p)}.$$

The blocks have different jobs:

- $\alpha^{(p)}$ is the product intercept.
- $a_{g[i]}^{(p)}$, $b_{\ell[i]}^{(p)}$, and $c_{t[i]}^{(p)}$ are centered, partially pooled geography, platform, and cohort-time effects.
- $\mathbf{q}_i = Q(\mathbf{z}_i)$ is the centred, penalty-compatible P-spline design of the Explicit Liquidity Features.

- $\tilde{\mathbf{h}}_i$ is the neural representation after out-of-fold residualisation against explicit liquidity and metadata.
- ψ_i contains a small pre-registered interaction set under strong heredity.
- $\mathbf{m}_i$ contains approved metadata and data-quality terms that are neither neural nor liquidity metrics.

The residual neural block is defined by

$$\tilde{\mathbf{h}}_i = \mathbf{h}_i - \hat{\mathbb{E}}_{-k(i)} [\mathbf{h}_i \mid \mathbf{q}_i, \mathbf{m}_i].$$

The projection is fitted without observation i 's validation fold. It does not guarantee perfect orthogonality in a new regime, but it sharply reduces the route by which the neural branch can reconstruct CFA, DLR, Earnings Velocity, Repayment Velocity, or reserve balance and count it again.

To keep a high-dimensional neural block from overwhelming the interpretable model, the implementation uses:

1. a fixed low-dimensional bottleneck selected inside training folds;
2. standardisation and whitening fitted on training data;
3. a regularized horseshoe or comparable block-shrinkage prior on β_h [3], [17];
4. separate shrinkage scales for explicit, neural, interaction, and metadata blocks;
5. strong heredity for interactions;
6. non-centred hierarchical effects with sum-to-zero identification; and
7. ablation gates requiring stable incremental value over the explicit-only model.

The score is calibrated out of time by product:

$$\text{logit} \left(PD_{i,\text{cal}}^{(p)} \right) = \kappa_p + s_p \eta_i^{(p)}, \quad s_p > 0.$$

Calibration is part of the model, not a cosmetic report. The decision service records the calibrated PD, interval, tail probability, model version, feature versions, fallback status, and reason-code inputs.

This specification deliberately avoids cluster-specific coefficient vectors over the full neural embedding. With limited defaults, that construction would multiply a high-dimensional identifiability problem across clusters. Partial pooling is concentrated in interpretable intercepts and a small number of pre-registered slopes where the data supports them.

1.4. Governed P-Splines for Nonlinear Liquidity Effects

Several key continuous covariates exhibit strongly nonlinear relationships with default probability that linear terms cannot capture:

- **DLR** ($\text{DLR}_i^{(h)}$): the relationship may steepen as obligations due over horizon h approach or exceed available liquidity. Values near one have economic meaning, but no universal default boundary or curve shape is imposed.
- **Repayment Velocity** (v_{repy}): movement from missed to scheduled repayment may be more informative than movement among large prepayments. The plateau and direction are tested by product and window.
- **Driver tenure**: an early learning or selection effect may flatten over time. A spline prevents an unbounded linear extrapolation while allowing the evidence to determine whether the relationship exists.

The diagnostic protocol begins with a regularised linear baseline. For a discrete Bernoulli outcome, a randomized quantile residual is

$$r_i = \Phi^{-1}(F_i(y_i^-) + u_i [F_i(y_i) - F_i(y_i^-)]), \quad u_i \sim \text{Uniform}(0, 1),$$

where $F_i(y_i^-)$ is the probability strictly below the observed outcome. Under correct specification these residuals should be approximately standard normal, subject to estimation and dependence [4]. Residual structure, calibration curves, partial-residual plots, and out-of-time performance can motivate a spline. Binning is used for visual diagnostics with uncertainty and adequate counts, not as the sole test.

The governed main effect is a P-spline: a moderately rich B-spline basis plus an explicit difference penalty [21]. For feature x ,

$$f(x) = \sum_{k=1}^K B_{k,d}(x) \zeta_k,$$

with second-order penalty

$$\Delta^2 \zeta_k = \zeta_k - 2\zeta_{k-1} + \zeta_{k-2}, \quad \Delta^2 \zeta \sim \mathcal{N}(\mathbf{0}, \tau_f^2 \mathbf{I}).$$

The basis controls representation; the penalty controls roughness. This reduces sensitivity to a single small knot set and gives the model a sparse Gaussian coefficient block that works naturally with the hierarchical inference described in Section 2. Knot number, degree, boundary knots, penalty order, centring constraint, smoothing-scale prior, validated support, clipping rule, and missingness treatment are fixed from development data and stored with the signed artifact.

The realised smooth is centred over the development observations so its average level does not compete with the intercept. Numerical conditioning may use a QR or eigen reparameterisation, but the transform must preserve the penalty's null space. Orthogonalising the basis without transforming the penalty would change the prior and is therefore not permitted. The reported effect is back-transformed to the original feature scale with posterior uncertainty.

Product-varying deviations are allowed only where the data support them:

$$f_p(x) = f_0(x) + g_p(x), \quad \Delta^2 \zeta_{g,p} \sim \mathcal{N}(\mathbf{0}, \tau_g^2 \mathbf{I}),$$

with partial pooling that pulls sparse product deviations toward the shared shape. Monotonicity is imposed only where a product owner and validation evidence justify it. Boundary observations are flagged and handled by a documented clipping or conservative extrapolation rule; smoothness is not treated as protection against an unobserved regime.

B-spline with RW1 and B-spline with AR(1) coefficient priors are another research challengers in discussed in Appendix B. The main body uses the P-spline because its difference penalty makes the smoothing objective explicit and computationally sparse, while the appendix preserves the alternative assumptions for sensitivity and validation.

PSIS-LOO can estimate observation-level expected log predictive density when its Pareto diagnostics are acceptable [7]. It does not by itself test transport to a new platform, geography, or vintage. Model selection therefore uses time-forward folds, driver-grouped folds, explicit held-out clusters, and, where the decision question demands it, refits or approximations that remove an entire group. A spline is retained only when shape stability, calibration, decision value, and held-out performance support it.

1.5. Time-Varying Coefficients: AR(1) Autoregressive Priors

IFRS 9 expected credit loss uses reasonable and supportable forward-looking information [12]. For underwriting, this motivates testing whether macro-related coefficients δ , including sen-

sitivity to fuel prices, platform commission, and ZEV transition, vary through time. A time-varying coefficient is adopted only if the data identify it and held-out performance supports it.

A coefficient estimated on a 2020 to 2023 window may not transport to a later portfolio under different energy prices, demand, platform rules, or ZEV policy. One candidate is to model selected macro coefficients as time-indexed parameters governed by first-order autoregressive priors:

$$\delta_{k,t} \sim \mathcal{N}(\mu_k + \rho_k(\delta_{k,t-1} - \mu_k), \sigma_{\delta_k}^2).$$

where:

- μ_k is the long-run stationary mean of the k -th macro coefficient, the baseline sensitivity when no regime shift has occurred.
- $\rho_k \in (-1, 1)$ is the persistence coefficient. A range such as 0.85-0.95 may be used in sensitivity analysis, but is not described as empirically appropriate before fitting and validating the relevant series.
- $\sigma_{\delta_k}^2$ is the innovation variance, the magnitude by which the coefficient can shift from one period to the next, controlling the sensitivity of the model to new macro information.

Under a stationary initial distribution and $|\rho_k| < 1$, the AR(1) variance is $\text{Var}[\delta_{k,t}] = \sigma_{\delta_k}^2 / (1 - \rho_k^2)$. This contrasts with the random-walk special case $\rho_k = 1$:

$$\delta_{k,t} \sim \mathcal{N}(\delta_{k,t-1}, \sigma_{\delta_k}^2)$$

The RW1 is non-stationary and has no mean-reverting level. It may represent gradual drift, while a stationary AR(1) represents variation around a long-run mean. Neither prior should be selected merely by labelling a shock permanent or cyclical. Prior predictive behaviour, forecast performance, and sensitivity determine which description is adequate.

The **time-varying cohort shock effect** γ_{jt} , an AR(1) prior on cluster-level baseline risk, follows the same structure:

$$\gamma_{jt} \sim \mathcal{N}(\rho_j \gamma_{j,t-1}, \sigma_{\gamma}^2)$$

When an unobserved common shock affects cluster j , such as a regional fuel disruption, labour action, or local platform change, the posterior for γ_{jt} may shift for the cluster. A driver with otherwise stable evidence can therefore receive a higher conditional PD because peers share an adverse state. This is one Bayesian representation of default clustering; it must not be interpreted as proof that every member was exposed to the same cause.

If the time series per cluster is long enough to identify separate persistence, one may use $\rho_j \sim \text{Beta}(a_\rho, b_\rho)$ on $(0, 1)$ with hierarchical pooling. Otherwise a shared or parsimonious persistence structure is safer. The posterior location is an empirical result; corridor labels do not predetermine it.

2. Scaling to Large Policyholder Portfolios: MCMC Architecture

2.1. Conditional Conjugacy for the Complete HLR

Pólya-Gamma augmentation introduces latent variables that express the logistic likelihood in a conditionally Gaussian form, enabling efficient blocked updates for the complete hierarchical logistic model [22]. Stack the global coefficients, P-spline coefficients, residual neural coefficients, selected interactions, and Gaussian hierarchy effects into ϑ , with design row $\mathbf{x}_i^\top$ so that $\eta_i = \mathbf{x}_i^\top \vartheta$. Introduce

$$\omega_i \mid \eta_i \sim \text{PG}(1, \eta_i).$$

Conditional on ω , the logistic likelihood is Gaussian in the linear predictor:

$$p(\mathbf{y} \mid \vartheta, \omega) \propto \exp \left\{ -\frac{1}{2} (\mathbf{X}\vartheta - \mathbf{z})^\top \Omega (\mathbf{X}\vartheta - \mathbf{z}) \right\},$$

where $\Omega = \text{diag}(\omega_1, \dots, \omega_N)$ and $z_i = (y_i - 1/2)/\omega_i$. With Gaussian priors or Gaussian priors conditional on scale parameters, the coefficient update is multivariate Normal. The P-spline block remains sparse because its prior precision contains the difference-penalty matrix. Geography, platform, cohort, selected time effects, and any well-identified driver effects retain partial pooling.

The conjugacy is conditional, not wholly closed form. Pólya-Gamma variables, hierarchy variances, spline smoothing scales, regularised-horseshoe local and global scales, LKJ correlation

structures, calibration parameters, and any time-persistence parameters still need their own updates. A candidate blocked Gibbs cycle is therefore:

1. update the Pólya-Gamma latent variables;
2. jointly or sparsely update Gaussian coefficient and hierarchy blocks;
3. update variance, covariance, and smoothing parameters;
4. update regularised-horseshoe scales; and
5. calculate calibration, posterior predictive, and convergence diagnostics.

This sampler is benchmarked against the approved NUTS reference on representative strata. The comparison covers posterior moments, interval and tail behaviour, group shrinkage, calibration, effective sample size per second, memory, and actual decisions. Pólya-Gamma sampling remains offline. The live underwriting request uses only an approved posterior scoring artifact; it never launches Gibbs sampling or MCMC.

2.2. The Bernoulli-to-Binomial Aggregation for Computational Scaling

As the Insurtech scales to portfolios of hundreds of thousands of active drivers, the computational cost of full MCMC over individual Bernoulli likelihoods becomes prohibitive. Each leapfrog step in Hamiltonian Monte Carlo requires evaluating the gradient of the log-posterior with respect to all model parameters. For a Bernoulli likelihood with N observations, this gradient has $O(N)$ computational complexity. At $N = 200,000$ drivers with daily updates, the gradient cost renders NUTS infeasible at the required update cadence.

One scaling option is Bernoulli-to-Binomial aggregation for observations with **exactly identical predictors, offsets, exposure definition, and applicable weights**. For an eligible group j :

$$K_j = \sum_{i=1}^{n_j} Y_{ij} \sim \text{Binomial}(n_j, \theta_j)$$

For exact equality, K_j is sufficient for the common probability. Quantising “near-identical” continuous covariates is an approximation and does not preserve the individual likelihood or Fisher information. Any such approximation must be benchmarked against an unaggregated holdout for calibration, tails, subgroup effects, and decision changes. The system should also compare minibatch variational inference, Laplace approximations, sequential updating, and periodic NUTS validation rather than assuming daily full-portfolio NUTS.

Evaluating splines at group averages does not recover within-group heterogeneity. Exact aggregation is reserved for identical design rows and a common probability. Quantised cells are explicitly labelled an approximation, benchmarked against individual likelihoods, and never described as sufficient. Individual scoring retains the original covariates.

2.3. LKJ Cholesky Decompositions and the Funnel Geometry

Hierarchical models can exhibit Neal’s funnel when a weakly identified group scale approaches zero and group effects concentrate near the global mean. This geometry can produce divergent transitions or inefficient exploration under HMC. Divergences warn that numerical exploration is unreliable; whether estimates are materially biased depends on where the sampler failed to explore.

For correlated random effects, the implementation separates marginal scales from a Cholesky factor of the correlation matrix and can place an LKJ prior on correlation. A non-centred form is compared with a centred form. This is preferable to invoking an inverse-Wishart prior without a reason, but the exact choice remains part of model design and sensitivity analysis.

The covariance can be written as $\Sigma = \text{diag}(\sigma)\Omega \text{diag}(\sigma)$, where Ω is a correlation matrix. This parameterisation separates their prior specification, but posterior dependence can remain because the data couple scale and correlation. The sampler transforms standard-normal latent variables using the Cholesky factor.

This usually improves posterior geometry, but it does not make the target perfectly isotropic or guarantee zero divergences. $\hat{R}$, effective sample size, energy diagnostics, divergences, tree depth, and posterior sensitivity must be reported from the actual fit.

The probabilistic programming pseudocode (language-agnostic conceptual representation for PyMC/NumPyro/Stan architectures) implementing the full MCMC specification:

ALGORITHM: Redundancy-controlled hierarchical Bayesian logistic regression

INPUTS:

<code>Y[N]</code>	Bernoulli outcomes at a defined product horizon
<code>Q[N, Kz]</code>	centred, scaled, QR-orthogonalised explicit spline basis
<code>H_RES[N, Kh]</code>	cross-fitted residual neural bottleneck
<code>R[N, Kr]</code>	small pre-registered interaction block
<code>X[N, Kx]</code>	essential contract and product metadata
<code>geo_id[N]</code>	geography hierarchy

```
platform_id[N]    platform or cohort hierarchy
time_id[N]        calendar hierarchy
```

PRIORS:

```
alpha            ~ Normal(0, 1.5)
sigma_geo        ~ HalfStudentT(nu=3, scale=s_geo)
sigma_platform   ~ HalfStudentT(nu=3, scale=s_platform)
sigma_time       ~ HalfStudentT(nu=3, scale=s_time)
geo_raw[J]       ~ Normal(0, 1)
platform_raw[S]  ~ Normal(0, 1)
time effect      ~ approved AR(1), RW1, or simpler pooled structure
beta_z, beta_h   ~ separate regularising block-shrinkage priors
gamma_interaction ~ strong-heredity shrinkage prior
delta_metadata   ~ regularising prior
```

TRANSFORM:

```
u_geo          = sigma_geo * geo_raw
v_platform     = sigma_platform * platform_raw
tau_time       = centred time process
```

LINEAR PREDICTOR FOR OBSERVATION i:

```
eta[i] = alpha
        + u_geo[geo_id[i]]
        + v_platform[platform_id[i]]
        + tau_time[time_id[i]]
        + dot(Q[i], beta_z)
        + dot(H_RES[i], beta_h)
        + dot(R[i], gamma_interaction)
        + dot(X[i], delta_metadata)
```

LIKELIHOOD:

```
Y[i] ~ BernoulliLogit(eta[i])
```

OPTIONAL EXACT AGGREGATION:

Only observations with an identical design row and common probability

may be collapsed to a Binomial count. Quantised rows are approximations.

POST-FIT:

Apply product-level out-of-fold calibration, then pass calibrated PD, uncertainty, data quality and reasons to the policy and compliance gate.

2.4. Production Inference and Approved Posterior Artifacts

BlackJAX is one candidate JAX-native implementation for the NUTS reference and can use compiled accelerators. Pólya-Gamma blocked Gibbs, Stan or another validated HMC implementation, Laplace approximation, and structured variational inference remain candidates. The example of 200,000 observations and 4,000 groups is a benchmark scenario to be reproduced with the production likelihood. No fixed speed-up is assumed; compilation, sparse linear algebra, memory transfer, chain count, precision, effective sample size, and approximation error are included in the benchmark.

- **Inference candidates:** Full-data NUTS, Laplace approximation, variational inference, sequential Monte Carlo, or other methods are benchmarked on the actual posterior. Subsampled HMC with control variates is not described as unbiased or deployed until its estimator and diagnostics establish that result for this implementation. High dimension increases cost, while divergence is primarily a posterior-geometry and integration issue rather than “gradient variance” in ordinary full-data NUTS.
- **Incremental updating:** A previous posterior can initialise a later fit, but material new data, drift, or model change may invalidate the old mass matrix and step size. Adaptation is not skipped by default. The approved cadence balances information gain, inference quality, model-change governance, and operational need; underwriting scores can update from new features without retraining the whole HLR each day.

The signed scoring artifact contains the model and feature schemas, posterior draws or an approved analytic approximation, P-spline basis and penalty metadata, residualisation and whitening transforms, hierarchy mappings for known and new groups, calibration parameters, support ranges, uncertainty rules, reason-code mappings, code and environment digests, and fallback instructions. New-group scoring integrates or draws from the approved population hierarchy rather than inventing a zero-risk local effect.

2.5. Hierarchical Piecewise-Exponential Timing Challenger

The HLR answers whether default occurs within a chosen horizon. A timing challenger asks when the event is expected to occur and uses right-censored episodes without labelling them as survivors. The governed challenger is a hierarchical piecewise-exponential proportional-hazards model, rather than an unqualified claim that a fully unspecified Cox model has been fitted [23], [24]. Split episode i into intervals j with at-risk exposure E_{ij} , event indicator d_{ij} , baseline log-hazard α_j , and time-valid covariates $\mathbf{x}_{ij}$:

$$d_{ij} \sim \text{Poisson}(E_{ij}\lambda_{ij}), \quad \log \lambda_{ij} = \alpha_j + \mathbf{x}_{ij}^T \beta + u_{g[i]} + v_{p[i]}.$$

The Poisson representation is a likelihood device for a piecewise-constant hazard. The risk-set builder records entry, exit, censoring, competing closure or prepayment, contract modification, intervention state, and the covariates available at each interval boundary. Product-specific baseline hazards and pooled platform or geography effects allow different clocks without creating a separate model for every sparse cohort.

For cumulative hazard $\Lambda_i(t)$,

$$S_i(t) = \exp[-\Lambda_i(t)], \quad PD_i(0, t) = 1 - S_i(t),$$

and the marginal default probability in month m is

$$q_{i,m} = S_i(t_{m-1}) - S_i(t_m).$$

These marginal probabilities feed collection and ECL schedules without multiplying a cumulative PD in every month. At any horizon shared with the HLR, $1 - S_i(H_p)$ is compared with the calibrated fixed-horizon PD. Differences are diagnosed through baseline hazard, censoring, episode construction, calibration, or population drift; the two outputs are not averaged into a synthetic probability.

A Gamma frailty is retained as an optional survival extension where repeated matured episodes identify persistent unobserved heterogeneity. Its conditional conjugacy belongs to the hazard challenger, not the foundation of the HLR. A driver-specific effect is included only when repeat experience and identification support it; product, platform, geography, cohort, and time pooling remain the primary hierarchy. Appendix C also retains a joint

longitudinal-survival challenger for a repeatedly measured latent liquidity trajectory whose measurement error and endogenous evolution justify joint modelling [25].

3. MCMC Convergence Diagnostics and Model Validation

Before a posterior summary informs portfolio analysis, accounting evidence, or a credit-policy input, inference quality must be verified. Diagnostics are rerun for every approved model version and after material data, feature, product, or method change. A PSI value of 0.25 is an internal investigation threshold, not by itself a retraining command.

3.1. $\hat{R}$: Rank-normalised Gelman-Rubin convergence statistic

An approved NUTS run uses multiple chains with dispersed initialisation. The classical variance-ratio intuition is

$$\hat{R} = \sqrt{\frac{\frac{N-1}{N}W + \frac{1}{N}B}{W}}$$

Modern diagnostics use split rank-normalised and folded $\hat{R}$, not only this classical expression [8]. Values close to one are necessary but not sufficient. A candidate acceptance threshold such as $\hat{R} < 1.01$ is paired with effective sample size, Monte Carlo standard error, divergences, energy and tree-depth diagnostics, trace review, and sensitivity. Non-centred parameterisation can help a scale parameter but does not routinely resolve every pathology.

3.2. Effective Sample Size (ESS)

The Effective Sample Size corrects for autocorrelation within MCMC chains. Highly autocorrelated chains produce fewer informative samples per iteration. ESS is estimated from lag- k autocorrelations ρ_k of the sample sequence:

$$\text{ESS} = \frac{S}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

Bulk and tail ESS are evaluated relative to the estimand and required Monte Carlo precision. A count such as 400 can be an initial screening convention, not a universal promise that every 95 percent interval is reliable. The model uses the stated Half-Student- t or other approved

scale priors consistently; it does not silently switch to Half-Cauchy priors in the diagnostic narrative.

3.3. Posterior Predictive Checks: Exhaustive Treatment

Posterior Predictive Checks (PPCs) answer the question: “If the model is correctly specified, can it reproduce the statistical properties of the observed data?” Each PPC generates simulated outcomes $\tilde{Y}^{(s)}$ from the posterior predictive distribution for $s = 1, \dots, S$ MCMC samples, compares their distribution to the observed outcomes Y^{obs} , and reports a Bayesian p-value $p_B = P(T(\tilde{Y}) \geq T(Y^{\text{obs}}))$ for a test statistic T .

- **PPC 1, central tendency:** Compare the posterior-predictive default-rate distribution with observed rates overall and by product, time, and relevant cohort. A discrepancy can arise from intercept, covariate, hierarchy, time, label, sampling, or calibration problems. Bayesian predictive tail areas are descriptive diagnostics and are not interpreted exactly like uniform frequentist p-values.
- **PPC 2, Tail-Risk Cascade Clustering:** Test statistic from the .rmd specification:

$$T(Y) = \max_j \left(\frac{1}{n_j} \sum_{i \in j} Y_{ij} \right)$$

This statistic probes extreme cluster default rates, subject to cluster size and multiple-comparison sensitivity. Failure of the marginal HLR to reproduce a cluster spike does not instruct the team to increase a Clayton parameter. First diagnose marginal calibration, omitted common factors, label and exposure definitions, hierarchy, and time structure. Dependence candidates are validated separately against joint outcomes and portfolio-loss behaviour.

- **PPC 3, Repayment Velocity nonlinearity:** Predicted PD is evaluated across observed v_{repay} and compared with empirical default rates in bins that retain adequate sample size. Residual curvature, including any transition near $0.8 < v_{\text{repay}} < 1.2$, is evidence to test a spline, not proof that a fixed “critical zone” exists. The check is repeated when the model or population changes.
- **PPC 4, cluster heterogeneity:** Compare observed and posterior-predictive between-cluster variation with uncertainty and minimum-count rules. Underdispersion can reflect excessive shrinkage, omitted predictors, time effects, labels, or the likelihood, while

apparent overdispersion can reflect small cells. Prior-scale sensitivity is one diagnostic, not an automatic instruction to double a parameter.

- **PPC 5, time-varying cohort effect:** Compare predicted and observed monthly rates by adequately sized clusters, then inspect residual autocorrelation. A pattern may implicate persistence, omitted seasonality or shocks, exposure mix, reporting lag, or label drift. Recent data enter through the governed update process, and a prior changes only after documented sensitivity and validation.

3.4. Discrimination, Calibration, and Stability Metrics

Kolmogorov-Smirnov statistic: Measures the maximum separation between score distributions for observed defaulters and non-defaulters. No universal KS above 0.40 is asserted as an IFRS 9 or Kenyan regulatory requirement. Thresholds are product- and use-specific and cannot replace calibration or accounting governance.

Gini coefficient and AUC-ROC: These measure ranking discrimination. An AUC target such as 0.75 can be a development objective, not a universal acceptability boundary. Partial pooling may reduce unstable extremes in sparse groups, but whether it improves held-out discrimination and calibration is measured.

Brier score:

$$BS = \frac{1}{N} \sum_{i=1}^N (p_i^{\text{cal}} - Y_i)^2.$$

The score measures squared probability error and reflects both calibration and resolution [9]. Partial pooling and shrinkage can help sparse groups, but superiority over a classical or explicit-only baseline is an out-of-time empirical result, not a mathematical consequence of being Bayesian.

Validation follows the deployment question. A random row holdout tests neither a new driver nor a new platform when the same entities appear on both sides. The approval suite therefore includes time-forward vintages, driver-grouped splits, platform- and geography-held-out tests, and product-horizon reconciliation between the HLR and timing challenger. Observation-level PSIS-LOO is useful for local influence when diagnostics permit it, but a claim about transport to an unseen group requires grouped log-likelihood construction, an explicit group-deletion approximation, or a refit that withholds the group. Posterior

predictive checks report total defaults, transition timing, survival curves, cure, EAD, recovery timing, and loss by material hierarchy.

Population Stability Index (PSI), production monitoring: PSI compares the model’s current score distribution A_b with a reference distribution E_b across B bins:

$$\text{PSI} = \sum_{b=1}^B (A_b - E_b) \ln \left(\frac{A_b + \varepsilon}{E_b + \varepsilon} \right).$$

Table 1: Discrimination, Calibration, and Stability Metrics Summary

PSI Range	Interpretation	Action
PSI < 0.10	Limited shift under this convention	Continue routine monitoring
0.10 ≤ PSI < 0.25	Monitoring convention: moderate shift	Investigate data, population, and performance
PSI ≥ 0.25	Monitoring convention: large shift	Escalate; consider restrictions and recalibration after diagnosis

A PSI threshold such as 0.25 is a common internal monitoring convention and may be adopted contractually; it is not a universal statutory trigger under SR 11-7, Basel, or Kenyan law. A breach can reflect population movement, seasonality, a pipeline defect, or a definition change. The response is diagnosis, performance review, and proportionate restriction. Automated re-training is not safe without corrected data, validation, approval, and deployment controls.

4. Real-Time Bayesian Decision Boundaries

4.1. The Loss Function Framework: Expected Value Maximization

A fixed rule such as “approve when $\hat{\theta} < 5\%$ ” discards posterior uncertainty. Two drivers can share a 4.5 percent posterior mean while one has a narrow credible interval and the other a wide interval caused by thin or conflicting evidence. A probability distribution bounded on $[0, 1]$, or posterior draws transformed from the logit scale, should be used rather than an unconstrained normal approximation for PD.

The Bayesian decision framework can integrate a payoff over posterior draws [1]. Define action a_1 as approving a limit extension and a_0 as declining it. For a linear payoff, expected value depends on the posterior mean, while separate uncertainty and tail rules retain information about width and shape. The asymmetric payoff for a_1 is:

- True outcome, no default: gain L_{gain} , net interest and permitted fees.
- True outcome, default: loss L_{loss} , unrecovered principal plus defined costs.

The expected financial value of approval is computed by averaging the payoff over all S MCMC posterior samples $\theta^{(s)}$:

$$\mathbb{E}[V(a_1)] = \frac{1}{S} \sum_{s=1}^S [(1 - \theta^{(s)})L_{\text{gain}} - \theta^{(s)}L_{\text{loss}}].$$

The expected cost of denial (opportunity cost of rejecting a creditworthy driver):

$$\mathbb{E}[V(a_0)] = -\frac{1}{S} \sum_{s=1}^S (1 - \theta^{(s)})L_{\text{opp}}.$$

Setting $\mathbb{E}[V(a_1)] = \mathbb{E}[V(a_0)]$ and solving for the break-even posterior mean PD yields the **Bayesian action threshold** p^* :

$$p^* = \frac{L_{\text{gain}} + L_{\text{opp}}}{L_{\text{gain}} + L_{\text{opp}} + L_{\text{loss}}}$$

This threshold is not a fixed number: it is calibrated separately for each product, since the loss structure differs materially across the triple product stack.

For microloans with a 7 to 30 day tenor:

- $L_{\text{gain}} = K \cdot r \cdot \text{days}/365$ (annualized fee on principal K)
- $L_{\text{loss}} = K(1 - R)$, principal net of a validated recovery rate and recovery cost;
- $L_{\text{opp}} = K \cdot (\text{Alternative Portfolio Yield}/26)$
- The former 12 to 18 percent threshold is retained only as an illustrative sensitivity range. The production value must be recomputed from lawful price, exposure, loss, recovery timing and cost, funding, operations, affordability, and policy. No “garnishment recovery” is assumed.

For Revolving Credit Lines:

- $L_{\text{gain}} = V_{\text{max}} \cdot U \cdot i + M$ (interest on utilized balance U at rate i , plus monthly maintenance fee M)
- $L_{\text{loss}} = V_{\text{max}} \psi(1 - R)$, potential loss after stress drawdown and recovery;
- $L_{\text{opp}} = V_{\text{max}}(1 - U)\text{CostOfCapital}$;
- The former 3 to 6 percent threshold is an illustrative sensitivity range. A revolving threshold depends on stress utilisation, EAD, LGD, recoveries, income, funding, operations, and portfolio constraints, then remains subject to the policy gate.

4.2. Three Automated Decision Rules

1. **Dynamic soft cap:** The economic comparison $\mathbb{E}[V(a_1)] > \mathbb{E}[V(a_0)]$ is an input to the Credit Policy and Compliance Gate, not an automatic approval. Affordability, product eligibility, licence, consent, exposure, and hardship rules remain binding.
2. **Uncertainty gate:** Even if the posterior mean is below p^* , the probability of exceeding a product risk cap may justify a smaller limit, explicit-only fallback, referral, request for another lawful signal, or freeze. The cap is an internal or contractual policy value, not automatically a regulatory threshold:

$$P(\theta_{ijt}^{(p)} > \theta_{\text{cap}}) = \int_{\theta_{\text{cap}}}^1 p(\theta|\text{Data}) d\theta > \epsilon_{\text{kill}}$$

for a configurable exceedance probability ϵ_{kill} . The value 0.10 remains a validation candidate, and the action must account for why uncertainty is high.

3. **Portfolio shock review:** A material rise in the cohort-time effect can trigger investigation and portfolio controls. A two-standard-deviation rule is illustrative. A geography-wide rollback requires evidence, fairness review, authority, and a defined cure because the signal can also reflect a data incident or an unmodelled benign shift.

4.3. From Default Probability to Cash Loss

The model's economic destination is not a PD alone. For each posterior and scenario draw, the loss engine separates six objects:

1. the fixed-horizon or marginal default probability;
2. EAD, including scheduled amortisation and stressed revolving drawdown;
3. cure and modification paths;
4. gross recovery and recovery timing;

5. IPF-specific eligible cancellation-refund cash, deductions, and delay; and
6. collection, servicing, legal, and workout costs.

For simulation draw s , a product cash loss can be written as

$$\mathcal{L}_{i,p}^{(s)} = D_{i,p}^{(s)} EAD_{i,p}^{(s)} - R_{i,p}^{(s)} - U_{i,p}^{(s)} + C_{i,p}^{(s)},$$

where D is the simulated default event, R is net ordinary recovery, U is an eligible IPF refund or other separately identified credit support, and C contains defined collection and recovery costs. Signs, currency, valuation date, horizon, and discounting are explicit. Cure is represented through state transitions and subsequent cash flows rather than as a second recovery credited on top of the same collection.

The EAD, cure, recovery, and refund components can be separate calibrated submodels or governed scenario functions. They do not automatically inherit the HLR's covariates or neural block. A shared systematic factor may affect several components, but it is introduced once in the simulation so that a fuel shock is not separately counted in PD, LGD, recovery delay, and a copula parameter without an identified causal or scenario mapping. Part 5 aggregates these draws by cohort and month, then applies reserves, fees, debt service, subordination, and the exact waterfall.

The HLR's pricing handoff ends with calibrated real-world PD, uncertainty, expected-loss inputs, and a documented decision horizon. Part 5, Section 1.2.1 carries those outputs into customer and investor pricing, keeps the physical measure distinct from any market-consistent valuation measure, and reconciles expected loss with the other components of K_{RBCP} . This leaves Part 2b focused on underwriting inference, dependence, validation, and decision boundaries.

5. Actuarial Modeling of Clustered Failures via Asymmetric Copulas

5.1. The limits of linear correlation and Gaussian copulas

The hierarchical Bayesian engine estimates marginal posterior PD distributions for each product, $\theta_{ijt}^{(\text{micro})}$, $\theta_{ijt}^{(\text{revol})}$, and $\theta_{ijt}^{(\text{IPF})}$. Their accuracy is an empirical result. A partner holding all

three exposures also needs a model of dependence to estimate joint default and portfolio loss; marginal probabilities alone are insufficient.

Dependence is introduced in a governed order. First, observed macro and operational factors enter the feature and scenario design. Second, platform, geography, product, cohort, and time effects absorb partially pooled common structure. Third, within-driver multi-product linkage is represented explicitly. Only the remaining joint behaviour is offered to a residual copula or another dependence challenger. This order prevents a tail parameter from becoming a container for omitted factors, duplicate borrower effects, and true transmission mechanisms at once.

The portfolio variance under Gaussian dependency assumptions:

$$\text{Var}[\mathcal{L}] = \sum_p \text{Var}[\mathcal{L}_p] + \sum_{p \neq p'} \rho_{pp'} \sqrt{\text{Var}[\mathcal{L}_p]} \sqrt{\text{Var}[\mathcal{L}_{p'}]}$$

A Gaussian copula can be a useful benchmark, but for correlation below one it has zero asymptotic upper and lower tail dependence. It can still show substantial finite-threshold association. Part 1 supplies an economic reason to test lower-tail asymmetry; it does not prove that every pairwise correlation converges to one or that Clayton is the correct family. The observed dependence must be estimated by product, horizon, cohort, and regime.

5.2. Sklar's Theorem and the Clayton Copula

By Sklar's theorem, a multivariate distribution can be represented using its marginals and a copula; uniqueness holds for continuous marginals [10]:

$$F(t^{\text{micro}}, t^{\text{revol}}, t^{\text{IPF}}) = C(F_1(t^{\text{micro}}), F_2(t^{\text{revol}}), F_3(t^{\text{IPF}}); \alpha_c)$$

where $u_p = F_p(t^{(p)}) \in [0, 1]$ are probability integral transforms of the marginals. The copula $C : [0, 1]^3 \rightarrow [0, 1]$ represents dependence separately from the marginal specifications. The HLR supplies calibrated marginal risk estimates; candidate copulas provide competing dependence structures. Clayton is not preselected merely because lower-tail dependence is economically plausible.

The d -dimensional Clayton copula is an Archimedean family for $\alpha_c > 0$:

$$C_{\alpha_c}(u_1, \dots, u_d) = \left(\sum_{j=1}^d u_j^{-\alpha_c} - d + 1 \right)^{-1/\alpha_c}.$$

For a bivariate margin, the lower-tail dependence coefficient is:

$$\lambda_L = \lim_{u \rightarrow 0^+} \Pr(U_2 \leq u \mid U_1 \leq u) = 2^{-1/\alpha_c}, \quad \lambda_U = 0.$$

This asymmetry makes Clayton a meaningful candidate. A fitted or scenario-conditioned increase in α_c raises lower-tail dependence. It does not demonstrate absolute lockstep default.

Conditional Clayton sensitivity: A dependence stress can change the marginals, copula family, parameter, or common-factor structure. The following values show how lower-tail dependence changes if Clayton is retained and α_c is increased; they are not the only permitted stress mechanism:

- **Scenario A, localised cluster shock:** α_c moves from 0.8 to 2.5; pairwise λ_L moves from approximately 0.420 to 0.758.
- **Scenario B, macroeconomic liquidity crunch:** $\alpha_c = 5.0$ implies pairwise $\lambda_L \approx 0.871$.
- **Scenario C, limiting dependence stress:** $\alpha_c \rightarrow \infty$ implies $\lambda_L \rightarrow 1$. This is a mathematical boundary, not a forecast.

The parameters above are illustrative. Production estimation compares competing families using held-out likelihood, probability-integral-transform diagnostics, formal or simulation-based goodness-of-fit checks, tail-event calibration, stability, and the resulting cash-flow decision [11].

5.3. Portfolio loss, unexpected loss, and capital boundaries

Let $\mathcal{L}^{(s)}$ be the simulated portfolio loss after dependent defaults, EAD, LGD, recoveries, and timing. Expected loss is $EL = \mathbb{E}[\mathcal{L}]$. “Unexpected loss” must be defined for the use case. Two common economic measures are

$$UL_{sd} = \sqrt{\text{Var}(\mathcal{L})} \quad \text{and} \quad UL_q = \text{VaR}_q(\mathcal{L}) - EL.$$

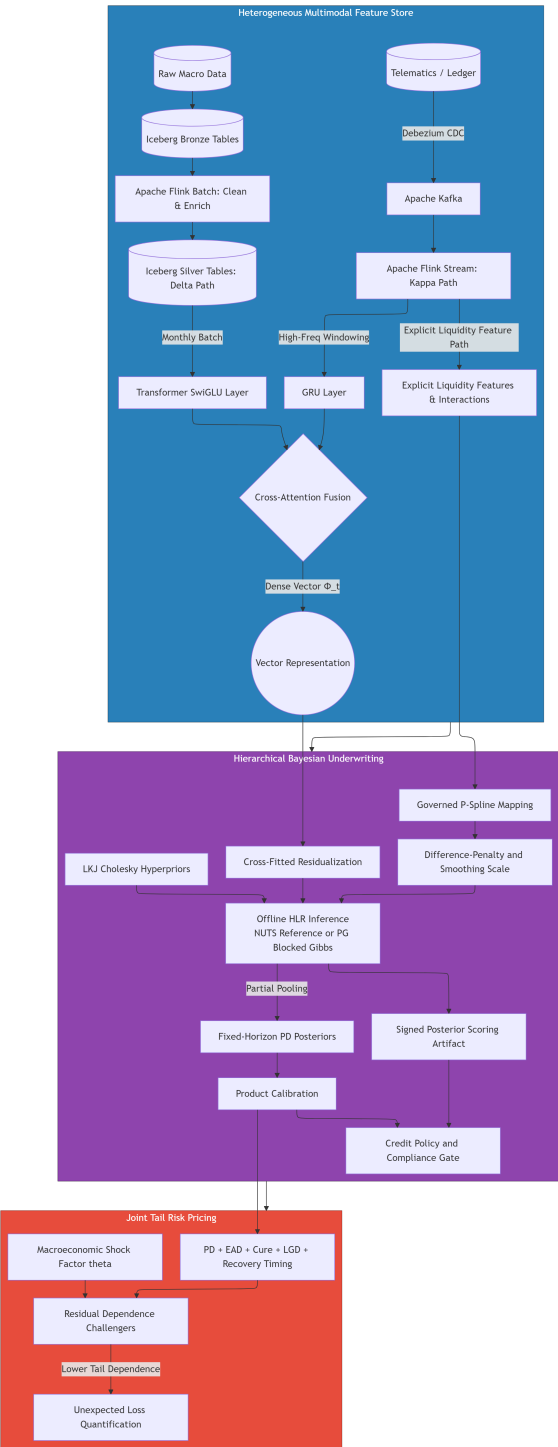
The copula simulation produces the joint distribution; a Clayton rank correlation is not inserted into a linear covariance formula as though it were Pearson loss correlation. Tail dependence can materially change high loss quantiles even when ordinary correlation moves little. Neither UL_{sd} nor UL_q automatically equals regulatory capital. The partner bank applies its authorised prudential approach, while the SPV uses contractually defined stresses and cash-flow tests.

The portfolio simulation separates parameter uncertainty from correlated outcome simulation:

1. Draw HLR parameters and calibrated marginal PDs $p_{i,p}^{(s)}$ from the posterior.
2. Draw a dependent uniform vector $\mathbf{U}_i^{(s)}$ from the selected copula with a jointly estimated or scenario-conditioned parameter.
3. Set $Y_{i,p}^{(s)} = \mathbf{1}[U_{i,p}^{(s)} \leq p_{i,p}^{(s)}]$.
4. Draw or scenario-set EAD, LGD, recovery timing, prepayment, and collection timing without double-counting the same systematic factor.
5. Compute monthly product and portfolio cash loss and pass it through the SPV waterfall.

The empirical loss quantile is a modelled VaR-type measure for the stated horizon and measure. A central 90% posterior predictive interval and a 95th loss quantile share an endpoint only under a particular definition; they are not generally equivalent and neither automatically constitutes regulatory capital. The cash-flow model must specify whether losses are real-world, accounting, economic-capital, contractual stress, or market-valued quantities.

Figure 1: End-to-End Underwriting Pipeline (Data Engineering to Joint Tail Risk Pricing)



6. Explainability and model-risk governance for K_{RBCP}

The HLR estimates one component of credit risk. Governance must still connect the feature record, calibrated PD, policy action, customer terms, override, and monitoring result. Explainability is therefore organised around the decision and its economic components, not around a claim that a latent vector is causal.

The CBK framework expresses the contractual lending rate through two separately traceable components:

$$R_{\text{customer},t} = \text{KESONIA}_t + K_{\text{RBCP},i,t},$$

with fees and charges added for total cost of credit [13], [14]. A compounded KESONIA amount may be used for accrual over a period, but CCR_t should not replace the reference-rate term in the customer-pricing identity. KESONIA requires source, dates, calendar, compounding, and fallback controls; it is not immune to operational error.

The institution should reconcile K_{RBCP} to expected loss, cost of capital, operating and liquidity costs, margin, and approved adjustments. A local model explanation cannot prove that the entire price is unbiased or objectively necessary. The audit package therefore combines component reconciliation, explicit reason codes, counterfactual checks, data lineage, subgroup outcomes, overrides, complaints, and independent validation.

6.1. Explanation as part of a challengeable decision

A GRU or Transformer representation is not self-explanatory to a credit officer, driver, validator, or auditor. Even an explicit variable can be misunderstood if its window, transformation, interaction, and decision context are hidden. The explanation design must therefore begin with the economic decision, the authorised rule, and the model component that materially influenced it.

SHAP and LIME are useful diagnostic tools, but neither is treated as a compliance certificate or a substitute for intelligible product reason codes. The implementation combines model-behaviour explanations with the explicit-feature record, rule result, price-component reconciliation, contract terms, override, and outcome.

6.1.1. SHAP (SHapley Additive exPlanations) Integration

Rooted in cooperative game theory, Shapley values define an additive attribution under a specified value function and background distribution. Exact computation is often infeasible, so practical SHAP implementations use model-specific methods or approximations. Results can change with feature dependence, background data, output scale, and model version.

For a material credit decision, the system may generate a local explanation of the calibrated PD or another clearly identified model output. It does not represent the entire customer premium as a model attribution. An illustrative explanation can separate:

- **Base Rate (Expected Value):** The unconditional mean risk premium for the driver's geographic cluster assuming no behavioral data is provided (e.g., $K = 3.50\%$).
- **Positive SHAP Contributions (Increasing Risk):**
 - an illustrative contribution from declining Repayment Velocity, sourced from the Explicit Liquidity Feature Path rather than the GRU;
 - an illustrative contribution from an approved platform-context term and a separately owned liquidity interaction;
- **Negative SHAP Contributions (Decreasing Risk):**
 - an illustrative protective contribution supported by a validated operational feature.

The production record stores model, feature, data, policy, price-component, and explanation versions under a governed retention schedule. SHAP is a model-behaviour explanation conditional on a background distribution. It cannot demonstrate that a decision was free of discriminatory proxies; outcome and counterfactual testing remain necessary.

For a chosen explanation scale, the SHAP values ϕ_i obey the additive representation:

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i$$

where ϕ_0 is the selected background expectation and M is the number of explained features or representation components. A neural approximation such as DeepSHAP can run asynchronously after inference, but its fidelity must be tested and its output stored with the prediction, background dataset, and explanation version. Bronze and Silver ingestion layers should not manufacture a post-decision explanation as though it were source data.

6.1.2. Local Interpretable Model-Agnostic Explanations (LIME)

LIME can provide a complementary local surrogate when its neighbourhood, perturbation process, kernel, and stability are appropriate. It is not assumed to be a reliable fast path merely because it is cheaper than an exact Shapley calculation.

LIME fits a simple local surrogate, such as a sparse linear model, to perturbed observations around a selected representation. Its coefficients approximate local behaviour under the chosen perturbation distribution and kernel; they are not necessarily the true gradient, a causal effect, or stable under correlated inputs.

A local surrogate may help a reviewer explore model sensitivity, but customer-facing reasons should come from stable mappings to material explicit factors, policy rules, and contract actions. The system must not tell a driver that an irregular shift caused a restriction when the explanation only describes a fragile local approximation. A more detailed model-behaviour analysis can run asynchronously for validation and complaints without being mislabelled as a mandated regulatory log.

6.2. Model-risk management with SR 11-7 as a comparative benchmark

Kenyan law, CBK requirements, partner policy, and the final institutional governance framework determine the applicable obligations. U.S. Federal Reserve SR 11-7 remains a useful comparative reference for conceptual soundness, validation, governance, and outcomes analysis [15]; it is not Kenyan law, and this paper does not claim compliance with an unverified successor circular.

The target operating model separates development, independent validation, approval, deployment, monitoring, and change control. Satisfying those controls is an evidence-based institutional conclusion, not an architectural property.

6.2.1. Separation of Development and Validation

The data science team responsible for training the GRU and Transformer layers is organisationally separated from the function responsible for independent challenge and approval. The validation scope includes:

1. **Conceptual soundness:** Challenge the Half-Student- t hyperpriors, hierarchy, spline basis, residualisation, interactions, likelihood, and candidate copulas. Test whether partial pooling understates or overstates risk in data-sparse corridors and document uncertainty rather than requiring an impossible proof of absence.

2. **Outcome Analysis (Backtesting):** Stress-testing the model against out-of-sample data, specifically evaluating how the model predicts defaults during extreme, unobserved tail events (e.g., a hypothetical 40% overnight fuel price spike or a total platform outage).
3. **Discriminatory Power Metrics:** Calculating the Receiver Operating Characteristic (ROC) curve, Area Under the Curve (AUC), and the Brier Score (to measure the calibration of the predicted probabilities against actual outcomes).

Only after the accountable committees approve the model, limitations, thresholds, monitoring, fallback, and deployment package may a signed artefact enter production.

6.2.2. Continuous Monitoring: Population Stability Index (PSI) and Gini Tracking

Gig-economy behaviour and data-generating processes can change when platform rules, fuel costs, regulation, seasonality, or labour supply change. Annual validation alone may therefore miss material drift. Monitoring frequency should match feature volatility, decision materiality, and the institution's response capacity.

The Population Stability Index (PSI) is calculated daily to measure distributional shifts between the original training data and the live production inference data. The PSI quantifies how much the underlying population has shifted:

$$PSI = \sum_{b=1}^B (\text{Actual}\%_b - \text{Expected}\%_b) \times \ln \left(\frac{\text{Actual}\%_b}{\text{Expected}\%_b} \right)$$

Where B represents the number of decile bins of the predicted probability score.

If input or score distributions shift, PSI may rise. The internal threshold of 0.25 triggers investigation, not an assumption about the cause and not automatic retraining. Controls may restrict affected decisions while data quality, calibration, discrimination, outcomes, and subgroup effects are reviewed.

Discriminatory power: Gini or AUC is monitored with calibration, Brier score, log loss, decision value, and realised outcomes. A value such as Gini below 0.35 can be an internal trigger; it is not labelled a universal regulatory minimum without an applicable source.

6.3. Algorithmic Fairness and Bias Mitigation

Predictive accuracy alone is not an adequate definition of a sound decision. A model can be well calibrated in aggregate and still allocate errors, limits, prices, or support unevenly. The governance objective is to identify unjustified disparities, proxy pathways, and product effects under applicable Kenyan law and partner policy, then redesign the system where a less harmful approach can achieve the legitimate objective.

The dual-regime neural architecture actively mitigates algorithmic bias through two stringent mechanisms:

6.3.1. Feature Orthogonalization and Blindness

Protected attributes are excluded from decision inputs unless lawfully needed for testing or an approved purpose, but “blindness” is not sufficient. Geography, device, platform, work schedule, vehicle, missingness, and wallet behaviour can retain proxy information. Adversarial probes, representation tests, conditional outcome analysis, calibration by group, and feature ablation estimate residual association. Zero mutual information is not promised.

6.3.2. Disparate Impact Ratio (DIR) Monitoring

Even with explicit blinding, complex neural models can reconstruct proxies. The governance layer continuously calculates the Disparate Impact Ratio across defined operational clusters. The DIR measures the ratio of favorable outcomes (e.g., micro-loan approval or a risk premium K below a certain threshold) for an unprivileged group compared to a privileged group.

An approval-rate ratio such as 80% can be used as a diagnostic convention. The U.S. four-fifths rule is not imported as a Kenyan legal test, and a geographic corridor is not automatically a protected group. The analysis should cover legally and ethically relevant populations, intersectional groups, error rates, calibration, price, limits, interventions, and complaints.

An alert requires the accountable team to investigate data coverage, label quality, model pathways, policy rules, partner practices, and outcomes. SHAP may help locate model sensitivity, but it cannot prove the disparity is justified. Where a less harmful design achieves the legitimate objective with comparable performance, the design should be changed.

Explainability, model-risk management, and fairness testing create a body of evidence that can be challenged by drivers, partners, validators, auditors, and supervisors. That challengeability, rather than invulnerability to scrutiny, is the standard the platform should seek.

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Appendix A: Underwriting Alternatives and the HLR Suitability Test

A.1. Structural models as a boundary case

Structural credit models associated with Merton infer default from the relationship between an asset-value process and a debt boundary [18]. They remain important for corporate-credit valuation, but a gig driver and vehicle do not provide the same observable traded firm value or replicating structure. Earnings Velocity is an engineered liquidity feature rather than a market value for the driver-vehicle system, and platform blocks, vehicle failures, and settlement interruptions create discontinuities that a simple geometric diffusion may not represent well.

Part 5, Appendix A retains the structural equations in abridged form and explains how a physical-measure underwriting estimate can remain separate from a market-consistent valuation overlay. Here, the relevant conclusion is narrower: the HLR and structural models address different empirical objects, and the underwriting champion must be selected by calibration, stability, decision relevance, and validation rather than mathematical prestige.

A.2. Reduced-Form Models and Intensity Processes

Recognizing the limitations of structural models, reduced-form (intensity-based) models treat default as an unpredictable Poisson jump process governed by a stochastic intensity (hazard rate) λ_t . The time of default τ is modeled as the first jump of a Cox process.

Under the $\mathbb{Q}$ -measure, the price of a defaultable zero-coupon bond with maturity T and zero recovery is:

$$P(t, T) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_t^T (r_s + \lambda_s) ds \right) \middle| \mathcal{F}_t \right]$$

Here, the credit spread is directly driven by the risk-neutral default intensity λ_s . In advanced implementations, λ_t is modeled as an affine jump-diffusion (AJD) process (e.g., a Cox-Ingersoll-Ross process):

$$d\lambda_t = \kappa(\theta - \lambda_t)dt + \sigma_\lambda \sqrt{\lambda_t} dW_t^{\mathbb{Q}} + dJ_t$$

Reduced-form models do not require an observed firm value. Basic formulations treat default time through an intensity adapted to available information; richer intensities can include covariates. The comparison should therefore be empirical and use-case specific.

A.2.1. The Exogenous Fallacy

For a ride-hailing driver, some defaults may be preceded by observable operating or liquidity changes; others arise from abrupt, unobserved, or genuinely exogenous events. The dual-regime architecture tests whether permitted sequences and explicit liquidity features improve warning time. Ten-hertz telematics does not prove determinism, and Earnings Velocity remains in the Explicit Liquidity Feature Path rather than the neural branch.

An intensity model and an HLR answer different questions. For short-duration underwriting with rich covariates, discrete-horizon real-world PD may be the more direct operational target. An intensity or market-consistent overlay can remain useful for valuation, survival analysis, or products whose timing and market inputs support it.

A.3. The suitability test for the HLR approach

The HLR is preferred only when it produces stable, calibrated, decision-relevant real-world probabilities and transparent uncertainty at the product horizon.

1. **Direct PD estimation:** The HLR models physical default probability $\mathbb{P}(\tau \leq T)$ over a defined product horizon. No claim is made that $\mathbb{P}$ and $\mathbb{Q}$ become equal because the tenor is short.
2. **Empirical pricing input:** A simplified one-period break-even credit component may use $k_{\text{credit}} \approx PD^{\text{cal}} LGD / (1 - PD^{\text{cal}})$. It does not guarantee realised loss coverage and is not the whole K_{RBCP} .
3. **Non-linear and hierarchical structure:** B-splines and cohort-time effects can represent real-world relationships without forcing them into a traded-asset diffusion, subject to the redundancy and calibration controls above.

The conclusion is one of scope, not mathematical taste. Use the HLR for calibrated physical-measure underwriting and portfolio decisions when it outperforms challengers. Use market-consistent methods only where valuation inputs and purpose justify them. Assemble K_{RBCP} transparently from risk and non-risk components. The model supports the institution's pricing and governance process; it does not, by itself, establish regulatory compliance, computational efficiency, or fairness.

Appendix B: Preserved Spline and Time-Prior Alternatives

The main specification uses governed P-splines, while B-spline, RW1, and AR(1) formulations remain explicit challengers. Their inclusion allows the validation team to test whether a difference penalty, a local random walk, or mean-reverting coefficient path better represents a particular effect without changing the champion by editorial fiat.

For a non-decreasing knot sequence $\xi_0 \leq \dots \leq \xi_{K+d}$, the B-spline basis retains the Cox-de Boor recursion [5]:

$$B_{k,0}(x) = \mathbf{1}[\xi_k \leq x < \xi_{k+1}],$$

$$B_{k,d}(x) = \frac{x - \xi_k}{\xi_{k+d} - \xi_k} B_{k,d-1}(x) + \frac{\xi_{k+d+1} - x}{\xi_{k+d+1} - \xi_{k+1}} B_{k+1,d-1}(x),$$

with a zero contribution when a denominator is zero. The smooth is $f(x) = \sum_k \zeta_k B_{k,d}(x)$. One local alternative uses an RW1 prior,

$$\zeta_k \sim \mathcal{N}(\zeta_{k-1}, \sigma_\zeta^2),$$

while the mean-reverting coefficient alternative uses

$$\zeta_k \sim \mathcal{N}(\rho \zeta_{k-1}, \sigma_\zeta^2), \quad |\rho| < 1.$$

These priors regularise adjacent basis coefficients but do not create an out-of-range safety rule. The comparison reports prior predictive shapes, effective degrees of freedom, boundary behaviour, posterior geometry, calibration, time-forward performance, and explanation stability. If a coefficient index lacks a defensible interpretation for mean reversion, the AR(1) challenger is not selected merely because it samples conveniently.

Selected macro or cohort effects may separately follow an AR(1) process around a long-run mean or an RW1 drift process. That is a time-process decision, distinct from the smoothing prior placed on a liquidity feature. The implementation records these two uses of AR1/RW1 under different parameter names so their economic meanings cannot be conflated.

Appendix C: Joint Longitudinal-Survival Research Challenger

Some liquidity variables are repeatedly observed with noise, change endogenously as credit actions occur, and may be poorly represented by carrying the last observed value into a hazard model. For a narrowly defined research challenger, let a latent liquidity trajectory be

$$m_i(t) = \mathbf{b}(t)^\top \alpha + \mathbf{z}_i(t)^\top \mathbf{u}_i,$$

with observations $w_{ij} = m_i(t_{ij}) + \epsilon_{ij}$. Link the trajectory to the timing model through

$$\lambda_i(t) = \lambda_0(t) \exp \{ \mathbf{x}_i(t)^\top \beta + \nu m_i(t) \}.$$

The shared parameter ν tests whether the latent trajectory contributes to event timing after the approved covariates [25]. This extension is considered only when measurement error, observation frequency, treatment timing, and informative missingness justify the additional joint likelihood. It must not recreate CFA, DLR, or Earnings Velocity as an ungoverned latent duplicate, and it does not replace the fixed-horizon HLR. Approval requires better time-forward calibration, survival prediction, and decision utility than a simpler time-updated hazard model.

Part 3

Joint Regulatory Capital Orchestration, Algorithmic Fairness, and Assistive Controls

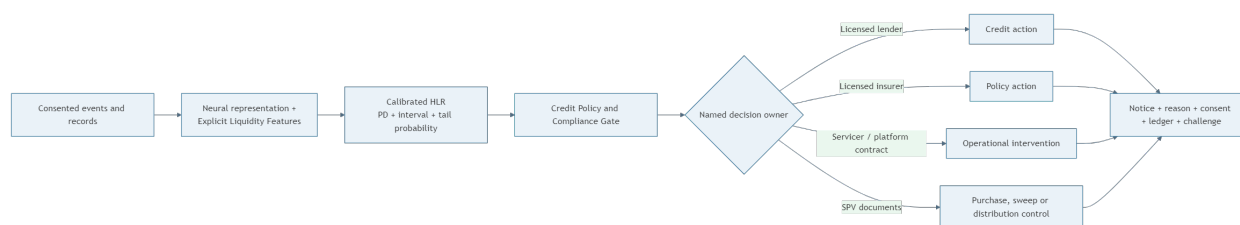
Mwendo Pamoja White Paper

1. From prediction to permission: a Kenya-first control architecture

Parts 1 and 2 described a system that can observe, estimate, and simulate. Part 3 asks a more human question: **who is allowed to act, on whose behalf, and what record must remain after the action?** A posterior PD cannot grant an insurance holiday. A telematics alert cannot amend a loan. A copula cannot set regulatory capital. Those decisions belong to licensed institutions, contractual parties, accountants, governance bodies, and, where applicable, supervisors.

The architecture is Kenya-first. IFRS 9 and IFRS 17 apply according to the reporting entity and instrument [13], [14]. Basel standards inform prudential treatment through the implementing rules applicable to the banking partner [15]. Solvency II, EBA guidance, and U.S. ECOA or SR 11-7 appear only as comparative design references unless the relevant entity or jurisdiction makes them applicable. This distinction does not weaken the technology. It tells every partner which propositions are models, which are internal policies, which are covenants, and which are legal requirements.

The canonical sequence is:



The sequence is implemented through a typed decision record. It distinguishes: **predictive evidence** from the HLR or timing challenger; a **calibrated risk estimate** for a stated horizon; an **accounting assessment** owned by the reporting entity; a **contractual covenant result** owned by the calculation agent; a **credit-policy action** owned by the licensed lender; a **compliance restriction** grounded in the applicable rule; an **assistive intervention** owned by the authorised product or platform party; and a **human override** with reason, authority, duration, and review date. The same numerical observation can inform several objects, but one object cannot silently stand in for another.

The Credit Policy and Compliance Gate receives a versioned schema rather than a generic risk score. Its minimum payload includes product and decision purpose; fixed-horizon PDs and, when approved, marginal survival probabilities; posterior interval and exceedance probab-

ities; current and stressed EAD; cure, LGD, recovery, and IPF-refund summaries; aggregate exposure and concentration; authoritative policy and insurance status; data quality, source age, support-range, and fallback flags; action and intervention history; model, calibration, policy, and contract versions; and the reason concepts available for communication. The gate returns a named action, owner, expiry, notice requirement, ledger instruction, review route, and evidence package.

2. Bank-counterparty accounting and prudential integration

2.1. IFRS 9 Impairment Pipeline under Joint Tail Distress

The core mandate of IFRS 9 is the forward-looking recognition of Expected Credit Losses before losses are realised in the income statement. The ECL model moves the reporting entity beyond an incurred-loss view by combining historical, current, and reasonable and supportable forward-looking information [7]. Stage 1 uses 12-month ECL, while Stages 2 and 3 use lifetime ECL under the applicable requirements [8]. In a short-duration gig-economy portfolio, contractual arrears can lag changes in operating cash flow. Governed telemetry and liquidity evidence can therefore enrich the reporting entity's SICR assessment when its incremental reliability, horizon, and relationship to lifetime credit risk have been validated. The accounting decision remains with the approved methodology and control process.

The Bayesian engine does not rewrite IFRS 9. The financial-asset holder's approved ECL methodology consumes relevant, reasonable, supportable, forward-looking evidence. The following mapping is an operating design:

- **Stage 1, 12-month ECL:** The exposure has not experienced a Significant Increase in Credit Risk under the approved framework. Neural state, explicit liquidity features, contractual performance, and macro scenarios may inform the estimate. CFA is an Explicit Liquidity Feature, not part of the GRU, and 0.6 is not a universal accounting threshold.

$$\text{ECL}^{(\text{Stage 1})} = \text{PD}_{ij}^{(12\text{M})}(\Phi_{it}) \times \text{LGD}_{ij}(v_t) \times \text{EAD}_{ij}$$

- **Stage 2, lifetime ECL after SICR:** The asset holder compares credit risk at the reporting date with credit risk at initial recognition using quantitative, qualitative, forward-looking, and backstop evidence. A twofold lifetime-PD movement may be one rebuttable

quantitative indicator. CFA and Earnings Velocity enter through the Explicit Liquidity Feature Path; the GRU supplies only the permitted sequence representation.

$$\text{SICRScore}_{i,t} = g\left(\frac{PD_{i,t}^{\text{life}}}{PD_{i,0}^{\text{life}}}, \Delta\text{rating}_{i,t}, \text{arrear}_{i,t}, \text{qualitative}_{i,t}, \text{macro}_t\right).$$

Upon an approved Stage 2 transition, the reporting entity recognises lifetime expected credit losses, measured as the applicable probability-weighted present value of cash shortfalls over the expected life. For a 30-day microloan, the 12-month and lifetime horizon can cover the same remaining contractual period, but revolving facilities require the entity's expected-life and exposure methodology, including drawdown and behavioural evidence.

- **Stage 3, credit-impaired:** Credit-impaired status follows the applicable default and impairment evidence. A single missed payment may be relevant but is not declared an automatic universal Stage 3 trigger. Interest revenue is calculated on the net carrying amount for credit-impaired financial assets as required by the applicable treatment.
- **Dynamic LGD:** LGD may depend on contractual collection routing, cure, recovery cost, recovery lag, policy refund rights, collateral, platform continuity, and calibrated Earnings Velocity. Mwendo Pamoja does not “garnish” earnings by unilateral operational control. Any deduction requires contractual and legal authority. Prudential floors depend on the exposure class, approach, and implementing rules; no 30%-45% range is asserted here without the banking partner's classification.

2.2. Basel prudential treatment as a bank-owned workstream

The banking partner determines whether exposures use a standardised or authorised IRB approach under applicable implementation. IFRS 9 allowance and prudential expected-loss treatment interact, but accounting provisions should not be described simply as “absorbing EL into pricing.” The following formula is an analytical reference, not a claim that the SPV or platform may use A-IRB.

The Basel IV regulatory asset correlation function for retail exposures (CRE31 formula) is PD-dependent:

$$R = 0.03 \frac{1 - e^{-35PD}}{1 - e^{-35}} + 0.16 \left(1 - \frac{1 - e^{-35PD}}{1 - e^{-35}}\right).$$

The relevant asset-class formula, PD definition, LGD, EAD, maturity, floors, scaling, and output-floor phase-in must be confirmed by the bank. No gig-portfolio range is treated as regulatory input before classification.

The capital requirement K per unit of EAD is:

$$K_{\text{cap}} = LGD \left[\Phi \left(\frac{\Phi^{-1}(PD)}{\sqrt{1-R}} + \sqrt{\frac{R}{1-R}} \Phi^{-1}(0.999) \right) - PD \right].$$

where Φ denotes the standard normal CDF and the 0.999 quantile reflects the Basel 99.9% one-year confidence level. Risk-Weighted Assets are then $RWA = K \times 12.5 \times EAD$, with the Basel IV **72.5% output floor** applying as a minimum relative to the Standardised Approach RWA. This output floor is the most significant Basel IV change for portfolios using Advanced IRB: internal models cannot reduce RWA below 72.5% of the Standardised Approach, limiting the capital relief from sophisticated internal PD estimation.

The copula bridge: The prescribed prudential formula and the institution's economic-capital stress model serve different purposes. A candidate Clayton model can inform concentration, tail stress, limits, and the SPV waterfall. It does not entitle the bank to replace prescribed correlation, alter R , claim capital relief, or prove an intervention effect. Any use in prudential modelling requires the bank's approved governance and supervisory process.

PSI in model governance: Weekly PSI and other drift metrics can support the bank's monitoring plan. A threshold of 0.25 is an internal convention or contractual trigger. It does not automatically require regulatory notification, model recertification, or suspension of IRB calculations under SR 11-7 or Basel. The bank's approved change policy determines escalation after the cause and performance impact are assessed.

Currency risk: The canonical SPV case is funded and serviced in KES, with USD used for headline reporting. The 130 KES/USD rate is an illustrative translation input, not a peg. If a USD note is issued against KES receivables, a forecast and excess KES reserve do not hedge it. The parties need an executed hedge or explicit unhedged loss allocation, with counterparty, collateral, tenor, basis, and break-cost treatment.

3. Insurer accounting and capital interfaces

The licensed insurer, not Mwendo Pamoja by default, maps policy cash flows into IFRS 17 and its applicable insurance-capital regime. If Mwendo Pamoja later becomes a licensed intermediary or risk carrier, that role requires its own analysis. Solvency II is presented as a comparative framework for a European-regulated carrier, not as Kenya's insurance-capital law.

3.1. IFRS 17 Accounting Protocols for Dynamic UBI

IFRS 17 establishes the accounting for insurance contracts within scope [14]. The insurer determines whether the Premium Allocation Approach is eligible for a group, including the coverage-period and material-difference criteria. The platform can supply exposure and policy-service data; it cannot elect the accounting method for the carrier.

3.1.1. The Building Block Approach (BBA) as Regulatory Backstop

Where an insurer validly applies the Premium Allocation Approach, the general measurement model is not described as an automatic backstop that the platform can trigger. The insurer assesses eligibility and the required measurement under IFRS 17. Under the general model, measurement includes probability-weighted fulfilment cash flows, discounting, a risk adjustment for non-financial risk, and the contractual service margin where applicable [3], [4].

Three design questions determine whether the simplified narrative remains appropriate:

- **1. Onerous Contract Identification:** IFRS 17 strictly prohibits masking loss-making policies within profitable portfolios. Even under PAA, groups of contracts must be monitored, and **Onerous Contracts** (groups of policies where the expected future cash flows and claims exceed the remaining premium, i.e., where expected FCF exceeds the Liability for Remaining Coverage (LRC)) must be identified and loss recognized immediately [5]. To test whether a cluster j of drivers constitutes an onerous group, the Insurtech cannot rely on PAA; it must invoke the first three BBA blocks to calculate true FCF:

$$\text{If } \mathbb{E}[\text{FCF}_j] > \text{LRC}_j \Rightarrow \text{Onerous Contract Group}$$

Model outputs can supply evidence to the insurer's fulfilment-cash-flow and onerous-group process, but the insurance actuarial model owns claim frequency, severity, ex-

penses, lapse, discounting, risk adjustment, coverage units, and grouping. A credit PD or driver-liquidity shock is not substituted directly for an insurance-claim forecast.

- **2. Multi-year policy assessment:** A multi-year group loses the automatic short-coverage route but may still qualify if the insurer demonstrates that PAA measurement would not materially differ from the general model. Telematics volatility and a candidate copula do not decide that test.
- **3. Risk adjustment evidence:** A candidate dependence model may inform the insurer's non-financial risk analysis, subject to actuarial validation and the insurer's disclosed confidence-level or cost-of-capital method. The credit-product copula does not automatically set the IFRS 17 risk adjustment.

3.2. Solvency II as a comparative internal-model case

Where a carrier is actually subject to Solvency II, it calculates SCR under the applicable standard formula or approved internal model. Mwendo Pamoja is not assumed to be that carrier.

The Solvency II standard formula aggregates prescribed risk modules and dependencies under its own calibration. A separate economic-risk analysis may test whether the platform's insurance and credit exposures exhibit tail behaviour not captured by a simpler dependence assumption. A Clayton copula is one candidate for lower-tail dependence, not a correction that can replace the insurer's prescribed calculation or prove that the standard formula understates this portfolio.

An approved partial internal model cannot be assumed or overridden by the platform. A carrier would need governance, data, use-test, calibration, validation, documentation, and supervisory approval. No capital relief is claimed for an intervention until the carrier and supervisor accept the evidence. For the current Kenya-first design, the copula remains an economic stress tool and the licensed insurer owns capital treatment.

4. Algorithmic Equity and Fairness Frameworks

4.1. The Structural Bias Risk in Deep Temporal Architectures

Continuous underwriting introduces a serious ethical and conduct risk: a model may reproduce historical disadvantage through geography, platform, device, work schedule, vehicle, road quality, missingness, or labels. The affected Kenyan populations, protected attributes,

and legal tests must be identified from applicable law and institutional policy. U.S. ECOA is a comparative example, not an assumed Kenyan cause of action.

The risk is structural, not incidental. The GRU's hidden state $\mathbf{h}_t^{\text{GRU}}$ will encode whatever patterns are predictive of default in the training data, including patterns driven by systemic infrastructural inequality rather than individual creditworthiness. No amount of removing explicit demographic features from the input resolves this if the telematics features (geohash, circadian fatigue accumulation, average trip distance) serve as effective demographic proxies. The solution must be embedded in the training objective itself [9].

4.2. The Mathematical Constraint: Equalized Odds

Equalized odds is retained as one diagnostic and possible optimisation objective, not a guarantee or a universal legal standard. Let A be a lawfully held audit attribute, Y a carefully defined observed outcome, and $\hat{Y}$ the decision. Protected attributes should not be inferred from geohash merely to complete a formula; the fairness data design requires lawful basis, access separation, minimisation, and an impact assessment.

A model satisfies Equalized Odds if its approval decisions are independent of the protected attribute A , conditional on the true outcome Y . Formally, both the True Positive Rate (TPR) and the False Positive Rate (FPR) must be identical across all protected groups:

$$P(\hat{Y} = 1 \mid Y = y, A = 1) = P(\hat{Y} = 1 \mid Y = y, A = 0), \quad \forall y \in \{0, 1\}$$

This constraint has two components with distinct policy implications [10]:

1. **Equal Opportunity (Equal TPR):** A creditworthy driver from a marginalized background ($A = 1$) must have the same mathematical probability of being approved for a revolving credit line as a creditworthy driver from a non-marginalized background ($A = 0$), conditional on both being genuinely creditworthy. This prevents the model from systematically denying credit to creditworthy marginalized drivers due to geohash-correlated proxy features.
2. **Equal FPR:** A non-creditworthy driver from a marginalized background must have the same probability of being *incorrectly* approved as a non-creditworthy driver from a non-marginalized background. This prevents the inverse bias: a model that incorrectly approves marginalized drivers at higher rates (a patronizing form of bias that leads to higher default rates in that group and subsequent portfolio-wide tightening).

4.3. Optimization Integration and Continuous Monitoring

Enforcing Equalized Odds creates a trade-off with absolute predictive accuracy (AUC-ROC), because the model is mathematically constrained from exploiting highly predictive but biased proxies. This accuracy-fairness frontier is not a binary choice: the fairness regularization parameter λ_{fair} controls the operating point on the frontier.

The Insurtech operationalizes this by adding a **Maximum Mean Discrepancy (MMD) regularization penalty** to the neural network’s training loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{predictive}} + \lambda_{\text{fair}} \mathcal{D}_{\text{fair}},$$

where $\mathcal{D}_{\text{fair}}$ may be an MMD-based representation discrepancy or another approved differentiable surrogate. TPR and FPR are rates rather than raw distributions, so an MMD term must specify which conditional score or representation distributions it compares.

A Bayesian analysis can express uncertainty around subgroup TPR, FPR, calibration, or outcome gaps when the sampling and model assumptions are appropriate. A point estimate can hide uncertainty, especially in small groups. A statement such as “the posterior probability that a defined TPR gap is below 3 percent is 95 percent, conditional on the data and model” is an analytical result, not a declaration of legal compliance. It must be accompanied by sample size, prior sensitivity, label quality, missingness, and practical impact.

Continuous fairness monitoring on Flink: The Flink pipeline computes metrics continuously and provides XAI (Explainable AI) governance:

1. **Explanations:** SHAP, local surrogates, explicit feature contributions, and counterfactual tests help diagnose model behaviour. They do not prove the absence of proxies.
2. **Outcome ratios:** DIR is monitored with confidence intervals and sample-size disclosure. The U.S. four-fifths convention may be used as a diagnostic benchmark, not a universal legal safe harbour or violation threshold.
3. **Conditional performance:** Calibration, TPR, FPR, false restriction, price, limit, intervention, complaint, and cure outcomes are measured by relevant groups and intersections.
4. **Drift:** PSI and data-quality measures trigger investigation. Drift is not itself unfairness and must be linked to outcomes and model pathways.

5. Operationalizing the Assistive Ecosystem

5.1. From Punitive to Non-Punitive Underwriting

The purpose of detecting a joint cascade early is not merely to measure it. It is to create time in which an authorised party may choose a less destructive path. The transition can be described simply: detect stress, diagnose its likely cause, offer an approved intervention, observe the result, and revise the decision. Collections and impairment recognition remain necessary when facts require them; assistive design must not become delayed recognition by another name.

An early intervention may cost less than an avoidable default and may preserve insured, productive time, but that is a hypothesis for cohort evidence. The evaluation must include the intervention cost, adverse selection, repeat use, modification accounting, driver welfare, opportunity cost, cure durability, ECL, and SPV cash timing. The ethical and financial case becomes stronger when both are measured rather than asserted.

Because an intervention changes what happens next, every action enters a **treatment and selective-label ledger**. The record includes the eligible population, offer rule, score and uncertainty before action, product state, owner, action offered, customer response, executed terms, cost bearer, exposure after action, outcome window, censoring, cure, default, recovery, and complaints or appeal. Approved declines and freezes retain the counterfactual data available at decision time even though their repayment outcome is unobserved. Validation then separates ordinary performance monitoring from treatment-effect estimation and rejection or action-selection bias. Randomised encouragement, phased rollout, matched controls, or another defensible design is used where lawful and practical.

When calibrated risk and uncertainty approach an action threshold, the Credit Policy and Compliance Gate selects only actions authorised for the product and counterparty. The following interventions are product hypotheses. Each requires a named owner, driver communication and consent where required, duration, exit rule, cost bearer, accounting treatment, fairness monitoring, and experimental evidence.

Intervention	Decision owner	Precondition	Primary evidence
Premium schedule support	Licensed insurer and finance provider	Policy wording and finance terms permit it	Cover continuity, claims, modification and cash-flow analysis

Intervention	Decision owner	Precondition	Primary evidence
Micro-reward bridge	Licensed lender plus contracted platform	Platform can lawfully offer work without discriminatory allocation	Driver take-home, arrears cure, complaints and outcome test
Fatigue mitigation	Platform or safety owner	Verified safety basis and lawful routing authority	Safety benefit, income impact, false restriction and appeal
ZEV routing support	Platform	Verified charger and range data	Uptime, income, data quality and geographic fairness
Margin compression support	Lender and funder	Contract and funded support budget	NIM, affordability, SPV cash flow and allocation of loss

5.2. The Financial Safety Net Interventions

- **Intervention 1: Dynamic Premium Holidays** A proposed trigger combines calibrated IPF risk with an acute decline in Explicit Liquidity Features such as Earnings Velocity, not a wallet metric hidden inside the GRU. A pause, extension, repricing, or continued cover occurs only if the insurer and finance provider approve it under the contracts. The model does not assume that stress will revert to the historical mean.

Unearned premium, coverage, policy liability, receivable modification, and SPV eligibility continue to evolve under the actual terms. The cost is allocated to Class C only if the SPV documents and funded model provide for it. The intervention objective is to preserve viable cover and work, not to declare the lender insulated.

- **Intervention 2: Micro-reward bridge and consensual restructuring** A bridge may be considered when approved arrears, liquidity, and viability criteria indicate that a short, funded opportunity could restore the payment path. The lender owns any restructuring. A contracted mobility platform may offer an opt-in incentive, additional earning opportunity, or sponsor-funded reward under transparent allocation rules. Any deduction from the resulting fare requires prior authority, a sufficiency control, clear communication, and a product-ledger receipt. The design must not reserve desirable

work exclusively for distressed borrowers in a way that unfairly shifts income or safety risk to other drivers.

The outcome must be measured rather than pre-written. A bridge may support cure, but it does not automatically avoid Stage 3 or change the accounting evidence. Preferential route allocation can affect other workers and may create safety, fairness, labour, platform, and conduct issues. The platform must opt in as the routing owner.

- **Intervention 3: Fatigue Mitigation Routing** Triggered by validated safety signals and operating rules, not a latent “desperation” label. A financial state must not be inferred from driving hours alone.

A platform may test a cooling-off, rest, or alternative-task option if it has authority and evidence. The test must measure collision proxy outcomes, actual claims where available, take-home income, false restrictions, and worker choice. No IFRS 17 or onerous-group result is guaranteed.

- **Intervention 4: ZEV-aware work and charging support** For drivers transitioning to zero-emission vehicles, charging availability, queueing, range, charger compatibility, and trip geography can affect productive time. A prolonged charging interruption during peak demand can reduce fare income and create a new operating-cost pattern that older vehicle-finance models may not capture.

Where lawful and contractually available, a context service can use verified charger location, compatibility, availability, price, queue, range, trip, and time data to recommend charging windows or work zones. The mobility platform remains the routing owner, and the driver should retain meaningful choice. The pilot measures charger-data quality, actual wait time, income, safety, geographic access, battery state, false recommendations, and effects on other drivers. The service seeks to reduce avoidable downtime; it cannot ensure income continuity or repayment.

- **Intervention 5: Funded rate-shock support** The governed benchmark service observes KESONIA, while treasury scenarios and the Transformer context branch can assess the wider economic regime. A contractual reset can affect driver affordability, but the timing and magnitude depend on product terms, notices, floors, caps, and lender policy. The platform does not characterise every lawful benchmark transmission as predatory or treat a model alert as authority to alter a contract.

During a funded support period, the lender may hold or cap a reset, use an agreed subsidy, or restructure the product. The model then shows who absorbs the difference and

whether SPV eligibility, yield, ECL, and waterfall change. Class C does not automatically pay the cost, and senior notes are not mathematically insulated until the cash-flow model demonstrates it.

6. Supervisory Auditing of Behavioral Interventions

The implementation of behavioral interventions (Dynamic Grace Periods, Micro-Reward Bridging, Fatigue Mitigation Routing, and ZEV Smart Routing) fundamentally shifts the platform from a passive observer of default cascades into an active, protective financial safety net. However, within a strictly regulated banking environment, these algorithmic interventions introduce profound compliance risks.

Automated changes to payment timing, limits, routing, or policy service can affect customer rights, accounting, credit classification, eligibility, and fairness. Kenya-first legal and accounting analysis governs the production design. EBA and ECOA materials are comparative references for partners subject to them, not direct Kenyan mandates.

6.1. Forbearance vs. Assistive Interventions (IFRS 9 and EBA Guidelines)

6.1.1. The Regulatory Threat: “Shadow Forbearance”

Under the EBA’s Guidelines on the application of the definition of default, and the overarching IFRS 9 Expected Credit Loss (ECL) framework, a critical distinction exists between standard contract modifications and **Forbearance**.

Forbearance commonly concerns a concession granted because a borrower is experiencing financial difficulty, but definitions and consequences depend on the applicable framework and institutional policy. Automated grace periods can create “shadow forbearance” risk if they delay recognition without changing the underlying condition. The lender must therefore classify, report, and provision from the facts, with clear modification, cure, and probation rules. Potential supervisory consequences should be stated only under an applicable Kenyan authority, not imported from a foreign framework as an automatic penalty.

6.1.2. The Architectural Solution: Proactive vs. Reactive Classification

The intervention register distinguishes preventive support, modification, concession due to financial difficulty, cure, and default. Timing is relevant but not conclusive. An ERP records

the classification and evidence; it does not mathematically prove the legal or accounting conclusion.

1. **Proactive support:** A validated operational or liquidity signal may justify offering information, a funded incentive, or another action already permitted by the contract before arrears. The classification still depends on the driver’s condition, the nature of the concession, and the applicable accounting and regulatory framework. An offline charger does not by itself establish financial distress, nor does calling an action proactive prove that it is outside forbearance analysis.
2. **Forbearance Interventions (Reactive):** Interventions triggered *after* a missed split-fare repayment (such as the Micro-Reward Bridge initiated at IFRS 9 Stage 2 or Stage 3).

6.1.3. Accounting integration and discounted-cash-flow testing

For a modified financial asset, the reporting entity applies its approved IFRS 9 modification, derecognition, EIR, and credit-impaired analysis. For a partner subject to EBA default rules, the diminished-financial-obligation calculation is a separate prudential test.

No universal IFRS 9 “1% rule” is used in this paper. Any EBA quantitative threshold is applied only by an institution within scope and with the complete qualitative default criteria. The system calculates the present-value difference without converting the result into an automatic accounting conclusion:

$$\Delta NPV = \frac{NPV_{\text{original}} - NPV_{\text{modified}}}{NPV_{\text{original}}},$$

using the appropriate original EIR and cash-flow definitions. A micro-reward bridge can change timing even without principal forgiveness, so ΔNPV is not assumed to be zero. The accounting owner decides modification, derecognition, staging, and interest treatment.

6.2. Algorithmic Fairness and Fair Lending (ECO/Disparate Impact)

6.2.1. The Regulatory Threat: Algorithmic Redlining

In jurisdictions governed by the Equal Credit Opportunity Act or related fair-lending doctrines, behavioural interventions can create discrimination risk when a facially neutral design produces unjustified adverse effects for a protected class. For this Kenya-first platform, ECOA

remains comparative material. Kenyan law, product duties, data-protection principles, contracts, and partner policy determine the production analysis.

Suppose the architecture offers valuable bridge opportunities more often to men in affluent Nairobi areas while imposing limit reductions more often on women in peri-urban Kisumu. That pattern demands investigation of exposure, need, data coverage, labels, product eligibility, platform behaviour, model pathways, policy rules, and outcomes. Excluding gender or ethnicity from decision inputs does not remove proxy risk: geography, shift time, device quality, vehicle, or platform can retain association. The strength and legal meaning of that association must be measured rather than assumed perfect.

6.2.2. The Architectural Solution: Less Discriminatory Alternative (LDA) Testing

To govern this, the Mwendo Pamoja architecture embeds a continuous Fair Lending validation engine directly into the Azure MLflow pipeline.

The Model Risk Committee uses a locally applicable fairness and data-protection framework. U.S. less-discriminatory-alternative analysis can be a comparative technique. Before deployment, the model is evaluated on lawfully governed audit data, with access separated from production decisioning where appropriate.

The system continuously calculates the **Disparate Impact Ratio (DIR)** for every specific behavioral intervention:

$$DIR_{Intervention} = \frac{P(Intervention|Unprivileged\ Group)}{P(Intervention|Privileged\ Group)}$$

If a diagnostic threshold is breached, release is blocked pending review. The team cannot “prove fairness” with SHAP or assume an apparently objective variable is harmless. It must investigate data, labels, features, model, policy, partner process, alternatives, outcome trade-offs, and driver impact.

6.3. Continuous control monitoring and reporting adapters

Transparency requires a complete internal intervention record before it requires a particular message standard. The canonical record contains:

- decision, trigger, model, feature, data-quality, and policy versions;
- calibrated PD, uncertainty, explicit reason codes, and overrides;
- decision owner, authority, driver communication, consent status, and challenge path;

- original and modified contractual cash flows;
- ECL, eligibility, waterfall, and accounting assessments;
- fairness and outcome-monitoring fields; and
- maker-checker approval, integrity evidence, and retention classification.

ISO 20022 message definitions and XBRL taxonomies can be implemented as **adapters** where a bank, payment rail, supervisor, or filing regime specifies them [16]. Governed, partner-specific adapters allow the canonical intervention record to travel through an approved schema while preserving its source, meaning, ownership, reconciliation status, and review history.

7. Conclusion

The gig economy changes the timing and observability of labour, cash flow, insurance exposure, and credit risk. The cascade in Part 1 is a plausible path created by shared dependencies, not a deterministic fate. The value of the architecture lies in making those dependencies visible early enough for a licensed and accountable party to choose a proportionate action.

The architecture developed across this series constitutes a mathematically rigorous, computationally scalable, and regulatorily compliant alternative to the legacy paradigm:

- **Part 1** located the shared productive asset and mapped how three products can transmit stress.
- **Part 2a** established event-time, point-in-time, and feature-ownership controls for neural and explicit paths.
- **Part 2b** calibrated a redundancy-controlled HLR and framed Clayton as one candidate for lower-tail dependence.
- **Part 3** separated model inference from policy permission, assigned decisions to the lender, insurer, platform, servicer, SPV, and accounting owners, and preserved foreign frameworks as comparative rather than invented local law.

The target system has six testable properties: calibrated posterior risk with uncertainty; partial pooling for sparse cohorts; explicitly owned liquidity features; a separately validated dependence model; a Credit Policy and Compliance Gate with named decision owners; and interventions whose human and financial outcomes are measured. It earns the description “assistive” only when the evidence shows that it preserves viable work, respects rights, avoids unfair restriction, and improves portfolio outcomes.

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Part 4

Enterprise ERP and Telemetry

Mwendo Pamoja White Paper

1. Institutionalization, Architectural Preservation, and the Microsoft Dynamics 365 Finance ERP Endpoint

1.1. Executive summary: when the model meets the books

At midnight, the driver's day must become a ledger that another institution can trust. A completed trip may have produced a fare, platform commission, protected operating amount, premium deduction, loan receipt, reserve movement, servicing fee, and SPV collection. The underwriting system may have changed a limit or recommended support. None of those facts becomes accounting merely because the model emitted JSON.

Parts 1 through 3 developed the economic, data, model, and intervention logic. Part 4 describes the institutional boundary: how approved product events become balanced journals, how legal entities remain separate, how a decision can be replayed, and how an ERP consumes financial primitives without ingesting raw 10 Hz telemetry.

The target stack uses Microsoft Dynamics 365 Finance as an enterprise general-ledger and entity-accounting endpoint, with Power BI as one reporting surface. Neither product is assumed to provide a complete lending subledger, IFRS 9 engine, IFRS 17 actuarial engine, asset-liability system, funds-transfer-pricing engine, trustee waterfall, or regulatory gateway. Those capabilities belong to specialised product services, controlled calculations, partner systems, or configured extensions.

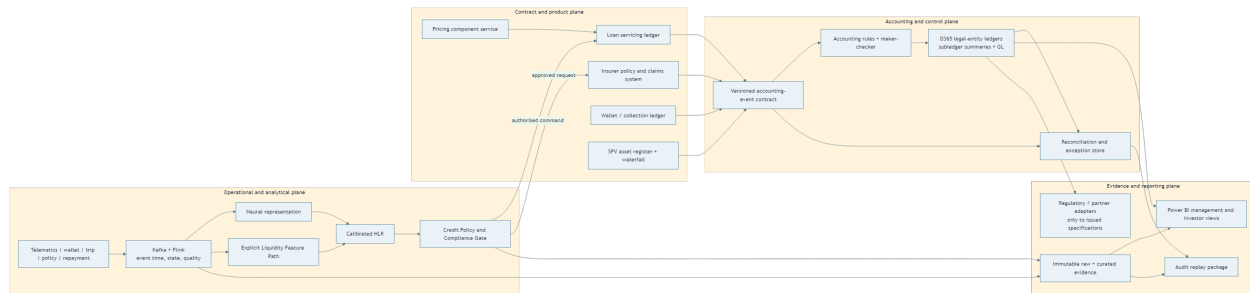
The architecture cannot prove compliance by design. It creates evidence: reconciled events, model and policy versions, balanced entries, approvals, data lineage, exception queues, retained source documents, and reports that can be validated. The HLR produces calibrated risk evidence. A governed pricing service assembles K_{RBCP} from risk and non-risk components. D365 records resulting accounting events after approval; it does not receive a model output as accounting truth.

1.2. Architectural Preservation and Augmentation Strategy

The most critical architectural constraint is preserving the high-frequency telemetry path without asking the enterprise ledger to become a stream processor. Banking ERPs are commonly optimised for controlled transactions, subledger integration, period close, and reporting. Even where an ERP can accept frequent events, raw 10 Hz telemetry, kinematic windows, and wallet-state computation belong in specialised event and analytical services. The underwriting path is therefore separated from the general ledger, with governed financial events crossing the boundary.

The architecture adopts a strictly decoupled, multi-layered approach:

Figure 1: Event, decision, product-ledger, accounting, and reporting boundaries



- Layer 1: The Telemetry and Data Engineering Core** The Apache Kafka and Apache Flink stream processing layer acts as the foundational ingestion engine. Debezium captures Change Data Capture (CDC) events from the raw telematics databases and platform ledgers. This layer executes stateful stream processing, ensuring point-in-time correctness for the machine learning models.
- Layer 2: The Neural-Bayesian Underwriting Engine** The neural and Explicit Liquidity Feature paths meet in the redundancy-controlled HLR. It emits calibrated PD, uncertainty, model version, and explanation inputs. The downstream Credit Policy and Compliance Gate emits an authorised action. Pricing is a separate service that reconciles K_{RBCP} to expected loss, cost of capital, operating and liquidity costs, margin, and approved adjustments.
- Layer 3: The Integration API and Transformation Middleware** The middleware validates commands from named product owners and translates product-ledger events into versioned accounting-event contracts. It does not invent an “IFRS-compliant” journal from a model action. A premium holiday first changes the insurer’s policy or finance contract, then the responsible accounting rule produces entries supported by that change.
- Layer 4: Microsoft Dynamics 365 Finance & Operations (GDI & SupTech Node)** D365 serves as the general-ledger and legal-entity accounting endpoint for configured journals. Product ledgers remain systems of record for contracts and transaction detail. Regulatory or partner submissions use separate adapters built only after an authoritative schema, taxonomy, frequency, and submission protocol are confirmed. No CBK GDI or SupTech interface is assumed.

- **Layer 5: Microsoft Power BI & The Regulatory Gold Tables** Power BI presents management, model-risk, servicing, SPV, and investor views from governed semantic models. A dashboard is not a regulatory filing or evidence by itself. Every metric retains source, owner, as-of time, definition, and reconciliation status.

1.3. Microsoft Dynamics 365 Finance: The Regulatory ERP Endpoint

The selection of D365 is a strategic ERP choice, subject to fit-gap, configuration, volume, licensing, security, and integration testing. D365 can be the authoritative general ledger for each configured legal entity. The loan servicer, insurer, wallet ledger, and SPV register remain authoritative for their own contracts and events, then reconcile to D365.

1.3.1. Product subledgers and the D365 integration boundary A scaled gig-economy portfolio can produce many daily credit, insurance, wallet, settlement, and reversal events. A single driver may generate a loan disbursement, several authorised fare deductions, a policy transaction, and a reserve movement in one day. The target volume must be benchmarked rather than fixed at an unsupported 100,000 drivers. Product systems retain contract-level operational state, while the D365 design receives the detail or summary needed for controlled accounting and reporting.

- **Loan servicing ledger:** Tracks contract principal, interest, fees, schedules, receipts, arrears, modifications, write-offs, and customer balances. D365 receives controlled summaries or detailed journals according to the chosen design.
- **Insurer policy and claims system:** Tracks premium, coverage, claims, cancellation, and insurer accounting inputs. An IPF receivable remains distinct from the insurer's unearned coverage liability.
- **Wallet and collection ledger:** Tracks settlement ownership, authorised deductions, reversals, residuals, driver reserves, and controlled SPV collections. It is not called escrow without the legal account structure.
- **SPV asset register and waterfall:** Tracks eligibility, purchase, note balances, reserves, tests, and priority of payments.

The credit-risk extension adds a common contract-and-risk spine to these systems. At minimum it carries driver, product, facility, contract, risk-episode, vehicle, policy, platform, geography, and cohort keys; contractual schedule and next due date; limit, utilisation, principal, interest, fees, arrears, modification, default, cure, and closure states; event, availability, decision, label-maturity, accounting, and cash-realisation times; current and stressed EAD;

recovery, refund, cost, and write-off cash flows; intervention and policy versions; and model, calibration, feature, hierarchy, and posterior-artifact versions. Each field has one authoritative owner and an effective-time history.

The risk platform retains the granular posterior and timing paths. The servicing ledger retains contractual balances and states. The ECL engine retains scenario-weighted accounting estimates. The SPV register retains purchased-asset and waterfall facts. D365 receives approved accounting events and reconciliation keys. This distribution allows every institution to use the same economic episode without asking the general ledger to store posterior draws or the model registry to become a receivables ledger.

Telematics does not become a journal. Financial product events are reconciled and transformed into balanced journal batches. Posting may occur intraday or at a controlled cut-off, with idempotency key, debit-credit balance, entity, currency, account, dimension, source event, reversal reference, approval, and period status. OData or the Data Management Framework is selected after throughput testing. Driver-level auditability remains in the product ledger and evidence store even when the GL receives a summary.

1.3.2. Mapping the governed pricing service to product and accounting systems

Under CBK's revised framework, the customer lending rate is $R_{\text{customer}} = \text{KESONIA} + K_{\text{RBCP}}$, with fees and charges added for total cost of credit [4], [5]. The HLR supplies calibrated risk inputs; the governed pricing service calculates and approves the full premium.

The authoritative KESONIA series is ingested with publication date, effective date, source, checksum, calendar, corrections, and fallback status [6]. Whether D365 can calculate the exact contractual convention natively is a fit-gap item; the loan servicing or pricing engine may calculate accrual and post the result.

The fit-gap is resolved contract by contract rather than by treating "D365 integration" as a complete design. The implementation team first records the KESONIA observation window, compounding method, day-count basis, margin, floor, cap, reset date, customer-notice rule, fallback, rounding, and correction treatment. It then configures representative IPF, microloan, and revolving contracts in a controlled environment and compares the native result with an independently approved calculation schedule. Finance, product, risk, accounting, and technology owners jointly approve the result and the system boundary [7].

Where native configuration reproduces the contract and required audit trail, D365 may calculate or control the event. Where it does not, the pricing or product subledger remains the

calculation engine and sends an approved accrual or accounting event to D365. The interface retains instrument identifier, source series, observation dates, factor, spread, effective date, currency, correction status, approval, and reconciliation key. This approach addresses the fit-gap without forcing the general ledger to imitate a product engine or allowing an external service to post unexplained totals.

A contractual rate changes only under the loan terms, customer notice, applicable rules, and authorised lender policy. Daily model movement must not silently reprice an existing contract. The record distinguishes a new-offer price, a permitted reset, a modification, and a monitoring-only risk movement. IFRS 9 classification follows the reporting entity's analysis, not an ERP timestamp.

1.3.3. Automating Assistive Interventions in D365 Assistive interventions become accounting events only after the licensed product owner authorises a contractual action and the responsible accounting function assigns treatment.

- **Premium schedule support:** The licensed insurer and finance provider first approve any contract or collection change. Their product systems emit the amended coverage, receivable, premium, and cash-flow events. Accounting rules then post the appropriate entries. Cover, earned premium, liability, ECL, and modification treatment are not fixed by a generic D365 flag.
- **Micro-reward bridge:** The platform and lender record the offer, driver acceptance, fare, deduction authority, and loan receipt. Cash is allocated according to the actual contract and law. Receipt may support cure but does not automatically avert Stage 3.

1.4. Sub-Ledger to General Ledger Pipeline Mechanics and IFRS 9 Staging

The ECL process can use higher-frequency evidence, but the accounting owner remains responsible. Integration automates data preparation, calculation inputs, journal workflow, and evidence retention while preserving review, overlays, and controls.

1.4.1. Higher-frequency IFRS 9 evidence and controlled staging Days past due remains important and may serve as a rebuttable backstop under an entity's IFRS 9 methodology, but it need not be the only evidence of Significant Increase in Credit Risk. For some gig-economy borrowers, cash flow can deteriorate before a contractual payment becomes past due. The platform tests whether permitted higher-frequency evidence provides reliable incre-

mental warning; it does not assume that every driver reaches zero cash or that a universal 20-day clock exists.

The HLR is an evidence source, not a staging oracle. The approved SICR service evaluates lifetime-PD movement, ratings, arrears, qualitative factors, macro scenarios, backstops, and overrides. DPD remains relevant evidence. The service writes a proposed stage, reasons, calculation version, and approval status; it does not bypass the methodology.

The base configuration uses a governed ECL engine outside the general ledger, with D365 controlling approved journals, consolidation, workflow, and reporting. A native or hybrid D365 route remains available if the fit-gap test proves that it reproduces the reporting entity's approved methodology, contract-level cash flows, scenario logic, effective-interest discounting, overlays, and audit evidence [8].

The governing quantity is the probability-weighted present value of cash shortfalls. For scenario s and cash-flow date τ ,

$$ECL_i = \sum_s w_s \sum_{\tau} \left(CF_{i,\tau}^{\text{contract}} - CF_{i,s,\tau}^{\text{expected}} \right) DF_{i,\tau}.$$

The expected cash flow includes scheduled exposure, drawdown, cure, prepayment, modification, ordinary recovery, eligible IPF refund, costs, and their timing. The factorised PD-LGD-EAD form below is a controlled computational representation only when its marginal default convention, exposure path, loss severity, cash timing, and discounting reproduce the approved cash-shortfall methodology.

For scenario (s) and future interval τ , the controlled calculation can be represented as:

$$ECL_i = \sum_s w_s \sum_{\tau} PD_{i,s,\tau} \times LGD_{i,s,\tau} \times EAD_{i,s,\tau} \times DF_{i,\tau},$$

with the horizon, marginal or conditional PD convention, cure and recovery treatment, scenario weights, and discount factor defined in the accounting methodology. The HLR contributes calibrated evidence; the approved SICR and ECL services determine how that evidence enters staging and measurement.

The fixed-horizon HLR and the timing challenger enter through an approved reconciliation layer. Cumulative PD is never multiplied afresh in each month. Where the survival model is used, monthly marginal default probability is $q_{i,m} = S_i(t_{m-1}) - S_i(t_m)$. Where only horizon

PD is approved, a documented term-structure allocation converts it to marginal cash-flow intervals and is calibrated back to the horizon total. The accounting engine records which route and artifact version generated each estimate.

The approved output passed to D365 contains the legal entity and reporting period; instrument or portfolio identifier; approved IFRS 9 stage; PD, LGD, EAD, scenario, model, rule, and overlay versions; opening and closing allowance; impairment movement; modification, write-off, recovery, and reversal identifiers; maker-checker status; posting profile; dimensions; journal date; currency; and reconciliation key. The loan subledger remains capable of reproducing the contract balance and ECL source data even when the general ledger receives an aggregated entry.

This creates three controlled deployment choices. A **native route** is used where D365 passes the fit-gap and independent recalculation. An **external route** calculates staging and ECL in a dedicated governed engine and sends approved journal instructions. A **hybrid route** retains the detailed calculation externally while using D365 for workflow, posting, consolidation, control evidence, and reporting. In every route, an unapproved model payload cannot unilaterally change an accounting stage.

1.4.2. General Ledger Consolidation and SPV Segregation Mwendo Pamoja operates multiple legal entities, primarily the main operating Insurtech entity and the proposed bankruptcy-remote Special Purpose Vehicle (SPV) that holds eligible purchased microloan, revolving-credit, and IPF finance receivables, excluding insurer-owned or unremitted premium cash. D365 Finance's multi-company architecture can represent the resulting multi-entity accounting structure after the legal and accounting design is approved.

The ERP provisions distinct legal-entity ledgers and dimensions. The licensed originator first records the loan under its approved accounting. A later receivable sale is recorded only after eligibility, purchase price, transfer, settlement, derecognition or continuing-involvement analysis, and supporting documents. The precise journals depend on those facts and are defined in an accounting-rules catalogue rather than hard-coded in this narrative.

Multi-company bookkeeping supports reconciliation but does not create bankruptcy remoteness, true sale, security, account control, or servicing continuity. Those are legal and operational conditions. When established, D365 records each entity's approved journals while the SPV asset register and waterfall calculate investor allocations. The USD 9 million equivalent structure is operated in KES and remains subject to the transaction tests described in the memorandum and term sheet.

2. Telemetry, SHAP/XAI, and Data Pipeline Enhancements

2.1. Explainability and the Algorithmic Fairness Mandate

The decision record must be explainable to the driver, lender, validator, auditor, and, where applicable, supervisor. “Total” or legally indisputable transparency is not promised. The record combines explicit feature contributions, neural diagnostics, pricing-component reconciliation, policy reasons, data quality, overrides, and outcome monitoring.

2.1.1. SHAP (SHapley Additive exPlanations) for Regulatory Governance The dual-regime architecture outputs a residual neural representation which, together with the Explicit Liquidity Features, enters the HLR and contributes to posterior PD. A SHAP pipeline may help explain a clearly identified model output. It cannot attribute the full customer premium because K_{RBCP} also contains cost, capital, liquidity, margin, and approved adjustments. SHAP values provide an additive attribution under a specified value function and background distribution. They describe model behaviour rather than causal blame, and their stability and approximation error must be tested. The evidence flow is:

1. **Inference time:** The service records the feature and model versions, calibrated output, uncertainty, policy result, and latency. A local explanation may be synchronous only when a tested method meets the service level.
2. **Evidence storage:** Explanation artefacts are stored with model, background distribution, feature, and decision versions. Earnings Velocity and Repayment Velocity remain explicit features and are not attributed to the GRU.
3. **Analysis and reporting:** Gold views may aggregate explanation diagnostics for authorised validation, risk, complaints, and audit users. Any supervisory access or submission follows an actual request and prescribed channel; a Power BI view is not presumed to be a CBK regulatory interface.

If a driver challenges K_{RBCP} , an authorised user can retrieve the customer price, component reconciliation, material reason codes, model limitations, and review route. SHAP explains model behaviour under assumptions; it does not by itself justify the entire price.

2.1.2. Equalized Odds and Continuous Fairness Monitoring The architecture monitors fairness as an ongoing risk rather than claiming to prevent it by formula.

- **Monitoring service:** A controlled analytics service calculates coverage, calibration, error, approval, price, limit, intervention, complaint, and outcome metrics on lawfully governed audit attributes. Protected status is not casually inferred from geography.
- **DIR alerts:** A value such as 0.80 is a diagnostic benchmark, not a universal Kenyan regulatory threshold.
- **Case management:** A material alert creates a case for review. Any corridor-level restriction requires evidence, authority, impact analysis, and a cure rule; an alert does not automatically penalise every driver in that place.

2.2. The Asynchronous Kappa-to-Delta Statutory Bridge

As established in **Part 2a**, the real-time underwriting decisions execute on a high-speed Kappa stream (via Redis and Flink) to bypass the latency of the data lake. However, Microsoft Dynamics 365 and Power BI require immutable, structured relational data to perform statutory accounting.

To resolve this latency mismatch, the architecture utilizes an **Asynchronous Kappa-to-Delta Bridge**. This strictly contrasts the real-time underwriting execution (the high-frequency Kappa path running on Redis/Flink) from the statutory compliance pipeline (the Delta/Medallion batch path required for historical audibility).

The integration layer consumes durable product and decision events. A loan issue, receipt, modification, write-off, policy event, asset purchase, waterfall allocation, or approved staging result can create an accounting or evidence record. A routine posterior update does not create a GL entry. Curated tables are derived data products, not “rigid accounting ledgers.”

This Gold Layer acts as the ultimate Statutory ERP Bridge, containing: - **The SPV Waterfall Table:** Pre-aggregated daily interest collections, micro-loan originations, and Class A/B allocations. - **The IFRS 9 Staging Table:** Driver-level probability of default, LGD, and SICR flags. - **The Pricing Engine Table:** The daily calculated Cumulative Compounded Rate (CCR) for KESONIA, merged with the driver’s “K” premium.

D365 receives validated journal interfaces and reference data; Power BI reads governed semantic models. Neither is required to use the same Gold table, and both reconcile to authoritative product ledgers and the GL.

3. The controlled reporting and evidence pipeline

Enterprise integration does not solve IFRS 9 or regulatory reporting by itself. The responsible institution first identifies each actual filing, partner report, investor report, frequency, schema, signatory, control, and delivery channel. The evidence pipeline then implements those requirements.

Automation prepares and validates reports, but accountable review remains where the filing or internal policy requires it. The five stages below are a control pattern, not a claim about a CBK submission interface.

3.1. Stage 1: Automated Validation Gates and Bounds Checking

Before a controlled report leaves the institution, validation checks the schema, source reconciliation, accounting period, currency, sign, range, completeness, duplicate status, exceptions, approvals, and report-specific rules. The ERP is one source and workflow endpoint, not an infallible sovereign.

The validation gates execute: 1. **Contract and legal bounds:** Apply counsel-approved interest, fee, notice, product, and in-duplum logic to the relevant balance and circumstance. 2. **Rate controls:** Reconcile KESONIA to the official publication and the contractual compounding convention; investigate material change without treating volatility as an error. 3. **Identity and KYC controls:** Validate required status through authorised sources and preserve error, unavailable, and review states. A hash match is not proof of identity or legal eligibility.

If a loan record fails a validation gate, it is instantly routed to an Exception Queue within D365. Crucially, the architecture utilizes non-blocking exception routing: the failure of Loan A does not prevent the transmission of compliant Loans B through Z. This prevents the “batch failure” pathologies common in legacy banking architectures.

3.2. Stage 2: Immutable WORM Repositories

Records that pass the validation gates are immediately serialized into an Immutable Write-Once-Read-Many (WORM) storage repository hosted on Azure Blob Storage.

Immutable or locked storage can support evidence retention when the applicable schedule and policy require it. This paper does not assert that CBK requires daily SHAP snapshots or a particular WORM service. Retention, legal hold, deletion rights, privacy, and key management must be reconciled.

Hash chaining can make later alteration detectable when anchors, keys, clocks, access logs, and verification are independently controlled. It does not provide absolute assurance or prevent authorised deletion under a misconfigured policy. Verification jobs and incident response are part of the design.

3.3. Stage 3: AES-256 GCM Encryption and Authenticated Telemetry

For an approved transmission, the security profile may use AES-GCM in accordance with NIST SP 800-38D [1], subject to nonce uniqueness, approved key management, rotation, access control, and protocol design.

AES-GCM provides authenticated encryption when implemented correctly. Security is not absolute, and performance must be benchmarked at the actual payload size and hardware. Transport, endpoint, credential, metadata, and operational risks remain.

3.4. Stage 4: Maker-Checker Segregation and X.509 PKI Digital Signatures

Regulatory submissions carry profound legal weight; submitting inaccurate capital or pricing data is a severe prudential violation. Therefore, the system enforces a strict “Maker-Checker” workflow.

In this architecture, the **Maker** is the automated RegTech Sweep pipeline (executing within Azure Data Factory and D365). It compiles, validates, and encrypts the data. However, the system cannot transmit the data autonomously.

The **Checker** is a designated human Chief Risk Officer (CRO) or Chief Compliance Officer. Through a secure Power Apps interface, the CRO is presented with a high-level statistical summary of the daily submission (e.g., Total Principal Outstanding, Average KESONIA Yield, Max/Min Risk Premium K , and Total Records Flagged in Exception Queue).

The approved signatory authorises the submission using the institution’s specified authentication and signing control. Hardware-backed X.509 signatures are one option where the receiving specification accepts them. A digital signature provides integrity and signer authentication under the trust framework; it does not make the figures accurate or create absolute non-repudiation.

3.5. Stage 5: Mutual TLS (mTLS) Transmission

The final stage of the RegTech sweep is the physical transmission of the signed, encrypted payload. The architecture utilizes a Mutual Transport Layer Security (mTLS) API tunnel to the central bank. Unlike standard TLS (where only the client verifies the server’s certificate),

mTLS requires both the central bank's server and the Mwendo Pamoja SPV's client server to cryptographically authenticate each other before any data is exchanged.

The five-stage pattern produces a reviewable chain from source event to approved report. Encryption, mutual authentication, signatures, reconciliation, model governance, fairness review, accounting controls, and legal validation operate as complementary layers. Transport controls protect the submission path and provenance, while the wider control framework establishes the quality and authorised meaning of the underlying information.

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Part 5

KESONIA Pricing and Capital Orchestration

Mwendo Pamoja White Paper

KESONIA Pricing and Capital Orchestration

1. From an overnight benchmark to a driver's daily obligation

1.1. Regulatory Paradigm Shift: The Central Bank of Kenya Risk-Based Pricing Framework

For a driver, a benchmark-rate reform becomes real only when it changes tomorrow's available cash. For a lender, the same movement changes asset yield and funding cost. For an SPV, timing and basis determine whether collections still cover notes. Part 5 follows KESONIA through those three perspectives and keeps customer price, funding rate, expected loss, capital, and investor return separate.

CBK's revised risk-based framework expresses the total lending rate as [14], [15]:

$$R_{\text{customer}} = \text{KESONIA} + K_{\text{RBCP}},$$

with fees and charges added for total cost of credit.

Where:

- **KESONIA** is CBK's transaction-based, volume-weighted overnight rate for unsecured KES interbank transactions [16]. It is a reference rate, not a universal risk-free rate.
- K_{RBCP} includes the institution's lending-related costs, shareholder return, and borrower risk profile under the framework. Expected loss is one input, not the whole premium.
- **Fees and charges** remain separate in total cost of credit and require their own basis and disclosure.

NOTE

Controlled notation

K_{RBCP} : Customer pricing premium under the CBK framework.

K_{cap} : Prudential capital requirement under the banking partner's applicable approach.

m_A : Class A SPV note margin.

m_B : Class B SPV note margin.

EL : Expected loss for the stated purpose and horizon.

CoC : Cost of capital.

$OpEx$: Allocated operating cost.

The bank may use only approved regulatory PDs for capital; the platform's underwriting PD does not automatically lower capital.

Mwendo Pamoja does not need to discredit every scorecard to make its case. Static and bureau-based models remain useful when they are well calibrated and governed. The platform's contribution is narrower and more valuable: it gives the lender a higher-frequency view of an income-generating vehicle, its operator, and the linked cash-flow obligations. That evidence can improve the expected-loss component of pricing and can show when an intervention is more rational than a punitive price increase.

Kinematics and latent representations are therefore not translated directly into price. The HLR produces calibrated PD and uncertainty; the pricing service reconciles expected loss with cost, capital, liquidity, and margin; and the licensed lender approves the customer rate. This division of labour turns model sophistication into an auditable decision instead of a mysterious surcharge.

1.2. The Bayesian engine as one governed pricing input

The redundancy-controlled HLR combines residual neural representation, Explicit Liquidity Features, hierarchical effects, and approved interactions to produce calibrated real-world PD and uncertainty. The output is one input to K_{RBCP} , not a direct price.

1.2.1. From Physical-Measure PD to Customer and Investor Pricing Part 2b outputs $PD_{i,\text{cal}}^{(p)}$ over a defined product horizon under the real-world, or physical, probability measure $\mathbb{P}$. This is the probability space used for underwriting, portfolio monitoring, stress evidence, intervention evaluation, and IFRS 9 inputs. It describes the frequency and uncertainty of default in the operating world represented by the data and approved scenarios.

A market-consistent valuation measure $\mathbb{Q}$ answers a different question: how a credit asset or contingent cash flow should be valued under stated market, calibration, discounting, and incompleteness assumptions. It is relevant only where the instrument, valuation purpose, and observable inputs support the transformation. Adding a premium to KESONIA does not convert a physical default probability into a risk-neutral probability.

The customer-pricing identity remains the formulation set out by CBK [14], [15]:

$$R_{\text{customer}} = \text{KESONIA} + K_{\text{RBCP}},$$

with applicable fees and charges added for total cost of credit. A transparent internal decomposition of the customer premium is:

$$K_{\text{RBCP},i}^{(p)} = EL_i^{(p)} + CoC_i^{(p)} + OpEx_i^{(p)} + Liq_i^{(p)} + M_i^{(p)} + Adj_i^{(p)},$$

where, using consistent annualised or period units,

$$EL_i^{(p)} = PD_{i,\text{cal}}^{(p)} \times LGD_i^{(p)} \times EAD_i^{(p)}.$$

A simplified one-period break-even credit component may be written

$$k_{\text{credit},i}^{(p)} \approx \frac{PD_{i,\text{cal}}^{(p)} LGD_i^{(p)}}{1 - PD_{i,\text{cal}}^{(p)}},$$

but it remains a physical-measure break-even approximation rather than the whole customer premium, a risk-neutral probability, or a guarantee of realised loss coverage.

Where:

- *CoC* is the allocated cost of capital under the lender's methodology.
- *OpEx* is allocated lending-related operating cost.
- *Liq* captures approved liquidity or funding cost not already in the benchmark.
- *M* is the shareholder-return or margin component.
- *Adj* contains documented product or customer adjustments, subject to policy, floors, caps, competition, and applicable rules.

The audit record reconciles each component to source, method, approval, effective date, and customer reason. A model explanation can show how risk evidence affected *EL*; it cannot justify cost allocations or shareholder margin on its own. The licensed lender owns the final customer price.

Four further quantities remain separate. The **asset yield** is the contractual return actually earned by the SPV on purchased receivables. The **SPV note margins** m_A and m_B are investor funding terms. The **FTP charge** is an internal matched-funding attribution used by a bank, or an analogous analytical allocation used by the SPV. The **investor hurdle or valuation discount rate** reflects the required return for the instrument and scenario. None of these is renamed K_{RBCP} , and regulatory capital remains K_{cap} .

Appendix A provides the abridged structural and equivalent-measure bridge for readers who need a market-consistent valuation overlay. The underwriting model itself remains calibrated under $\mathbb{P}$.

1.2.2. Coping with KESONIA Volatility: Forward-Looking Adjustments KESONIA is a backward-looking rate. However, gig-economy drivers are intensely sensitive to forward-looking liquidity shocks. The architecture must cope with two distinct directional risks stemming from KESONIA volatility:

1. **Upward-rate shock:** CBR and KESONIA can move differently and with timing differences. A permitted customer reset may reduce affordability. A funded cap, subsidy, or modification can be tested, but the cost falls to Class C only if the contract and model allocate it there.
2. **Downward and basis risk:** If assets and liabilities reset on different indices, lags, floors, or frequencies, NIM can move in either direction. The canonical Class A note is also KESONIA-linked, reducing but not eliminating basis and repricing risk.

The Transformer can contribute a macro context signal, but it is not an interest-rate oracle or a hedge. Treasury scenarios use observed curves, contractual repricing, and shocks. New-offer pricing may change through the lender's policy; the system may not raise customer margin automatically to protect the SPV.

1.3. The KESONIA Compounding Engine: Mathematics and SQL Implementation

Where the contract elects compounded KESONIA in arrears, daily observations are compounded over the accrual period. Other products may use a different lawful convention. Realised observations reduce forecast dependence but do not eliminate source, calendar, correction, implementation, or conduct risk.

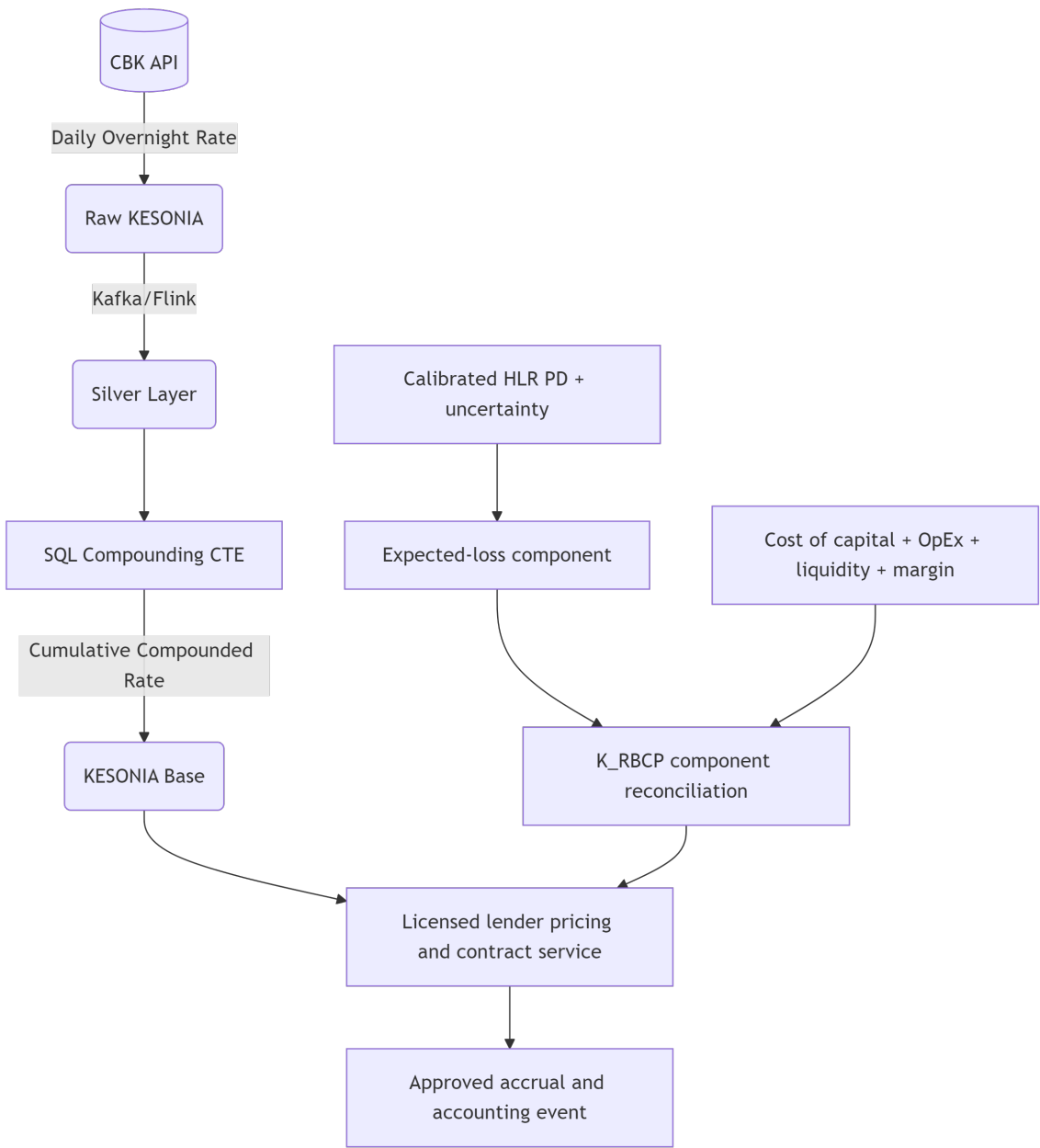
1.3.1. The Cumulative Compounded Rate (CCR) Formula The mathematical formulation for the Cumulative Compounded Rate over an accrual period is derived from first principles:

$$CCR = \left[\prod_{i=1}^{d_b} \left(1 + \frac{KESONIA_i \cdot n_i}{365} \right) - 1 \right] \times \frac{365}{d}$$

Where: * $KESONIA_i$: The overnight rate published on business day i . * n_i : The number of calendar days for which rate i applies. * d_b : The number of business-day observations in the accrual period. * d : The total calendar days in the interest period.

The multiplicative structure of the product operator ($\prod$) reflects the daily reinvestment of accrued interest: each observation compounds on the accumulated interest of all preceding days.

Figure 1: From benchmark and risk evidence to an approved customer price



1.3.2. Calendar Modeling and Observation Lookbacks Business Days versus Calendar Days: The CBK publishes KESONIA on business days. Loans, however, may accrue interest across calendar days. The CCR formula accommodates this through the weighting coefficient n_i . For example, a rate published on a Friday may receive $n_i = 3$ where the contract applies that observation across Friday, Saturday, and Sunday. If a holiday falls on Wednesday, Tuesday's rate may receive $n_i = 2$. This is modelled in the `holiday_calendar` schema so that the calculation applies the selected convention without inventing interpolated observations.

Observation convention: CBK explains publication and non-business-day treatment but does not impose a universal five-business-day lookback for every loan [16]. The definitive contract must choose lookback, observation shift, lockout, payment delay, or another convention and specify business calendar, corrections, floor, fallback, and day count. The calculation engine implements that election and tests it against independently calculated examples.

1.3.3. The 6-Layer SQL Architecture (Mapped to the Medallion Data Lake) To process millions of gig-worker loan contracts daily without human intervention, the architecture executes a 6-layer SQL pipeline directly mirroring the Medallion Data Lake:

1. **Data Ingestion (Bronze Layer):** Retrieval, timestamp validation, and immutable storage of raw CBK publications (`kesonia_raw_feed`).
2. **Data Staging (Silver Layer):** Merging raw observations with the `holiday_calendar` to calculate the n_i weights.
3. **Validation (Silver Layer):** Duplicate detection and business-calendar consistency checks.
4. **Interest Computation (Gold Layer):** SQL-based compounding using Common Table Expressions.
5. **Product-ledger event:** Sending a versioned calculation result to the authoritative lending or SPV subledger after validation and approval.
6. **General-ledger and reporting integration:** Posting approved accounting events to D365 and reconciling calculated balances in the reporting layer.

1.3.4. Production SQL Compounding Logic (Log-Sum Transformation) At portfolio scale, the logarithm-exponential identity converts the multiplicative product into an additive aggregation within $\text{EXP}(\text{SUM}(\text{LN}(\dots)))$. This is a practical implementation pattern,

but production code must still test precision, rate units, negative-domain constraints, missing observations, corrections, and database-specific behaviour.

*-- Conceptual SQL. contract_accrual_days is produced from the executed
-- convention and maps each accrual day to the permitted KESONIA observation.*

```
WITH AccrualDaily AS (
    SELECT
        a.contract_id,
        a.accrual_period_id,
        a.accrual_date,
        a.rate_observation_date,
        r.rate_decimal,
        a.day_weight,
        a.day_count_basis
    FROM Silver.contract_accrual_days a
    JOIN Silver.kesonia_publications r
        ON r.publication_date = a.rate_observation_date
        AND r.approved_version = a.rate_version
    WHERE a.validation_status = 'APPROVED'
        AND r.validation_status = 'APPROVED'
        AND 1.0 + (r.rate_decimal * a.day_weight / a.day_count_basis) > 0
),
CompoundedRates AS (
    SELECT
        contract_id,
        accrual_period_id,
        EXP(SUM(LN(
            1.0 + (rate_decimal * day_weight / day_count_basis)
        ))) - 1.0 AS compounded_factor,
        SUM(day_weight) AS accrual_days,
        MAX(day_count_basis) AS day_count_basis
    FROM AccrualDaily
    GROUP BY contract_id, accrual_period_id
)
SELECT
    contract_id,
```

```

accrual_period_id,
compounded_factor,
compounded_factor * day_count_basis / NULLIF(accrual_days, 0)
    AS annualised_compounded_rate
FROM CompoundedRates;

```

The calculation service emits the factor and annualised rate with lineage and validation status. The product ledger then applies the contract’s treatment of premium, floor, cap, rounding, and day count to produce an authorised accrual event; D365 receives the resulting accounting entry. A contractual premium may accrue on a simple, compounded, or other specified basis, so it is not blindly added to an already annualised number. The customer premium is K_{RBCP} , not an unqualified generic “K.”

2. SPV Financial Modeling and Capital Orchestration (Basel IV, IFRS 9, IFRS 17)

2.1. From Borrower Posterior to the SPV Cash-Flow Distribution

The SPV is a ring-fenced financing vehicle proposed to purchase eligible KES receivables. Its total capitalisation is USD 9 million equivalent, not USD 9 million of senior debt. Customer pricing, note pricing, and the waterfall remain distinct.

The SPV does not purchase a PD. It purchases an eligible receivable whose contractual cash flow can be accelerated, delayed, cured, modified, prepaid, defaulted, refunded, recovered, or written off. The bridge from underwriting to project finance therefore preserves the whole path from the driver-product-risk episode to monthly cash.

For each posterior and economic scenario draw, the engine:

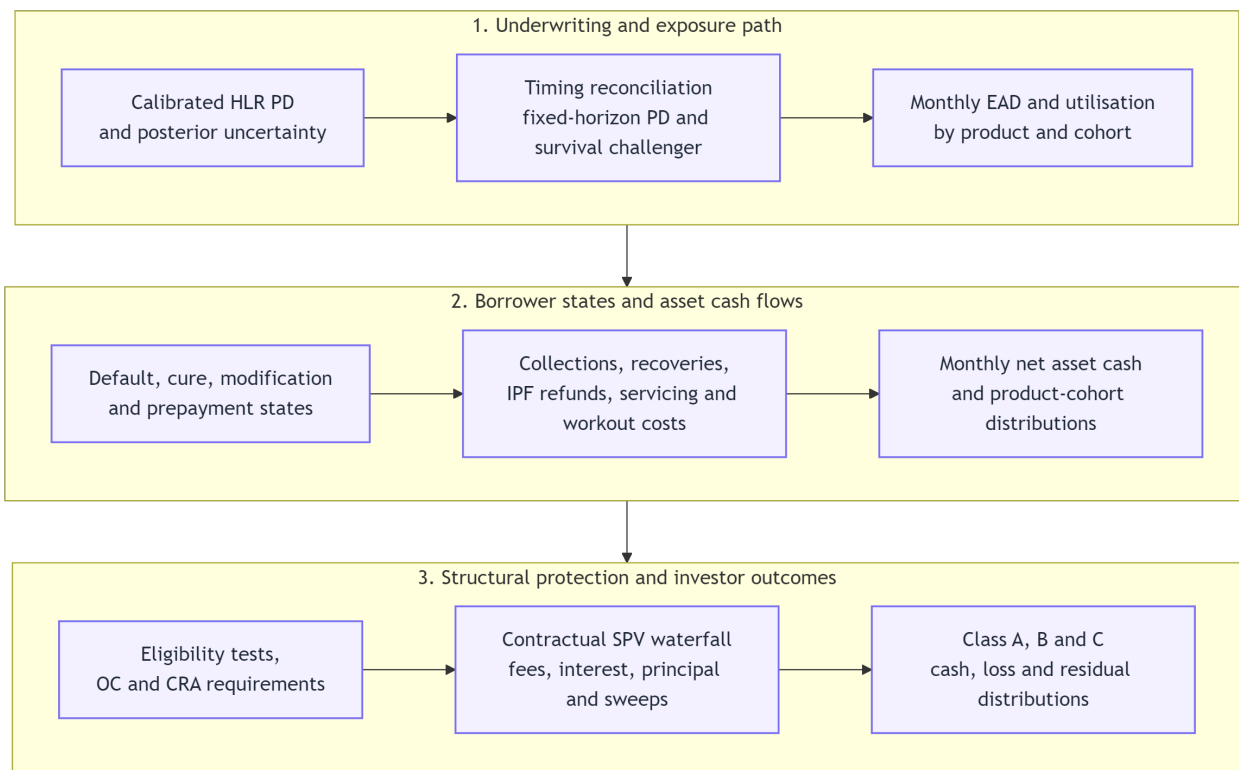
1. starts with the product-specific fixed-horizon HLR probability and, where approved, a reconciled marginal survival path;
2. projects scheduled balance, undrawn commitment, amortisation, prepayment, and stressed revolving drawdown to obtain EAD by month;
3. simulates or scenario-sets default, cure, modification, and closure transitions;
4. projects ordinary collections, recovery amount, recovery expense, and recovery lag;
5. models IPF cancellation and an eligible net refund separately from ordinary recovery and from insurer-owned premium cash;

6. aggregates principal, interest, fee, recovery, refund, and cost cash flows by product cohort; and
7. passes the cohort cash through eligibility, OC, CRA, fees, debt service, triggers, and the contractual waterfall.

For draw s , monthly net asset cash is

$$\begin{aligned}
 NCF_m^{(s)} = & \text{Collections}_m^{(s)} + \text{Recoveries}_m^{(s)} \\
 & + \text{IPFRefunds}_m^{(s)} - \text{Purchases}_m^{(s)} \\
 & - \text{Servicing}_m^{(s)} - \text{WorkoutCosts}_m^{(s)}.
 \end{aligned}$$

The waterfall converts $NCF_m^{(s)}$ into note interest, principal, reserve cure, sweeps, and residual distributions. Loss is measured at several destinations: borrower-level expected loss for pricing, probability-weighted discounted cash shortfall for IFRS 9, one-year loss for capital planning, ultimate loss for full cohort economics, and tranche loss after structural protection. These quantities share source cash flows but retain separate purpose, horizon, measure, discounting, and owner.

Figure 2A: From Borrower Posterior to SPV Cash-Flow Distribution

Portfolio dependence is applied after observed common factors and hierarchical effects have been represented. The simulation distinguishes parameter uncertainty, idiosyncratic outcome risk, systematic scenario risk, residual dependence, and model-form uncertainty. A common fuel or platform shock is not independently loaded into PD, LGD, recovery delay, and the copula without a documented mapping.

The lender pack reports both **one-year loss**, which supports annual capital and liquidity decisions, and **ultimate loss**, which follows each purchased cohort through final recovery. For tranche k with attachment A_k and exhaustion E_k , scenario loss allocation is

$$L_k^{(s)} = \min \left\{ \max \left(L_{\text{pool}}^{(s)} - A_k, 0 \right), E_k - A_k \right\}.$$

Reverse stress solves for combinations of default timing, drawdown, recovery delay, refund haircut, collection interruption, basis shock, and concentration that first exhaust Class C, then impair Class B, breach the CRA or OC tests, and finally expose Class A principal. The result is a decision surface for eligibility, reserve, advance rate, concentration, and intervention funding rather than a single severe-case percentage.

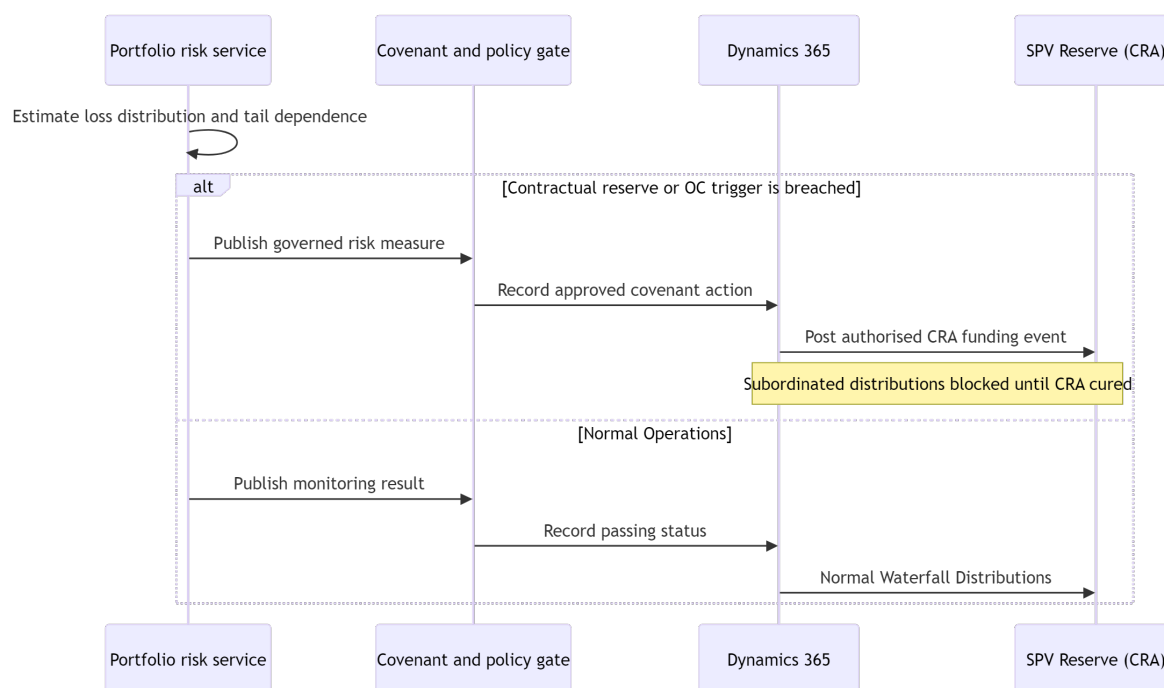
2.2. Orchestrating Capital Under KESONIA Volatility

2.2.1. Net Interest Margin (NIM) Compression Risk vs. Equity Absorption The canonical asset side is KES customer pricing, while Class A is compounded KESONIA plus m_A . Class B has a negotiated KES return. Differences in reset frequency, floors, product mix, collections, defaults, fees, and cash drag still create NIM risk:

1. **Upward-rate shock:** Model contractual asset and liability resets, affordability, support costs, defaults, and Class C availability. Senior protection is calculated, not presumed.
2. **Downward and basis shock:** Model the actual floors, lags, fixed-rate products, Class B cost, reserve income, and asset-liability repricing.

The treasury response: Scenario analysis, pricing governance, product mix, cash buffers, note terms, and, where economical, actual hedging manage basis risk. A model forecast or customer repricing rule is not an endogenous hedge.

2.2.2. Dynamic Cash Reserve Accounts (CRA) The CRA has a contractual floor, proposed as three months of defined debt service. A negotiated risk-based overlay may increase the requirement when portfolio stress rises, but no model has unilateral control of the account.

Figure 2: Dynamic SPV Capital Defense

The calculation agent determines the amount under the documents; the waterfall funds or cures it; the account bank holds the cash; and D365 records approved events.

A Clayton copula can be one candidate for lower-tail dependence, such as simultaneous distress across microloans, IPF, and revolving lines after a fuel shock. The selected family, parameter, and portfolio loss measure must survive back-testing and comparison with alternative copulas. The result becomes governed evidence for a contractual rule, not a direct command from a data lake. Software does not establish bankruptcy remoteness or guarantee principal protection.

2.2.3. Project Finance Structure & Capital Stack The primary case is a **USD 9,000,000 equivalent SPV funded and serviced in KES**, subject to true-sale, security, account-control, servicing, tax, and insolvency analysis. A 75% total-capital advance-rate convention implies USD 12 million equivalent of eligible receivables at closing, while OC is measured separately against Class A plus Class B.

Table 1: SPV Capital Stack and Target Yields

Tranche	Size (USD)	Share	Pricing basis	Priority and protection
Class A	\$6,750,000	75%	Compounded KESONIA + m_A	Senior priority; OC and CRA protection
Class B	\$1,350,000	15%	20.00% illustrative	Subordinate to Class A
Class C	\$900,000	10%	Residual return	First-loss and residual position
Total SPV	\$9,000,000	100%	Formula-driven hurdle	Subject to executed documents

Class A's 1.25x minimum OC and three-month CRA are proposed contractual protections rather than unconditional guarantees. Class C is held by the Insurtech in the planning case and absorbs losses and margin compression before the debt classes to the extent its funded capital remains available. Tax shields, withholding, issue costs, and deductibility remain outside the table until entity-level advice confirms their treatment.

*Separately from the SPV waterfall, the planning case includes a **USD 1,000,000 equivalent HoldCo facility**, creating a USD 10 million consolidated funding narrative.*

2.2.3.1. Weighted funding hurdle For the SPV cash-flow model, the useful quantity is a formula-driven weighted cash cost by class, kept distinct from a corporate WACC:

$$H_t = w_A R_{A,t} + w_B R_{B,t} + w_C R_{C,t}^{\text{target}}, \quad w_A + w_B + w_C = 1.$$

With the canonical stack, $w_A = 0.75$, $w_B = 0.15$, and $w_C = 0.10$. $R_{A,t}$ is compounded KESONIA plus m_A ; R_B is the negotiated Class B rate; and R_C^{target} is a scenario return, not a contractual expense. Taxes, deductibility, withholding, fees, idle cash, reserves, and issue costs enter separately after tax advice. The former fixed 12.06% result combined mezzanine with equity, assumed a live KESONIA rate, and applied a tax shield without entity-level support; it is therefore retained only in the backup history.

The underwriting hurdle for assets is stronger than H_t :

$$Y_{\text{asset},t}^{\min} = H_t + EL_t + OpEx_t + Fees_t + CashDrag_t + Buffer_t.$$

2.2.4. Macroeconomic Stress Testing and Asset Resilience The minimum OC covenant is 1.25x. The initial illustrative ratio of 1.4815x follows mechanically from USD 12 million equivalent of eligible receivables divided by USD 8.1 million equivalent of Class A plus Class B, not from Bayesian underwriting. Later ratios must be produced by the monthly cohort and waterfall model. The table retains scenario targets pending that calculation:

Table 2: Macroeconomic Stress Scenarios & Expected Credit Loss

Scenario	Illustrative ECL input	OC treatment	Covenant consequence
Base Case	4.5%	Initial 1.4815x; then model output	Test against 1.25x
Mild Shock	9.1%	Monthly cohort output	Calculate waterfall and cure
Severe Contagion	22.0%	Monthly cohort output	Apply trigger formula

The scenario percentages are model inputs, not portfolio forecasts. Early amortisation begins only when the executed covenant formula is breached. The cohort model must calculate collections, defaults, recoveries, reserve movements, and debt balances before assigning the later-period OC ratio or investor loss.

2.3. Regulatory Capital Synergies (Basel IV and IFRS 17)

The platform can provide governed data, models, and reports to support partner analysis. It does not sell guaranteed regulatory-capital optimisation.

2.3.1. Basel IV Advanced IRB and Output Floors The bank classifies exposures and determines the applicable standardised or authorised IRB approach. The platform's product label does not settle prudential classification.

Mwendo Pamoja can provide the bank with versioned posterior PDs, uncertainty, loss estimates, data lineage, calibration results, and monitoring reports. Explanation tools such as SHAP can assist investigation, but they do not prove stability, absence of bias, or regulatory

acceptability. The partner bank's independent validation function determines whether any model can be used in a prudential process. SR 11-7 is cited as comparative model-risk guidance rather than Kenyan law.

Intervention evidence can inform internal stress and model validation. It does not automatically justify a lower prescribed correlation or capital requirement.

2.3.2. IFRS 17 and Onerous Contract Testing For a licensed insurer issuing the underlying commercial motor policies, IFRS 17 requires groups of insurance contracts to be assessed and accounted for under the applicable measurement model, including recognition of losses for onerous groups where the standard's conditions are met.

The insurer's actuarial process may use validated telematics and exposure evidence, but credit PD and a latent fatigue representation do not directly predict an onerous group. The insurer owns grouping, fulfilment cash flows, risk adjustment, PAA eligibility, loss-component assessment, approval, and journal instruction. D365 records the approved result.

(see **Figure 1** and **Figure 2** in Sections 1.2 and 2.2.2 respectively).

3. Enterprise Accounting Mechanics: IFRS 9 EIR and Contract Modifications

KESONIA-linked contracts require a clear distinction among scheduled benchmark accrual, a permitted contractual reset, a new-offer price, and a negotiated modification. Model movement does not automatically change an existing customer's K_{RBCP} .

Daily accrual under an existing floating-rate formula is not the same as a daily contract modification. The product ledger calculates contractual cash flows and the accounting owner applies IFRS 9. The IBOR Phase 2 amendments are not treated as a permanent general exemption for every KESONIA reset [13].

3.1. Contractual floating-rate accrual and benchmark transition

An instrument written from inception with a floating benchmark already contains contractual variability. Ordinary benchmark resets under those terms are not automatically treated as renegotiated modifications. A change outside the original terms may require modification or derecognition analysis, and a non-derecognised modification may require recalculation of the gross carrying amount using the original effective interest rate, or the original credit-adjusted effective interest rate where applicable. The accounting owner determines the treatment from

the instrument and facts, rather than treating every daily KESONIA movement as a new modification.

For a new contract written directly on KESONIA, ordinary movements of the chosen floating benchmark follow the contractual EIR mechanics. If an existing instrument transitions from another benchmark, the reporting entity assesses whether the specific IBOR-reform relief applies and whether changes are economically equivalent. The conclusion is instrument-specific.

The calculation service emits the rate source, observation period, compounded factor, contractual spread, effective date, correction status, and instrument identifier. An accounting rules engine classifies the event as scheduled accrual, reset, transition, or modification and routes exceptions for approval. A reason-code tag cannot bypass accounting analysis.

3.1.1. Illustrative journal pattern for an authorised KESONIA accrual After the subledger has validated the contractual accrual and the accounting rules engine has approved its classification, D365 may receive an aggregated entry such as:

Debit: Loan Receivables - Accrued Interest (Balance Sheet) **Credit:** Interest Income - KESONIA Component (P&L) *(Amount: calculated under the instrument's contractual accrual method and reconciled to the subledger. This illustrative entry does not determine impairment, modification, fair-value, tax, or consolidation treatment.)*

3.2. Substantial vs. Non-Substantial Modification Accounting for K

The contractual benchmark and K_{RBCP} each follow the instrument terms. A credit-driven change outside those terms is a modification proposal requiring customer, legal, accounting, and policy analysis.

The HLR may show higher risk. Earnings Velocity belongs to the Explicit Liquidity Feature Path, not the GRU. For an existing loan, that signal can change monitoring, limit, support, staging evidence, or a future offer. It does not silently raise the contracted margin.

IFRS 9 does not establish a universal 10% bright-line test for financial-asset modifications equivalent to the commonly used financial-liability test. The asset holder applies qualitative and quantitative derecognition and modification analysis under its approved accounting policy.

1. **Modification without derecognition:** Recalculate the gross carrying amount using the original EIR as required, recognise the applicable modification result, and reassess credit risk and ECL.

2. **Derecognition:** Apply the entity's criteria to whether contractual rights expire or the terms are substantially different, then recognise the new asset and related result under the applicable requirements.

The integration preserves the evidence needed for audit. Compliance and securitisability remain conclusions of the responsible accountants, auditors, counsel, and investors.

4. Asset-Liability Management (ALM) and Matched-Maturity FTP

The financing structure contains two connected treasury problems. The partner bank owns prudential balance-sheet risk and internal FTP under its applicable framework [12]. The SPV must separately prove that receivable cash flows, reserves, expenses, and recoveries can meet note obligations through the waterfall. Mwendo Pamoja supplies reconciled contractual and behavioural cash flows to both processes without treating underwriting software as a treasury system.

4.1. SPV ALM and Matched-Funding Attribution

The SPV runs ALM as a cash-flow discipline. Its asset file contains receivable-level or cohort-level schedules for contractual principal, interest, fees, repricing, prepayment, default, recovery, cancellation refund, and collection timing. Its liability file contains Class A and Class B interest and principal, Class C support, CRA movements, servicing and trustee fees, taxes where applicable, hedge cash flows, and every priority-of-payment rule. Currency, reset convention, day count, floor, cap, behavioural tenor, and source date remain explicit.

For each time bucket b , the first view is the net liquidity gap:

$$Gap_b = \sum_{a \in b} CF_a^{\text{in}} - \sum_{l \in b} CF_l^{\text{out}}, \quad CumGap_B = \sum_{b \leq B} Gap_b.$$

The SPV calculation agent produces the ladder daily for near-term cash control and monthly for the lender model. A negative bucket is not automatically a default: opening cash, the CRA, permitted liquidity support, later collections, and contractual cure periods may cover it. The analysis must nevertheless identify when a temporary mismatch becomes a reserve draw, purchase stop, cash sweep, or payment shortfall.

The SPV then assigns a matched-funding charge to each product cohort. A 14-day microloan, a monthly IPF instalment stream, and a revolving line with stress-driven behavioural extension

should not receive the same funding charge merely because all are quoted over KESONIA. For cohort c , a transparent attribution is:

$$FTP_{c,t}^{\text{SPV}} = r_{\text{match}}(d_c, q_c, \rho_c, o_c) + Liq_{c,t} + Opt_{c,t} + Basis_{c,t},$$

where d_c is behavioural duration, q_c is repricing frequency, ρ_c is recovery and collection timing, and o_c represents prepayment, drawdown, cancellation, or other embedded options. The matched rate is anchored to the executed funding structure rather than an invented universal spread.

Economic spread is then measured after the costs that the asset must actually support:

$$NIM_{c,t}^{\text{economic}} = Y_{c,t}^{\text{asset}} - FTP_{c,t}^{\text{SPV}} - EL_{c,t} - OpEx_{c,t} - Fees_{c,t} - CashDrag_{c,t}.$$

The benchmark cancels only when the asset and funding conventions match exactly. Residual margin reflects product design, servicing, selection, intervention, recoveries, liquidity, capital support, and operations. It is not attributed solely to model sophistication.

The operating cycle is practical. The servicer and product subledgers deliver reconciled cash flows. The calculation agent builds daily and monthly asset and liability ladders. Treasury allocates matched funding by cohort and runs benchmark, basis, collection-lag, recovery-delay, behavioural-tenor, FX, and concentration shocks. The trustee and account bank confirm controlled cash and reserve balances. The risk or investment committee then adjusts purchase eligibility, product mix, advance rate, CRA target, hedging, or cash sweeps only through the authority provided by the transaction documents.

For the pilot, the validated cohort and waterfall workbook can perform this analysis. At larger scale, Oracle Financial Services Asset Liability Management can model instrument-level cash flows and interest-rate, liquidity, and foreign-currency risk, while Oracle Financial Services Funds Transfer Pricing can apply matched-maturity pricing methods to instrument or aggregated positions [18], [19]. Oracle also documents the integration of FTP rules into ALM forecasts and a shared cash-flow engine for repricing, option costs, scenario analysis, and audit output [20], [21].

OFSAA is therefore an optional institutional landing point rather than a prerequisite. It may run inside the financing bank using SPV cash-flow feeds, or within another authorised institutional environment if licensing, volume, data governance, and operating ownership justify it.

It should receive reconciled data from the product subledgers and SPV register rather than become the source of contractual truth. D365 continues to receive approved accounting events. If the pilot workbook remains more transparent and proportionate, OFSAA adoption can wait until scale and control needs justify the additional platform.

4.2. IRRBB, OFSAA, and a Repricing Use Case

IRRBB means **interest rate risk in the banking book**. It is the risk that changing interest rates alter a bank's earnings or the economic value of its banking-book positions. The partner bank owns the prudential IRRBB assessment. The SPV faces analogous repricing and basis risk, but the paper does not label the SPV itself a regulated banking book.

Consider a tangible Mwendo Pamoja case. The SPV holds one-month IPF and microloan instalments plus revolving balances whose behavioural maturity can extend during stress. Class A pays compounded KESONIA plus m_A . If KESONIA rises by 200 basis points immediately while some customer assets reset only at the next contractual date, Class A funding cost increases before all asset income adjusts. If stressed revolving borrowers repay more slowly at the same time, the asset duration extends precisely when funding is more expensive. The 12-month NII effect can be expressed as a loss-positive Earnings at Risk measure:

$$EaR_{12m} = NII_{base,12m} - NII_{shock,12m}.$$

The longer-horizon economic-value view compares the net present value of assets and liabilities under the shocked and base curves:

$$\Delta EVE = [PV(A_{shock}) - PV(L_{shock})] - [PV(A_{base}) - PV(L_{base})].$$

BCBS 368 describes NII and EVE perspectives and standardised shock scenarios, including parallel and non-parallel curve movements [12]. The bank maps those principles to the current applicable local framework and its approved methodology. OFSAA can ingest the instrument cash flows, behavioural assumptions, currencies, curves, and repricing rules, run the approved scenarios, and preserve cash-flow and interest-rate audit results [18], [21].

The result supports a concrete decision. A material negative EaR may justify a different asset-reset mix, cash buffer, note convention, hedge, or origination pace. A material EVE sensitivity may reveal that apparently short contractual assets have become longer behavioural exposures. The platform supplies accurate, versioned cash flows and risk states; the bank applies pru-

dential IRRBB and internal FTP, while the SPV applies the corresponding contractual ALM, reserve, and waterfall controls.

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Appendix A: Structural and Market-Consistent Valuation Bridge

The HLR estimates default risk under the physical measure $\mathbb{P}$. A separate market-consistent overlay may be useful when an investor values a transferable credit asset against observable instruments, a stated discounting framework, and explicit market-price-of-risk assumptions. The overlay changes the valuation measure; it does not rewrite the operating history used to train the underwriting model.

In the Merton structural model, terminal default occurs when firm asset value V_T falls below a debt boundary D [17]. A stylised asset process under $\mathbb{P}$ is:

$$dV_t = \mu V_t dt + \sigma V_t dW_t^{\mathbb{P}}.$$

Under an equivalent pricing measure $\mathbb{Q}$, and subject to the model's tradability, completeness, and calibration assumptions, the drift is replaced by the pricing rate r :

$$dV_t = r V_t dt + \sigma V_t dW_t^{\mathbb{Q}}.$$

With a constant market price of risk $\lambda = (\mu - r)/\sigma$, the change of measure can be represented as:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\lambda W_T^{\mathbb{P}} - \frac{1}{2}\lambda^2 T\right).$$

The corresponding stylised terminal pricing-measure default probability is:

$$P^{\mathbb{Q}}(\text{Default}) = \Phi\left(-\frac{\ln(V_0/D) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}\right).$$

For Mwendo Pamoja, these equations provide a valuation boundary rather than the operational underwriting specification. A driver and vehicle do not offer an observable traded firm-value process, and the market is incomplete. Any investor-pricing overlay must therefore identify the valued cash flow, calibration instruments, discount curve, liquidity premium, recovery assumption, and incompleteness treatment. The customer rate remains $K_{\text{ESONIA}} + K_{\text{RBCP}}$, the SPV note margin remains m_A or m_B , and the physical HLR probability remains the underwriting input.

Appendix B: Deterministic SPV Planning Model

The following deterministic planning model makes the core financing assumptions explicit and provides a reproducible bridge to the lender-grade cohort and waterfall workbook. It calculates the 36-month origination path, capital allocation, pricing components, and scenario inputs. The workbook extends this foundation with monthly defaults, recoveries, collections, reserves, debt schedules, taxes, covenants, and priority-of-payment mechanics.

```
import pandas as pd
import numpy as np

# Core Assumptions
facility_size_usd = 9000000
holdco_venture_debt_usd = 1000000 # Optional overlay outside the SPV
class_a_pct = 0.75
class_b_pct = 0.15
class_c_pct = 0.10
fx_rate = 130 # KES per USD
advance_rate = 0.75

# KES values
facility_size_kes = facility_size_usd * fx_rate
class_a_kes = facility_size_kes * class_a_pct
class_b_kes = facility_size_kes * class_b_pct
class_c_kes = facility_size_kes * class_c_pct
target_oc = 1.25

# Illustrative rates only; replace with sourced and approved inputs
kesonia = 0.105
expected_loss_component = 0.045
cost_of_capital_component = 0.020
allocated_opex_component = 0.020
shareholder_margin_component = 0.040
k_rhcp = (
    expected_loss_component
    + cost_of_capital_component
```

```

    + allocated_opex_component
    + shareholder_margin_component
)
total_lending_rate = kesonia + k_rhcp

class_a_rate = kesonia + 0.02
class_b_rate = 0.20
class_c_target_return = 0.25

# Weighted funding hurdle; tax effects require separate entity-level advice
weighted_funding_hurdle = (
    class_a_pct * class_a_rate
    + class_b_pct * class_b_rate
    + class_c_pct * class_c_target_return
)

# Fee and tax placeholders; validate legal basis, payer, base, rate and effective date
origination_fee_rate = 0.03
excise_duty_rate = 0.00

# Originations
start_drivers = 12000
growth = 0.04
ticket = 45000
spv_share = 0.85

months = range(1, 37)
originations_kes = []
active_drivers = start_drivers

for m in months:
    orig = active_drivers * ticket * spv_share
    originations_kes.append(orig)
    active_drivers *= (1 + growth)

```

```

df = pd.DataFrame({'Month': months, 'Originations_KES': originations_kes})
total_originations = sum(originations_kes)

# Illustrative fee bridge
gross_fee_revenue = total_originations * origination_fee_rate
kra_excise_duty_remittance = gross_fee_revenue * excise_duty_rate
net_fee_revenue = gross_fee_revenue - kra_excise_duty_remittance

# Stress scenarios and expected credit loss. OC must be calculated by the cohort cash-
# and waterfall model; it is deliberately not hard-coded in this illustrative appendix
scenarios = {
    'Base Case': {'ECL_Input': 0.045, 'OC_Output': None},
    'Mild Shock': {'ECL_Input': 0.091, 'OC_Output': None},
    'Severe Contagion': {'ECL_Input': 0.220, 'OC_Output': None}
}

summary = {
    'Class_A_USD': facility_size_usd * class_a_pct,
    'Class_B_USD': facility_size_usd * class_b_pct,
    'Class_C_USD': facility_size_usd * class_c_pct,
    'HoldCo_Debt_USD': holdco_venture_debt_usd,
    'KESONIA_Rate': kesonia,
    'Weighted_Funding_Hurdle': round(weighted_funding_hurdle, 4),
    'K_RBCP': k_rbcpr,
    'Total_Lending_Rate': total_lending_rate,
    'Gross_Origination_Fees_KES': gross_fee_revenue,
    'KRA_Excise_Duty_Remittance_KES': kra_excise_duty_remittance,
    'Net_Origination_Fees_KES': net_fee_revenue,
    'Total_Originations_36M_KES': total_originations
}

# Output Rich Tables
summary_df = pd.DataFrame(list(summary.items()), columns=['Metric', 'Value'])
print("\n=== SPV DETERMINISTIC PLANNING RESULTS ===")
print(summary_df.to_markdown(index=False))

```

```
scenarios_df = pd.DataFrame.from_dict(scenarios, orient='index').reset_index()
scenarios_df.rename(columns={'index': 'Scenario'}, inplace=True)
print("\n=== MACROECONOMIC STRESS SCENARIOS ===")
print(scenarios_df.to_markdown(index=False))
```

Part 6

Implementation Roadmap and Execution

Mwendo Pamoja White Paper

Implementation Roadmap and Execution

1. The first live decision

At 6:17 on a Tuesday morning, a driver accepts the first airport trip of the day. The vehicle is insured, yesterday's fuel purchase has already reduced the wallet balance, and an instalment is due before the evening peak. Nothing has yet gone wrong. The platform's value is not that it can label the driver risky faster than a conventional lender. Its value is that it can recognise a tightening operating margin early enough to preserve the driver's ability to work, while still protecting the people who funded the receivable and the insurer who carries the motor risk.

That apparently simple moment contains almost the whole implementation problem. The platform must know which driver, vehicle, policy, facility, wallet, platform account, and consent record belong together. It must distinguish a delayed platform settlement from a genuine income collapse. It must compute an interpretable liquidity signal without leaking future information into the past. It must combine that signal with a constrained neural representation without counting the same financial fact twice. It must send a recommendation to the correct decision owner. If an action is approved, it must preserve the original evidence, communicate fairly with the driver, change the right product ledger, post the right accounting event, and eventually reconcile the cash that reaches the SPV waterfall.

The roadmap therefore does not begin with an ERP installation or a neural network. It begins by proving that one decision can be made lawfully, intelligibly, and reversibly from source event to cash result. Enterprise software, structured finance, and automated reporting enter only when the underlying product truth can survive reconciliation.

IMPORTANT

This is a stage-gated implementation plan, not a promise that calendar time alone produces readiness. The proposed horizon is 18 months for a controlled pilot and transaction-ready evidence pack. A gate may take longer if data rights, legal opinions, model validation, accounting policy, or partner approvals are incomplete.

1.1. The implementation thesis

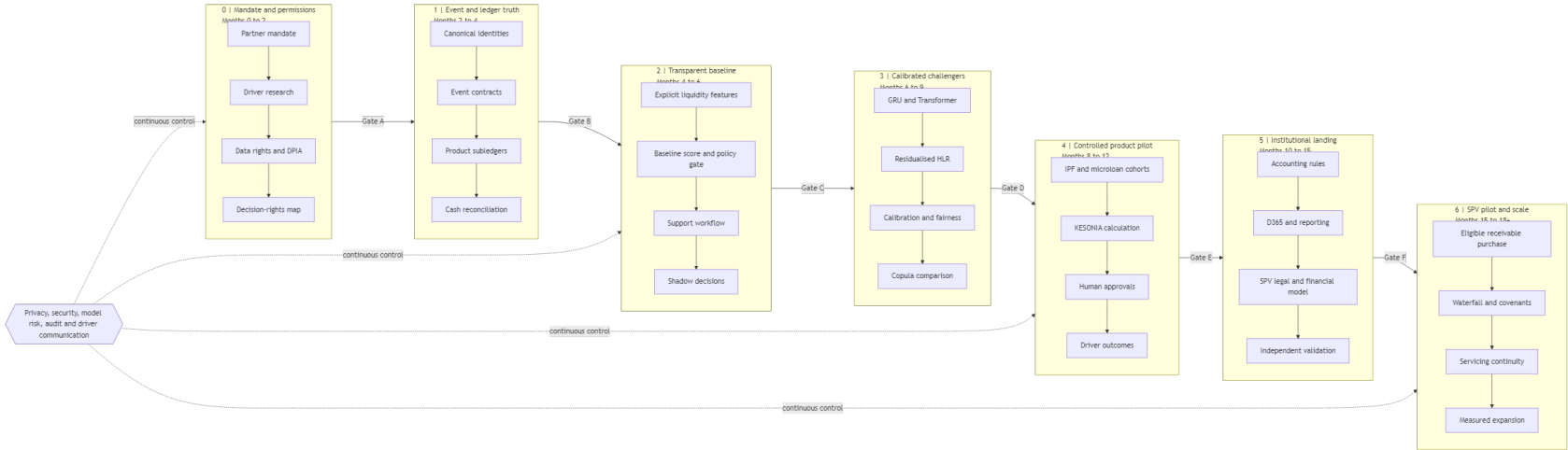
Seven principles govern the sequence.

1. **A decision has an owner.** The model produces evidence. A licensed lender, insurer, platform, servicer, trustee, or SPV calculation agent performs the action allocated to it.

2. **A ledger has an authority.** Product subledgers and the SPV receivables register hold contractual detail. Dynamics 365, if selected, is the general-ledger and enterprise-control destination, not a substitute for every product system.
3. **A feature has one engineered owner.** CFA, DLR, Earnings Velocity, Repayment Velocity, wallet volatility, reserve balance, time since depletion, utilisation, and approved interactions belong to the Explicit Liquidity Feature Path. Raw wallet-event sequences may inform a neural representation, but those engineered metrics are not duplicated there.
4. **A model must earn complexity.** The explicit-only baseline goes live first. Neural and copula components begin as challengers and are promoted only when out-of-time and out-of-group evidence shows incremental value, stability, and fairness.
5. **Accounting follows contracts and evidence.** An HLR output is not an IFRS 9 stage, an IFRS 17 loss component, a journal, or a prudential capital number. The responsible reporting entity applies its policy and approval.
6. **A covenant is not a dashboard threshold.** Advance rate, OC, CRA, concentration, and early-amortisation rules become operational only through executed transaction documents and a controlled calculation process.
7. **Scale follows a reconciled pilot.** The programme will not onboard hundreds of thousands of drivers merely because the streaming layer can process their events.

1.2. The stage-gated journey

Figure 1: An 18-month path from product truth to controlled scale



The phases overlap deliberately. Legal structuring does not wait until Month 10, and data engineering does not stop at Month 4. The labels show where each workstream becomes the critical path and where a formal decision should prevent premature escalation.

2. Phase 0: mandate, lived experience, and permissions

The first deliverable is not code. It is a signed operating hypothesis shared by the platform, licensed lender, insurer, mobility platform, sponsor, and proposed servicer. Each party should be able to explain the same customer journey in plain language: what the driver receives, who owns the receivable, who may change a limit, who may cancel or reinstate insurance, who holds cash, and who answers a complaint.

2.1. Begin with the driver and the operating asset

Interview drivers across vehicle type, platform, city, gender, tenure, financing history, and income volatility. The research should follow a full working week rather than a single survey. It should observe how fuel, commission, repairs, insurance, household needs, and debt compete for the same wallet. The purpose is not to turn personal hardship into another feature. It is to identify where the proposed intervention is useful, where it is intrusive, and where the product design itself creates avoidable fragility.

The research team converts those journeys into testable service blueprints. A premium holiday, for example, must specify who funds the temporary shortfall, whether cover continues, how the insurer approves it, what the driver is told, how it affects future instalments, and what happens if income does not recover. A micro-reward bridge must specify whether it is a grant, rebate, credit, or platform incentive. Ambiguity at this stage becomes customer harm and accounting error later.

2.2. Establish legal and data authority

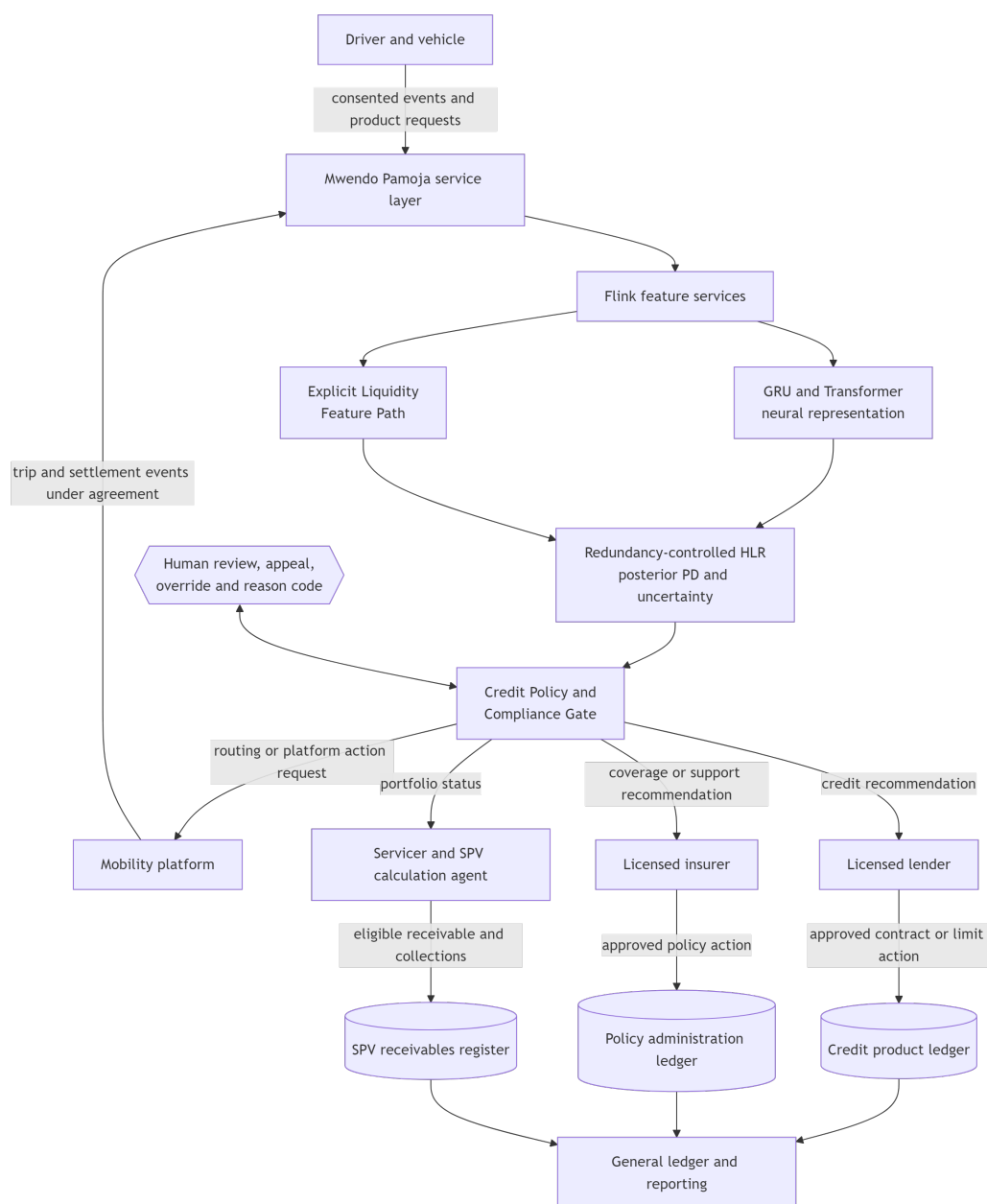
Create a source-to-use register for every proposed field. Each row records the source controller, collection purpose, lawful basis, notice language, retention period, approved users, model role, policy role, sharing path, security class, and deletion or anonymisation rule. Because continuous telematics and financial behaviour can create high risks to data-subject rights, the programme should complete a Data Protection Impact Assessment before pilot processing and update it when purpose, scale, or model use changes [3], [10].

Consent must not be treated as a universal cure. The analysis should identify where processing relies on contract, legal obligation, legitimate interest, consent, or another permitted ba-

sis, and whether the power relationship makes a purported choice meaningful. Driver-facing notices should explain material data uses and consequences without presenting a neural architecture diagram as an explanation.

2.3. Freeze the first decision-rights map

Figure 2: Evidence may be shared; authority is not



Gate A is passed only when the parties approve the service blueprint, entity and decision-rights matrix, data map, DPIA plan, complaints and appeal design, initial accounting position papers,

and a narrow pilot mandate. A cloud subscription or memorandum of understanding is not a substitute.

3. Phase 1: make events and money agree

The second phase constructs a small but trustworthy spine. The system should be able to trace one trip, one wallet settlement, one premium instalment, one loan repayment, and one reserve movement from source event to product balance and bank cash.

3.1. Canonical identity and bitemporal event contracts

Create immutable identifiers for the driver, vehicle, device, policy, credit agreement, wallet, platform, receivable, SPV purchase, and accounting event. Identity resolution should preserve effective-from and effective-to dates so that a vehicle change or policy replacement does not rewrite history.

Every material event requires an event identifier and type; source system and source record; event time, source-recorded time, ingestion time, and processing time; subject and contract identifiers; amount, currency, sign, unit, and measurement basis where relevant; schema version and quality status; correction, reversal, and predecessor identifiers; purpose and permitted-use tags; and cryptographic integrity metadata where appropriate.

Point-in-time joins must use only information available at the decision cutoff. If t_i^{event} is the economic event time, t_i^{known} is when the platform could have known it, and t_d is the decision time, a training feature is eligible only when

$$t_i^{\text{known}} \leq t_d.$$

This condition prevents a late-arriving repayment or corrected trip record from leaking future knowledge into a historical underwriting example.

3.2. Product subledgers before the enterprise GL

Build or nominate authoritative subledgers for IPF, microloans, revolving credit, wallet reserve pockets, insurance policies, and SPV receivables. These systems hold schedules, contractual rates, limits, drawdowns, arrears, cancellation state, allocations, and reversals. Dynamics 365 may later receive summarised and approved accounting events, but it should not be forced to impersonate every operational product engine.

For each subledger, daily reconciliation should satisfy

$$B_t^{\text{close}} = B_{t-1}^{\text{close}} + O_t + C_t + I_t + F_t - P_t - W_t - R_t,$$

where the terms are defined for that product and signs remain consistent. For a receivable, for example, O_t can be originations, I_t interest, F_t fees, P_t principal collections, W_t write-offs, and R_t reversals or recoveries as the accounting policy requires. The exact bridge matters more than the acronym.

Gate B requires reconciled sample journeys, versioned schemas, data-quality thresholds, correction procedures, lineage evidence, and proof that a product balance can be reproduced from its events.

The same gate now requires five connected credit-state ledgers: identity and exposure; obligations and contractual state; decisions and actions; outcomes and label maturity; and recoveries and cash flows. Each record carries event, availability, decision, label-maturity, accounting, and cash-realisation times. The team must reproduce both a matured fixed-horizon label and a right-censored at-risk episode from the same source history before advanced modelling begins.

4. Phase 2: launch the transparent baseline first

The first shadow underwriting service should use the Explicit Liquidity Feature Path and a deliberately simple baseline model. This gives operations a system they can understand while the programme learns which data are reliable and which interventions actually help.

4.1. Canonical feature ownership

Flink computes CFA, DLR, Earnings Velocity, Repayment Velocity, wallet volatility, reserve balance, time since depletion, utilisation, and the approved interactions. Definitions include window, denominator, missingness rule, winsorisation, currency treatment, event cutoff, refresh frequency, and owner. The online and training implementations must share the same specification and be tested for skew.

The Credit Policy and Compliance Gate remains downstream. It applies debt caps, affordability or sufficiency rules, uncertainty limits, product eligibility, insurance status, data-quality fallback, concentration controls, and intervention permissions. A feature is evidence; a policy rule is an authorised response to evidence.

4.2. A baseline that can lose gracefully

Begin with a regularised logistic or monotonic additive model using the explicit features, existing bureau information where lawfully available, and essential product metadata. Run it in shadow mode against the partner's current process. Record disagreement, not merely average accuracy.

Nonlinear explicit effects begin with governed P-splines: a moderately rich B-spline basis, a documented difference penalty, centred effects, validated support, and product deviations only where partial pooling and evidence justify them. The knot protocol, penalty order, smoothing-scale prior, boundary rule, and back-transformation are versioned. B-spline RW1 and AR1 formulations remain appendix challengers so that the team can compare shape and time assumptions without silently discarding the earlier research.

The baseline is acceptable only if it passes out-of-time and out-of-group validation, shows usable calibration by product and material subgroup, and supports operational reason codes. Useful diagnostics include the Brier score

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2,$$

and a calibration model

$$\text{logit Pr}(Y_i = 1) = a + b \text{logit}(p_i),$$

where ideal held-out calibration is approximately $a = 0$ and $b = 1$. These are diagnostics, not statutory thresholds.

Gate C requires stable feature computation, a documented model card, product-specific calibration, error analysis, adverse-impact investigation, manual-review workflows, fallback logic, and evidence that a support action does not silently become a credit sanction.

5. Phase 3: let complexity compete for its place

Only after the explicit baseline is stable should the programme introduce the GRU, Transformer, hierarchical Bayesian layer, and tail-dependence models. They begin in shadow mode as challengers.

5.1. The redundancy-controlled HLR

The canonical underwriting equation is

$$\text{logit}(p_i) = \alpha + u_{g(i)} + v_{s(i)} + \tau_{t(i)} + \mathbf{q}_i^\top \beta_z + \tilde{\mathbf{h}}_i^\top \beta_h + \gamma^\top \mathbf{r}_i + \delta^\top \mathbf{x}_i.$$

Here, $\mathbf{q}_i$ is a centred, scaled, and QR-orthogonalised spline basis for the Explicit Liquidity Features; $\mathbf{h}_i$ is a low-dimensional neural bottleneck; $\mathbf{r}_i$ contains a small, pre-registered set of economically meaningful interactions; and $\mathbf{x}_i$ contains essential contract metadata. The hierarchy terms u , v , and τ pool geography, platform or cohort, and time effects rather than creating thousands of unrelated local models.

To reduce duplicate signal, estimate the neural residual only inside each training fold:

$$\tilde{\mathbf{h}}_i = \mathbf{h}_i - \hat{\mathbb{E}}_{-k(i)} [\mathbf{h}_i \mid \mathbf{q}_i, \mathbf{x}_i].$$

The expectation is trained without fold $k(i)$ and applied to observations in that fold. This cross-fitted residualisation makes the neural block compete on information not already represented by explicit liquidity and essential metadata. It does not mathematically guarantee zero collinearity, so the programme also uses a narrow neural bottleneck and whitening within the training fold; separate regularising priors for explicit, neural, hierarchy, and interaction blocks; strong heredity for interactions; posterior-correlation and variance-inflation diagnostics; explicit-only, neural-only, and combined ablations; and out-of-time calibration and stability testing before promotion.

If the combined model does not improve decision-relevant held-out performance after complexity and operational cost, the explicit-only model remains champion. Innovation is preserved by testing it honestly, not by making deployment inevitable.

5.2. Offline inference and the timing challenger

Full posterior estimation remains offline. The reference implementation uses NUTS or HMC, while Pólya-Gamma augmentation is benchmarked as a blocked Gibbs strategy for the complete HLR. Conditional on the Pólya-Gamma variables and scale parameters, the global, spline, residual-neural, hierarchy, interaction, and selected time blocks receive sparse Gaussian updates. Partial pooling remains intact. The comparison reports posterior agreement, calibration, tail behaviour, group shrinkage, effective sample size per second, and operational reproducibility.

Production loads a signed posterior artifact containing the approved feature schema, basis and penalty, residualisation and whitening transforms, posterior representation, hierarchy mappings, calibration, support ranges, reason concepts, and fallback. It performs no MCMC during a credit request.

A hierarchical piecewise-exponential proportional-hazards model enters as a timing challenger. The risk-set builder uses entry, event, censoring, intervention, modification, closure, and time-valid covariate intervals. Its survival curve must reconcile to the HLR at shared horizons. A Gamma frailty or shared longitudinal-liquidity component remains an advanced challenger, used only when repeated matured episodes and measurement-error evidence support the additional structure.

5.3. Complete the loss chain before dependence

Before a copula is promoted, the programme builds separate EAD, cure, LGD, ordinary recovery, IPF-refund, recovery-cost, and recovery-delay components. Each component has its own target, horizon, source, validation, and owner. The cash-flow engine converts fixed-horizon or marginal default probabilities into monthly product cohorts, then one-year loss, ultimate loss, ECL cash shortfalls, economic-capital views, and SPV waterfall results.

Dependence is layered in order: observed common factors, hierarchical platform and geography effects, within-driver multi-product linkage, and only then residual copula dependence. This ordering reduces the chance that one tail parameter absorbs omitted factors and duplicated signal.

5.4. Copulas as portfolio challengers

Fit Clayton, survival Clayton, Gumbel, Gaussian, Student- t , and other defensible candidates where data support them. Compare tail fit, parameter stability, conditional loss estimates, and stress behaviour. The selected model informs portfolio scenarios and a negotiated reserve overlay; it does not automatically change a customer's price, the advance rate, or the CRA.

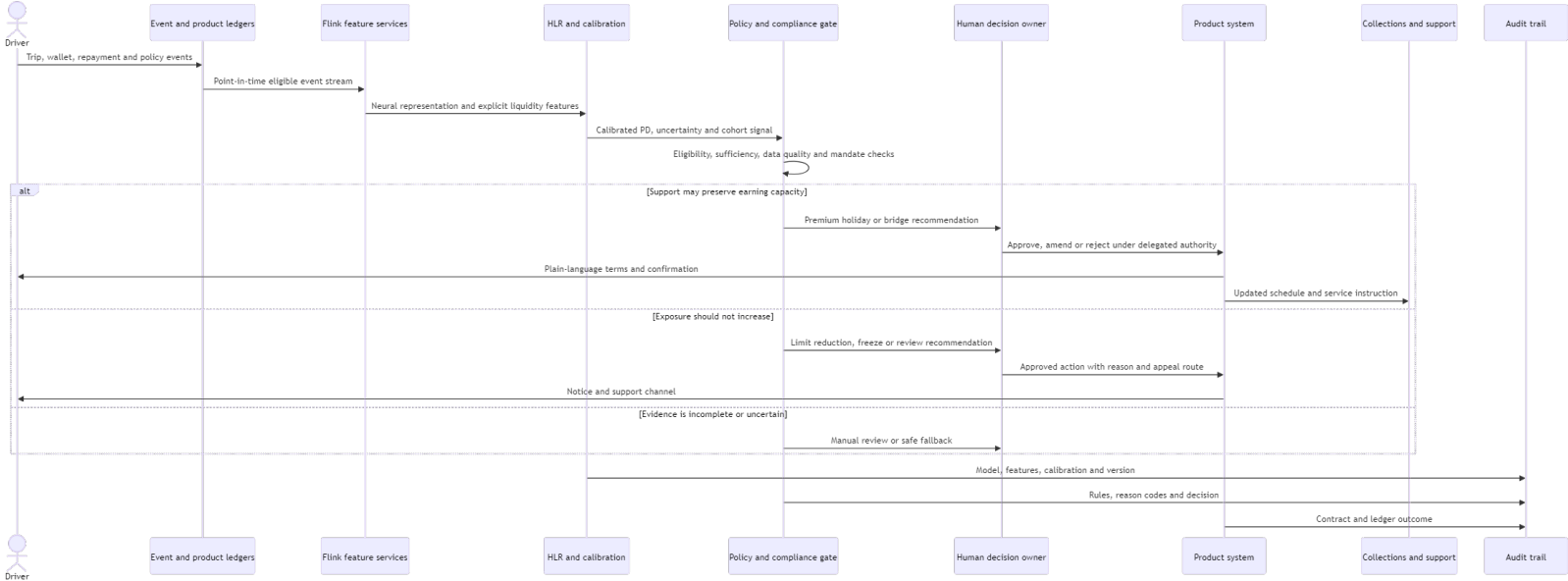
Gate D requires independent model review, reproducible training, model and feature registries, P-spline and residualisation validation, agreement testing between NUTS and any Pólya-Gamma implementation, censoring and risk-set review, HLR-survival horizon reconciliation, EAD and recovery validation, approval of the champion-challenger result, fairness analysis, data-drift response, uncertainty limits, and a documented decision on whether residual copula evidence is mature enough for transaction use.

6. Phase 4: a controlled product pilot

The live pilot should be small enough to supervise and broad enough to reveal operational failure. A staged cohort can begin with one or two mobility partners, defined geographies, limited product limits, and pre-agreed insurer and lender capacity. It should not begin with every product and every intervention.

6.1. The intervention loop

Figure 3: One risk signal, several accountable outcomes



Measure outcomes at driver, product, and portfolio level. Driver measures include uninterrupted insured days, days able to work, net cash sufficiency after essential commitments, complaints, overrides, and repeat distress. Product measures include arrears cure, roll rates, cancellation, utilisation, repayment, loss, recovery, and modification cost. Portfolio measures include concentration, liquidity, cash timing, and expected versus realised loss.

Every offer, restriction, referral, override, and intervention enters an action ledger with the pre-action score and uncertainty, eligible population, decision owner, customer response, executed terms, exposure change, cost bearer, outcome window, censoring, cure, default, recovery, complaint, and appeal. The evaluation design accounts for selective labels created by approvals, declines, freezes, and treatments. Where feasible, phased rollout, randomised encouragement, or another lawful comparison design estimates the effect of support rather than attributing every later cure to the model.

6.2. KESONIA without hidden repricing

For applicable new variable-rate customer contracts, the pricing bridge follows the CBK formulation [1]:

$$R_{\text{customer},i,t} = \text{KESONIA}_t + K_{\text{RBCP},i},$$

with fees and charges disclosed separately in total cost of credit. The calculation service stores publication source, observation period, business-day convention, compounded factor, correction status, contractual premium, floor, cap, and effective date. An HLR update does not silently change an existing contractual premium. New-offer pricing, contractual reset, and modification remain distinct.

The pilot independently recomputes sample accruals using CBK examples and reconciles them to the product ledger [1], [2]. A missing or corrected benchmark observation enters an exception queue under a documented fallback rather than being guessed.

Gate E requires demonstrated customer communication, safe manual operations, reconciled cash and accruals, complaint handling, intervention funding, performance and fairness evidence, incident drills, and a formal go or no-go decision by every licensed product owner.

7. Phase 5: institutional landing and transaction readiness

Enterprise integration begins when the product systems can already explain themselves. This avoids configuring an expensive ERP around assumptions that later change.

7.1. Accounting architecture

The lender or SPV asset holder owns IFRS 9 classification, effective-interest-rate treatment, modification and derecognition analysis, staging policy, and expected credit loss [4]. The insurer owns IFRS 17 grouping, measurement, fulfilment cash flows, risk adjustment, onerous-group analysis, and journal instruction [5]. The platform supplies governed evidence and approved events but does not collapse those standards into one model output.

D365 receives approved journal events with contract, product, legal entity, currency, accounting date, source ledger, model version where relevant, rule version, and reversal lineage. It does not accept an IFRS9_Stage_Override merely because middleware sent one. A stage recommendation enters an accounting control that preserves days-past-due backstops and other approved evidence.

For every posting batch b , the control objective is

$$\sum_{j \in b} \text{Debit}_j = \sum_{j \in b} \text{Credit}_j,$$

and the aggregate general-ledger movement must reconcile to the underlying product sub-ledger within an approved tolerance and timing window. Exceptions remain visible until resolved; they are not cleared merely to make a dashboard green.

7.2. The SPV is a legal and cash-flow project

The primary structure is USD 9 million equivalent of KES-funded SPV capital: USD 6.75 million equivalent Class A, USD 1.35 million Class B, and USD 0.90 million Class C. A separate USD 1 million HoldCo facility creates the optional USD 10 million consolidated funding view. HoldCo operating cash does not enter the SPV waterfall.

Before a receivable is purchased, counsel and transaction parties must settle true sale, eligibility, perfection, account control, servicing, commingling, set-off, data transfer, tax, insolvency, replacement servicing, and enforcement. The financial model then represents the documents rather than inventing them.

At closing, if the transaction adopts a 75 percent total-capital advance convention, eligible receivables are

$$ER_0 = \frac{9.0}{0.75} = 12.0 \quad \text{million USD equivalent.}$$

OC is separately measured against debt notes:

$$OC_t = \frac{ER_t}{A_t + B_t},$$

which gives an initial illustrative ratio of $12.0/8.1 = 1.4815$ before haircuts, timing, ineligibility, or other transaction adjustments. The covenant floor is proposed at 1.25 times. The CRA floor is proposed as three months of contractually defined debt service:

$$CRA_t^{\min} = 3 \times DS_t^{\text{defined}}.$$

The cohort model must calculate whether collections, recoveries, reserve cures, and subordination protect Class A under each scenario. “Senior principal protected” is an output conditional on assumptions, not a slogan.

7.3. Reporting adapters, not invented regulatory messages

Power BI may present management, lender, servicing, and audit views. Regulatory submissions must use formats actually prescribed by the relevant authority. ISO 20022 is a governed financial-message standard with a formal catalogue and community-specific usage [8]. It can be used for payment or partner-message adapters when an applicable scheme requires it. The programme will not invent tags, call them native ISO 20022, or claim that CBK accepts a continuous model-risk feed without a published specification and onboarding process.

Gate F requires signed accounting position papers, audited reconciliations, independent model validation, draft transaction documents, a validated 36-month cohort and waterfall model, servicing and business-continuity tests, investor diligence materials, and clear outstanding conditions precedent.

8. Phase 6: SPV pilot and measured scale

The first SPV purchase should use a limited eligibility box and a portfolio whose data history has already been observed during the product pilot. The opening month is treated as a controlled closing, not a marketing launch.

8.1. Daily, monthly, and event-driven controls

Daily controls include cash receipt matching, benchmark ingestion, failed payments, insurance status, data quality, product limits, ledger balance, and unresolved exceptions. Monthly controls include receivables roll-forward, waterfall, debt schedule, CRA, OC, concentration, ECL, collections vintage, and investor reporting. Event-driven controls cover servicer failure, data outage, policy cancellation, suspected fraud, benchmark unavailability, material model drift, cyber incident, concentration breach, and early amortisation.

The waterfall follows the executed priority of payments. At a minimum, the model distinguishes statutory and trustee costs, servicing, Class A interest and principal, Class B interest and principal, reserve cure, contractual sweeps, and Class C residual. Equity distributions stop whenever the documents require them to stop.

8.2. Scale is earned by stable cohorts

Expansion should be authorised in increments by platform, geography, product, and vehicle type. Each increment requires enough observation to assess data completeness, customer outcomes, loss emergence, recovery, concentration, and servicing capacity. If a new platform cannot provide settlement timing with the required quality, the answer may be a lower limit, a slower ramp, or no purchase, even if aggregate demand is attractive.

Gate G is not a final certification. It establishes a repeatable scale decision. The investment and risk committees receive a pack showing cohort economics, scenario survival, covenant headroom, model monitoring, fairness investigation, operational incidents, complaints, overrides, reconciliation breaks, and unresolved diligence. Approval states the permitted expansion and the conditions that would reverse it.

9. Workstreams, owners, and definitions of done

Workstream	Accountable owner	Pilot definition of done	Evidence retained
Driver proposition	Product committee with driver-research lead	Terms, support, appeal, and communications tested with target users	Research protocol, findings, approved service blueprint
Credit product	Licensed lender	Eligibility, pricing, limits, collections, modification, and complaints approved	Product paper, delegated authorities, sample decisions
Insurance product	Licensed insurer	Policy, IPF, cancellation, support, and claims interfaces approved	Policy wording, actuarial approval, interface tests
Data protection	Each controller, coordinated by DPOs	DPIA and source-to-use register approved; data-subject controls operational	DPIA, notices, agreements, retention tests
Data engineering	Platform technology owner	Point-in-time event spine and reconciled product feeds meet service levels	Schemas, lineage, skew tests, quality reports
Underwriting	Lender model owner with platform model team	Champion approved; challenger evidence documented; fallback tested	Model card, validation, calibration, fairness analysis
Policy gate	Product and compliance committees	Rules map to authority, contract, rationale, test, and reason code	Versioned rulebook and approval log
Accounting	Reporting entity finance owners	Product subledgers reconcile to approved GL postings	Position papers, journals, reconciliations
SPV	Sponsor, arranger, counsel, trustee, servicer	Legal structure and cash mechanics match the financial model	Opinions, documents, model, servicing plan

Workstream	Accountable owner	Pilot definition of done	Evidence retained
Security and resilience	Security and operations owners	Threat model, key management, recovery, incident, and vendor controls tested	Test results, runbooks, incident exercises
Independent assurance	Validation, audit, counsel, actuarial and tax reviewers	Material assumptions and controls reviewed by the correct discipline	Reports, findings, remediation and sign-offs

RACI labels alone are not enough. Each owner must have authority, competence, access to evidence, and a defined escalation route. A platform cannot assign an accounting conclusion to a data scientist or a credit decision to a dashboard.

10. Engineering, security, and operating standards

10.1. Service-level objectives and safe degradation

Set service levels by decision need, not by technological ambition. Fraud or draw-control signals may need seconds; an IFRS 9 ECL close may need hours with stronger reconciliation. Track event freshness, feature availability, model latency, product-ledger posting time, cash-match completion, recovery-point objective, recovery-time objective, exception age, and manual fallback capacity.

A degraded neural service should fall back to an approved explicit-only model or manual process. A degraded data feed should not convert missingness into evidence of concealment. A failed KESONIA feed should invoke the contractual fallback and exception procedure. Every fallback has an owner, start time, customer effect, and recovery test.

10.2. Security and privacy by design

Segment raw telematics, identity, wallet, model, product, and accounting zones. Use least privilege, managed identities, encryption in transit and at rest, secret rotation, monitored privileged access, tamper-evident audit records where justified, and tested restoration. AES-GCM is an authenticated-encryption mode with strict nonce requirements; naming it does not by itself create secure evidence [9].

Model registries should retain code version, configuration, training-data snapshot or reproducible reference, feature definitions, evaluation, approval, deployment, and rollback. Cryptographic hashes can demonstrate that an artefact has not changed since hashing, but they cannot prove that its source data were accurate or its use lawful.

10.3. Testing as a portfolio of evidence

Testing includes unit and property tests for formulas, calendars, signs, rounding, and boundaries; contract tests for source APIs and event schemas; point-in-time and online-offline feature consistency tests; model calibration, discrimination, uncertainty, stability, subgroup, and stress tests; policy-rule decision tables and counterfactual edge cases; ledger, bank, GL, waterfall, debt, reserve, and covenant reconciliations; security, privacy, penetration, backup, restoration, and incident exercises; user acceptance with drivers, operations, complaints, finance, risk, and partner teams; and independent reproduction of KESONIA accruals and SPV scenario outputs.

No single dashboard proves readiness. Readiness is the consistent intersection of legal permission, product clarity, data truth, calibrated prediction, governed decisions, reconciled accounting, resilient operations, and financeable cash flow.

11. The first eighteen months in human terms

By Month 2, the programme should know whose problem it is solving and what it has permission to observe. By Month 4, it should reproduce a small number of driver and product journeys from event to cash. By Month 6, it should have a transparent shadow decision that operations can challenge. By Month 9, the innovative model should have shown whether it adds signal beyond the explicit path without reintroducing the same variables through a neural side door. By Month 12, a controlled cohort should have experienced real support, real repayment, real exceptions, and real appeals. By Month 15, accountants, validators, counsel, insurers, lenders, and the SPV parties should be looking at the same reconciled evidence. By Month 18, the transaction should be capable of purchasing a limited pool and reporting what actually happened.

The endpoint is not a machine that predicts distress beautifully. It is a system in which an early sign of strain can lead to a proportionate intervention, the driver can understand the consequence, the product owner can defend the decision, the ledger can reproduce the balance, and an investor can follow collections through the waterfall. That is how the microeconomic

story, the insurance proposition, the embedded-finance product, the computation, and the project-finance structure become one operating institution.

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Reader Guide

Interdisciplinary Field Map and Technical Glossary

Mwendo Pamoja White Paper

Reader's Guide: Disciplines, Terms, and Mathematical Notation

How to use this guide

Mwendo Pamoja follows one economic event through several institutional worlds. A fuel-price increase, vehicle interruption, platform commission change, or reduction in trip demand first affects the driver's operating cash flow. It then appears in wallet and telematics data, enters a governed underwriting process, activates a policy response, changes accounting and portfolio measures, and may ultimately affect the SPV's collections and lender cash flows. No single professional discipline owns that entire chain.

This guide has two layers. The first is an interdisciplinary field map showing where the white paper draws from economics, insurance, finance, statistics, technology, law, accounting, and operations. The second is an alphabetical technical glossary defining the paper's principal terms and symbols in their Mwendo Pamoja context. A definition here is a navigational aid, not a substitute for the fuller discussion, qualifications, equations, or IEEE sources in Parts 1 through 6.

Terms are defined according to their intended role in this architecture. In particular, the neural representation, Explicit Liquidity Feature Path, Bayesian underwriting layer, and Credit Policy and Compliance Gate are separate components. Similarly, customer pricing, SPV note pricing, accounting impairment, prudential capital, and market-consistent valuation are related but distinct calculations.

Interdisciplinary field map

Economics and the driver's operating reality

The paper begins in microeconomics rather than in model architecture. It treats a gig driver as a small income-producing enterprise whose labour, vehicle, insurance, liquidity, and platform access are mutually dependent. Household finance explains reserve depletion and short-term borrowing; labour and platform economics explain variable work opportunities and commissions; behavioural economics informs repayment and intervention design. Macroeconomics, monetary economics, fuel-price exposure, urban mobility, and the transition toward lower-emission vehicles provide the external conditions under which the driver operates. Development economics and financial inclusion supply the broader question: whether finance can preserve productive capacity without converting continuous observation into continuous punishment.

Lending, insurance, and embedded-finance product design

Mwendo Pamoja combines digital lending, microfinance, revolving credit, insurance premium financing, and usage-sensitive insurance within an embedded-finance experience. These disciplines meet at origination, affordability assessment, limit management, premium funding, collections, restructuring, and servicing. Insurance and lending remain legally and economically distinct. The insurer receives premium under the applicable insurance arrangement; the IPF lender holds a finance receivable against the driver; and the SPV may acquire only eligible receivables and rights that can legally be transferred. Continuous underwriting updates risk between conventional application or renewal dates, while the policy layer determines which operational response is permitted.

Project finance, structured finance, and institutional capital

The financing proposition draws on project finance, structured finance, securitisation, fixed-income analysis, and corporate finance. HoldCo owns and develops the operating capability, while a ring-fenced SPV purchases eligible receivables and allocates collections through a priority of payments. Senior debt, mezzanine debt, and first-loss equity form the capital stack. Overcollateralisation, a cash reserve account, subordination, asset eligibility, concentration limits, recovery rights, covenants, and early amortisation provide structural protection. Lender underwriting evaluates debt service, liquidity, losses, recovery timing, and stress performance rather than relying on an unsupported claim of principal protection.

Treasury, pricing, and balance-sheet management

KESONIA connects monetary conditions to customer and institutional pricing. The paper therefore covers benchmark compounding, risk-based credit pricing, expected loss, funding margins, net interest margin, liquidity ladders, asset-liability management, and matched-funding attribution. It distinguishes the customer's lending rate from the SPV asset yield, Class A and Class B note margins, internal or analytical funds-transfer prices, investor hurdle rates, and valuation discount rates. Interest-rate risk, foreign-exchange exposure, reserve cash drag, behavioural maturity, and collection timing are treated as cash-flow questions. IRRBB remains a prudential banking concept even when the SPV faces analogous repricing and basis risks.

Mathematics, statistics, and quantitative risk

The underwriting design combines probability theory, hierarchical Bayesian statistics, logistic regression, nonlinear splines, time-varying effects, shrinkage priors, and posterior com-

putation. Linear algebra appears through QR orthogonalisation, whitening, projections, and Cholesky factorisation. MCMC diagnostics, posterior predictive checks, calibration, discrimination, and out-of-time validation govern whether the model is usable. Copulas and tail-dependence measures address correlated product losses. Structural and reduced-form credit models appear as valuation and methodological comparators. Decision theory converts posterior risk and uncertainty into an action only after costs and policy constraints are made explicit.

Artificial intelligence and responsible model use

Deep learning is used for temporal representation, not as an unconstrained final decision-maker. GRUs encode short-horizon operational sequences; Transformers encode longer-horizon context; attention mechanisms align information arriving at different frequencies. Explicit financial variables have a separate owner and enter the Bayesian layer directly. Cross-fitted residualisation, a narrow bottleneck, whitening, block-specific priors, ablation tests, and stability checks reduce duplicate signal. Explainability, fairness assessment, customer reason codes, model-risk governance, challenger testing, and human review address how predictive systems should be governed rather than merely how they can be built.

Data engineering, telematics, and computation

The data architecture spans edge capture, vehicle telematics, mobile and wallet events, change-data capture, durable streaming, stateful event-time processing, and lakehouse storage. Kafka, Flink, Debezium, Redis-style serving, and Bronze, Silver, and Gold layers illustrate the required functions. Bitemporal records and as-of joins preserve point-in-time correctness. Missingness, late events, identity resolution, data lineage, quality controls, and replay determine whether a model result can be reproduced. SQL supports benchmark-rate compounding and accounting interfaces, while GPU or TPU computation is a possible acceleration mechanism for posterior inference rather than an architectural requirement.

Accounting, prudential risk, law, and assurance

IFRS 9 connects instrument-level evidence to staging, expected credit loss, modifications, write-offs, and recoveries. IFRS 17 remains an insurer-accounting interface rather than an SPV receivable rule. Basel capital and IRRBB are bank-owned prudential workstreams. Kenyan banking, digital-credit, insurance, capital-markets, consumer-protection, and data-protection requirements determine what the parties may originate, transfer, process, explain, and report. True sale, enforceability, perfection, account control, insolvency analysis,

and servicing continuity support the SPV structure. Audit trails, maker-checker approval, immutable records, encryption, signatures, and reconciliation provide complementary forms of assurance.

Enterprise systems, cybersecurity, and operating controls

Enterprise architecture links product services and subledgers to Microsoft Dynamics 365 Finance without making the general ledger perform every instrument-level calculation. Oracle OFSAA is discussed as an optional institutional mechanism for cash-flow modelling, ALM, and FTP where scale and governance justify it. Cybersecurity includes encryption, public-key infrastructure, mutual authentication, access control, evidence retention, and safe degradation. Reliability engineering, reconciliation, service levels, backup servicing, and business continuity ensure that a valid model result can become a controlled operational and accounting event.

Product delivery, governance, and institutional change

The implementation roadmap draws on product management, service design, human-centred research, operational-risk management, programme delivery, pilot design, change management, and due diligence. It assigns decision rights among the driver, platform, lender, insurer, servicer, SPV, trustee, account bank, investors, model developers, validators, finance teams, and regulators. Scale is stage-gated. A more complex model or larger financing structure is promoted only after the baseline, data lineage, customer treatment, cash movements, accounting, and validation evidence remain stable.

Mathematical notation at a glance

- $\mathbb{P}$: The physical or real-world probability measure used for underwriting, portfolio risk, stress evidence, and IFRS 9 inputs.
- $\mathbb{Q}$: A risk-neutral or market-consistent measure used only for an appropriate valuation purpose with a defensible calibration basis.
- $\mathbf{z}_i$: The original Explicit Liquidity Feature vector for observation i .
- $\mathbf{q}_i = Q(\mathbf{z}_i)$: The centred, scaled, and QR-orthogonalised spline representation of the Explicit Liquidity Features.
- $\mathbf{h}_i$: The low-dimensional neural embedding before residualisation.
- $\tilde{\mathbf{h}}_i$: The cross-fitted residual neural embedding after removing the component predicted by explicit liquidity and approved metadata.

- $\mathbf{m}_i$ or $\mathbf{x}_i$: Approved contract, product, data-quality, and other essential metadata, according to the notation used in the relevant specification.
- β_z and β_h : Coefficients for the explicit-liquidity and residual-neural blocks.
- PD , LGD , and EAD : Probability of default, loss given default, and exposure at default.
- EL : Expected loss over a stated horizon, ordinarily derived from consistent PD , LGD , and EAD definitions.
- K_{RBCP} : The customer-pricing premium over KESONIA under the applicable risk-based credit-pricing formulation.
- K_{cap} : Regulatory or economic capital notation, deliberately distinct from K_{RBCP} .
- m_A and m_B : Negotiated margins for the Class A and Class B notes.
- λ_L and λ_U : Lower-tail and upper-tail dependence coefficients.

Alphabetical technical glossary

A

Advance rate. The proportion of an eligible receivable pool that a financing facility is willing to fund. It must be distinguished from overcollateralisation because the two tests can use different denominators. In the SPV model, the applicable asset definition, haircut, and debt basis must be stated explicitly. See Parts 1 and 5.

Algorithmic matching elasticity. A measure or proxy describing how responsive a platform's trip allocation or matching activity is to changes in driver supply, passenger demand, geography, time, or platform policy. It is contextual information, not a protected characteristic and not automatically a causal estimate. See Part 2a.

Algorithmic redlining. A pattern in which a model or policy produces unjustified exclusion or materially worse treatment for protected or vulnerable groups, potentially through proxy variables such as geography. The paper addresses it through feature governance, fairness testing, reason review, less discriminatory alternatives, and human oversight. See Parts 2b and 3.

AR(1) prior. A first-order autoregressive prior in which a current time effect depends on the preceding effect plus innovation noise. It can represent persistence in cohort or calendar effects and can smooth adjacent spline coefficients. It is a modelling choice that requires stability and sensitivity testing. See Part 2b.

As-of join. A temporal join that matches each observation only to information available by the decision timestamp. It is essential when telemetry, wallet, platform, macroeconomic, and regulatory data arrive on different clocks. See Part 2a.

Asset-liability management (ALM). The measurement and management of timing, repricing, currency, liquidity, and behavioural differences between assets and funding obligations. For the SPV, it means a receivable and waterfall cash-flow ladder; for the partner bank, it may also form part of prudential balance-sheet management. See Part 5.

Assistive intervention. A controlled action intended to preserve the driver's productive capacity or prevent avoidable deterioration, such as a temporary payment adjustment, premium holiday, micro-reward bridge, routing support, or draw restriction. Its eligibility, accounting effect, customer communication, and policy authority must be recorded. See Part 3.

Asymmetric copula. A dependence model that permits stronger association in one tail of a joint distribution than in the other. The Clayton copula is considered because joint distress may be more pronounced than joint favourable outcomes. Selection remains an empirical portfolio question. See Part 2b.

B

Bankruptcy remoteness. The degree to which SPV assets and cash flows are insulated from the insol-

vency of the originator or sponsor. It depends on legal transfer, enforceability, perfection, separateness, account control, servicing continuity, commingling and set-off analysis, and applicable insolvency law. The label SPV does not create remoteness by itself. See Parts 1, 5, and 6.

Basel framework. International prudential standards for bank capital, risk management, disclosure, and related controls. In this paper, Basel calculations and approvals remain responsibilities of the regulated partner bank. They are not automatically obligations of the SPV or technology provider. See Parts 3 and 5.

Behavioural maturity. The expected timing of an asset or liability's cash flows after considering borrower behaviour such as prepayment, redraw, delinquency, restructuring, or tenor extension. It may differ from contractual maturity and is important to ALM and stress testing. See Part 5.

Bitemporal data. Data carrying both valid time, when a fact was economically effective, and system time, when the platform recorded or learned it. Bitemporality supports reproducibility, correction handling, and point-in-time model reconstruction. See Parts 2a and 6.

Brier score. The mean squared difference between a predicted probability and the observed binary outcome. It reflects both calibration and resolution and should be interpreted alongside product horizon, class balance, discrimination, and operational decision value. See Part 2b.

B-spline. A locally supported piecewise-polynomial basis used to model a smooth nonlinear relationship without imposing abrupt bins. The paper applies centred and regularised spline bases to selected Explicit Liquidity Features. Knot placement, extrapolation, and smoothing remain governed modelling decisions. See Part 2b.

C

Calibration. Agreement between predicted probabilities and observed outcome frequencies over a defined product, population, and horizon. A model may discriminate well but remain poorly calibrated. Mwendo Pamoja therefore tests calibration by product, cohort, time, and relevant groups. See Part 2b.

Capital stack. The ordered combination of Class A senior debt, Class B mezzanine debt, and Class C first-loss equity used to finance the SPV. Subordina-

tion determines which class absorbs loss or receives cash first under the waterfall. See Parts 1 and 5.

Cash Reserve Account (CRA). A controlled SPV account maintained to cover a defined number of months or amount of debt service, fees, or other senior obligations. Its target, permitted uses, replenishment order, release conditions, and investment treatment must be contractual. See Part 5.

Cash-flow asymmetry (CFA). An Explicit Liquidity Feature measuring the imbalance between unusually large positive wallet inflows and the driver's ordinary earnings base. Its exact numerator, denominator, window, winsorisation, and treatment of reversals must be specified before use. CFA bypasses the GRU and Transformer encoders and enters the Bayesian layer directly. See Part 2a.

Champion-challenger framework. A governance process in which a production model is compared with credible alternatives under the same data, horizon, and acceptance criteria. The explicit-only model remains a valid champion unless neural or dependence challengers demonstrate stable incremental value. See Parts 2b and 6.

Change data capture (CDC). The capture of database inserts, updates, and deletes as an ordered event stream. Debezium is used as an architectural example. CDC supports auditability and replay but does not by itself guarantee correct economic timestamps or complete source data. See Part 2a.

Clayton copula. An Archimedean copula with lower-tail dependence and no upper-tail dependence in its standard bivariate form. It is a candidate for modelling joint deterioration across products, subject to held-out fit, tail calibration, stability, and cash-flow relevance. See Part 2b.

Collections account. A controlled bank account into which borrower or platform deductions are directed before allocation through servicing and SPV rules. Account ownership, control, reconciliation, commingling protection, refund handling, and waterfall treatment must be documented. See Parts 1 and 5.

Continuous underwriting. The governed reassessment of risk as new authorised operational, wallet, repayment, and contextual evidence becomes available. It supplements rather than erases contractual rights, customer notice, fairness review, accounting governance, and policy controls. See Parts 1 through 3.

Copula. A function that joins marginal distributions into a multivariate distribution and isolates a chosen form of dependence. A copula does not repair mis-calibrated marginal PDs, prove causality, or automatically determine regulatory capital. See Part 2b.

Credit Policy and Compliance Gate. The downstream rule layer that receives posterior risk, uncertainty, explicit features, exposure, concentration, reserve, and covenant information and returns an authorised action. It contains contractual, regulatory, policy, and operational controls. It is neither a neural model nor another feature service. See Parts 2b, 3, and 6.

Cross-attention. An attention mechanism in which one representation supplies queries and another supplies keys and values. It is used conceptually to align short-horizon driver state with longer-horizon contextual information. See Part 2a.

Cross-fitted residualisation. The removal from a neural embedding of the component predicted by explicit liquidity features and approved metadata, using a projection trained without the observation's validation fold. It reduces duplicate signal and leakage but does not guarantee permanent orthogonality under a new regime. See Parts 2b and 6.

D

Data lineage. The traceable path from a source event through transformations, features, model versions, policy decisions, accounting entries, and reports. Lineage supports reproducibility, audit, complaints handling, and controlled correction. See Parts 2a, 4, and 6.

Debt service coverage ratio (DSCR). Cash available for debt service divided by the debt service due over the same period and under a stated waterfall definition. The numerator, reserve use, taxes, fees, and principal schedule must be explicit. See Part 5.

Debezium. An open-source change-data-capture platform used in the paper as an example of how operational database changes can enter a durable event stream. It is a component choice, not the source of legal authority or economic meaning. See Part 2a.

Default cascade. A feedback process in which an external shock reduces driver cash flow, causes arrears or insurance disruption, threatens platform access, and increases the likelihood of default across several products. It is a systems hypothesis to be measured rather than assumed universal. See Part 1.

Disparate Impact Ratio (DIR). A group fairness statistic comparing the rate of a favourable outcome between groups. It can flag potential disparity but does not, by itself, establish legal discrimination, fairness, or causal mechanism. See Parts 2b and 3.

Dynamic Debt-to-Liquidity Ratio (DLR). Short-horizon debt obligations divided by a governed measure of available wallet liquidity. It is an Explicit Liquidity Feature and may enter the HLR through a nonlinear spline. Its obligations, liquidity exclusions, horizon, and zero-denominator treatment require specification. See Parts 2a and 2b.

Dynamics 365 Finance (D365). The enterprise resource-planning endpoint used for approved journals, legal-entity accounting, dimensions, consolidation, workflow, and reporting. Product subledgers and governed engines may retain instrument-level calculations that D365 cannot represent natively. See Part 4.

E

Earnings Velocity. An Explicit Liquidity Feature comparing recent net earnings with a longer historical baseline. It indicates acceleration or deterioration in earnings capacity. It belongs to the direct liquidity path rather than being duplicated as a defined GRU input. See Part 2a.

Effective interest rate (EIR). The IFRS 9 rate that exactly discounts estimated contractual cash flows over an instrument's expected life to its initial gross carrying amount, subject to the standard's requirements. It is not interchangeable with KESONIA, the customer rate, or the investor hurdle. See Parts 4 and 5.

Effective sample size (ESS). The amount of independent-equivalent posterior information contained in autocorrelated MCMC draws. ESS is assessed for relevant parameters and quantities together with convergence, divergences, and Monte Carlo error. See Part 2b.

Eligible receivable. A receivable satisfying contractual purchase criteria relating to ownership, documentation, performance, data quality, currency, product, concentration, policy status where relevant, and absence of prohibited encumbrances. Insurer-owned premium cash is not an eligible loan receivable. See Parts 1, 5, and 6.

Embedded finance. The delivery of a financial product within a non-financial operating journey,

such as a driver or mobility platform. Embedding changes distribution and data access but does not remove licensing, conduct, accounting, or contractual responsibilities. See Part 1.

Equalized odds. A fairness criterion requiring equal true-positive and false-positive rates across specified groups, conditional on the observed outcome. It can conflict with calibration or other objectives when base rates differ and must be applied in a legally and operationally grounded governance process. See Parts 2b and 3.

Expected credit loss (ECL). The probability-weighted present value of credit cash shortfalls under IFRS 9, including reasonable and supportable forward-looking information. ECL is an accounting measurement, not simply the underwriting PD multiplied by a fixed loss rate. See Parts 3 through 5.

Expected loss (EL). The statistical expectation of credit loss over a stated horizon, often represented as a consistent combination of PD, LGD, and EAD. Its horizon, timing, discounting, cure, and recovery assumptions must align with the use case. See Parts 2b and 5.

Explicit Liquidity Feature Path. A low-latency Kappa and Redis-style feature service that computes interpretable financial variables and routes them directly into the HLR through splines and controlled interactions. These engineered metrics bypass the GRU and Transformer encoders. See Parts 2a, 2b, and 6.

Explicit Liquidity Features. Governed variables including CFA, DLR, Earnings Velocity, Repayment Velocity, wallet volatility, reserve balance, time since depletion, utilization, and approved interactions. Each feature requires a definition, owner, timestamp, lineage, and monitoring rule. See Part 2a.

Exposure at default (EAD). The expected amount exposed when default occurs, including the treatment of drawn balances and, where relevant, future drawings or credit-conversion effects. It must be product- and horizon-specific. See Parts 2b, 3, and 5.

F

Feature ownership. The rule assigning each engineered variable to one canonical computation and model path. Raw wallet events may inform more than one branch, but CFA, DLR, Earnings Velocity, and Repayment Velocity are not independently reconstructed and counted in both neural and explicit blocks. See Parts 2a and 6.

Flink. Apache Flink, used as the principal example of stateful stream processing with event-time semantics, windows, late-event handling, and continuously updated features. See Part 2a.

Funds-transfer pricing (FTP). A method for attributing funding, liquidity, and sometimes optional costs to assets, products, or business units. In the SPV context, the paper uses matched-funding attribution by product cohort rather than implying a full bank-wide internal FTP system. See Part 5.

G

Gated Recurrent Unit (GRU). A recurrent neural architecture with update and reset gates used to encode short-horizon sequences such as kinematic and operating behaviour. Its output is a low-dimensional representation, not a final credit decision. See Part 2a.

Geohash. A hierarchical encoding of geographic location. It can support exposure aggregation, routing, and partial pooling, but fine-grained geography can create privacy, sparsity, proxy-discrimination, and re-identification risks. See Parts 2a through 3.

Gini coefficient. A rank-order discrimination measure related to the area under the ROC curve. It does not measure calibration, fairness, causal validity, or customer benefit and should not be used alone for model approval. See Part 2b.

General Ledger (GL). The authoritative enterprise accounting record for approved financial postings. Detailed contractual and behavioural calculations remain in governed product subledgers or specialised engines. See Part 4.

GNSS. Global navigation satellite systems used to derive location, speed, route, and temporal movement information. Collection and use require data minimisation, quality controls, lawful purpose, and protection against inappropriate inference. See Part 2a.

H

Hamiltonian Monte Carlo (HMC). An MCMC method that uses gradients and auxiliary momentum to explore a posterior distribution more efficiently than a simple random walk. Reliable use still requires convergence, divergence, effective-sample-size, and posterior predictive diagnostics. See Part 2b.

Hierarchical Bayesian logistic regression (HLR). The principal underwriting model combin-

ing partially pooled geography, platform or cohort, and time effects with explicit liquidity splines, residual neural information, controlled interactions, and approved metadata. It produces posterior PD and uncertainty rather than an unqualified deterministic score. See Part 2b.

Horseshoe prior. A global-local shrinkage prior that pulls many weak coefficients strongly toward zero while permitting a smaller number of signals to remain substantial. The regularized horseshoe adds a finite slab. In this paper it is a candidate for the residual neural block, with separately governed priors for other blocks. It does not orthogonalise correlated predictors. See Parts 2b and 6.

I

IFRS 9. The financial-instruments standard governing classification and measurement, impairment, EIR, modification, write-off, and related accounting for relevant receivables and financial liabilities. The responsible reporting entity must approve its methodology. See Parts 3 through 5.

IFRS 17. The accounting standard for insurance contracts. It applies to the insurer's relevant contracts and reporting, not automatically to the SPV's purchased loan receivables. Interfaces arise through premium, coverage, cancellation, and insurer cash-flow information. See Parts 3 and 5.

Insurance premium financing (IPF). Credit used to fund an insurance premium that would otherwise be paid upfront or on a less convenient schedule. Once premium is validly paid to the insurer, the lender may hold a separate finance receivable against the driver. See Part 1.

IPF finance receivable. The lender-owned right to collect funded premium principal, interest, and authorised charges from the borrower under the IPF agreement. It is distinct from insurer-owned premium cash and may be transferred to the SPV only if legally assignable and eligible. See Parts 1 and 5.

Interest rate risk in the banking book (IR-BBB). The bank's exposure to adverse changes in earnings or economic value caused by interest-rate movements affecting banking-book assets and liabilities. The SPV has analogous repricing and basis risk, but it is not itself labelled a regulated banking book. See Part 5.

K

Kafka. Apache Kafka, used as the durable distributed event-streaming backbone for telemetry, wallet, platform, and operational events. Durability and replay support recovery and reproducibility, but semantic correctness still depends on event contracts and controls. See Part 2a.

Kappa architecture. A stream-first data architecture in which current state and derived features are computed from a durable event log rather than maintained through separate batch and speed code paths. Mwendo Pamoja combines this low-latency role with a historical lakehouse. See Part 2a.

K_{cap} . Notation reserved for regulatory or economic capital where required. It prevents the symbol K from being reused for customer pricing, capital, and unrelated concepts. See Parts 2b and 5.

K_{RBCP} . The borrower-specific premium added to KESONIA under the applicable risk-based credit-pricing formulation. It may reflect lending-related costs, shareholder return, and the borrower's risk profile. It is not identical to PD or an SPV note margin. See Parts 2b and 5.

KESONIA. The Kenya Shilling Overnight Interbank Average, used as the Kenyan shilling overnight reference benchmark in the paper's pricing and funding architecture. Contractual use requires an approved compounding, observation, day-count, publication, correction, and fallback convention. See Part 5.

L

Lakehouse. A data platform combining durable object-style storage with table formats, transactions, schema controls, and analytical access. It supports replay, point-in-time reconstruction, feature development, validation, accounting evidence, and reporting. See Part 2a.

Late-arriving event. An event that becomes available after its economic event time or expected processing window. The architecture must decide whether to update current state, revise history, initiate an accounting correction, or preserve the original decision while recording the later evidence. See Parts 2a and 6.

Less discriminatory alternative (LDA). A feasible alternative model, feature set, threshold, or policy that achieves the legitimate objective with less adverse group effect. LDA testing requires compara-

ble performance, operational feasibility, legal review, and documented trade-offs. See Part 3.

LIME. Local Interpretable Model-Agnostic Explanations, a technique that fits a local surrogate around a selected observation. Its output depends on the perturbation distribution, neighbourhood, and kernel and should not be mistaken for a causal explanation. See Part 2b.

Loss given default (LGD). The proportion of exposure lost after recoveries, costs, timing, collateral or refund proceeds, and discounting. IPF, microloan, and revolving-credit LGDs require separate evidence and recovery mechanics. See Parts 2b, 3, and 5.

M

Maker-checker control. A segregation-of-duties control requiring one authorised person or service to prepare an action and another to approve it. It is applied to model promotion, accounting entries, reporting submissions, and material overrides. See Parts 3, 4, and 6.

Markov-chain Monte Carlo (MCMC). A family of methods that generates dependent samples from a posterior distribution. The paper discusses computational scaling, sampling geometry, and validation because merely completing a chain does not establish reliable inference. See Part 2b.

Medallion architecture. A tiered data design commonly described as Bronze, Silver, and Gold. Bronze preserves source events, Silver conforms and validates them, and Gold supplies governed analytical, accounting, or reporting outputs. See Parts 2a, 4, and 5.

Missingness mechanism. The process determining why information is absent. Missing completely at random, missing at random, and missing not at random have different implications. In production, missingness can also indicate device failure, consent withdrawal, network interruption, platform change, or deliberate manipulation. See Part 2a.

Model-risk management. The governance of model purpose, data, assumptions, development, validation, approval, implementation, monitoring, limitations, changes, and retirement. It converts a mathematically plausible model into a challengeable institutional process. See Parts 2b, 3, and 6.

Multi-head self-attention. An attention operation using several learned projections to represent different relationships within a sequence. The Trans-

former branch applies it to longer-horizon contextual information. See Part 2a.

Multimodal fusion. The combination of information from different modalities, such as telematics, wallet events, repayments, platform operations, and macroeconomic context. Fusion must respect timestamps, permissions, missingness, feature ownership, and validation boundaries. See Part 2a.

Mutual TLS (mTLS). Transport Layer Security in which both parties authenticate using certificates. It protects the communication channel and counterparty authentication; it does not independently prove the correctness, fairness, or legal meaning of the transmitted data. See Part 4.

N

Net interest margin (NIM). Asset interest and related income less funding and specified financing costs, expressed as an amount or rate over a consistent asset base. The paper examines NIM sensitivity to KESONIA, losses, servicing costs, reserves, and product mix. See Part 5.

Neural embedding. A learned low-dimensional vector representing temporal or contextual information from the GRU, Transformer, and fusion process. It is an intermediate model input rather than a customer-facing reason, policy rule, or final credit decision. See Parts 2a and 2b.

Neural residual. The embedding remaining after cross-fitted prediction from explicit liquidity features and approved metadata has been subtracted. The HLR uses this residualised block to reduce double-counting. Whitening and shrinkage provide further controls. See Parts 2b and 6.

Non-performing loan (NPL) trigger. A contractual portfolio test based on a defined non-performing exposure ratio. If the agreed threshold is breached, it may stop new purchases, block equity distributions, or initiate early amortisation. The definition, cure, numerator, denominator, and measurement date must be contractual. See Part 5.

O

OBD-II. On-board diagnostics data from a vehicle, such as fault or diagnostic events. It may help identify operating interruptions but requires device compatibility, quality controls, lawful use, and careful interpretation. See Part 2a.

Oracle OFSAA. Oracle Financial Services Analytical Applications, discussed as an optional institutional platform for instrument cash-flow modelling, ALM, FTP, and related analytics. A validated cohort and waterfall model may remain more proportionate during the pilot. See Part 5.

Out-of-time validation. Evaluation on a later period excluded from model fitting and tuning. It tests temporal transportability, calibration, drift, and operational stability more credibly than a random row-level split when drivers and conditions repeat through time. See Parts 2b and 6.

Overcollateralisation (OC). The amount by which eligible collateral exceeds the relevant debt claim, usually expressed as a ratio. The receivable definition and debt denominator must be explicit. A 125 percent threshold is not the same calculation as a 75 percent advance rate. See Part 5.

P

Partial pooling. Hierarchical estimation in which sparse groups borrow information from the wider portfolio while data-rich groups may deviate when supported by evidence. It lies between one model for everyone and unrelated models for every cluster. See Part 2b.

Physical measure $\mathbb{P}$. The probability measure describing real-world event frequencies. Underwriting PD, portfolio monitoring, stress evidence, and IFRS 9 inputs generally begin under this measure. It must not be relabelled risk-neutral merely by adding a funding or credit spread. See Parts 2b and 5.

Point-in-time correctness. The requirement that a training record, backtest, or reconstructed decision use only information actually available at that decision time. It prevents look-ahead bias and requires temporal source and revision lineage. See Part 2a.

Population Stability Index (PSI). A binned measure of change between a current and reference distribution. Thresholds such as 0.10 or 0.25 are governance conventions unless established otherwise by applicable authority or contract. PSI is a monitoring signal, not proof of model failure or discrimination. See Parts 2b and 5.

Posterior predictive check (PPC). A comparison between statistics or patterns observed in real data and those generated from the fitted posterior predictive distribution. PPCs test whether the model repro-

duces decision-relevant features, not whether it is universally true. See Part 2b.

Premium cash. Money payable to or received for the licensed insurer under the applicable insurance arrangement. It must not be confused with an IPF lender's finance receivable or treated as an ordinary SPV loan asset merely because the lender facilitated payment. See Part 1.

Probability of default (PD). The probability that a defined default event occurs within a stated horizon for a specified product and exposure. PD requires a precise default definition, observation window, cure treatment, and calibration population. See Parts 2b, 3, and 5.

Project finance. Financing evaluated primarily through the cash flows, contracts, controls, and risks of a defined project or ring-fenced vehicle. In Mwendo Pamoja it informs the SPV, waterfall, reserves, covenants, and lender diligence. See Parts 1, 5, and 6.

Q

QR orthogonalisation. A linear-algebra transformation that converts a design matrix into orthogonal basis directions. The paper applies it to the centred and scaled spline basis of Explicit Liquidity Features. It improves conditioning but does not establish causal independence or permanent out-of-sample orthogonality. See Parts 2b and 6.

R

Rank-normalised $\hat{R}$. An MCMC convergence diagnostic comparing within-chain and between-chain variation after rank normalisation. Values near one are necessary but not sufficient evidence of reliable posterior computation. See Part 2b.

Real-time decision boundary. A governed mapping from posterior risk, uncertainty, exposure, costs, and constraints to a candidate action. The Credit Policy and Compliance Gate remains responsible for whether the action is permitted. See Part 2b.

Recovery. Cash or economic value received after delinquency or default, net of timing, costs, haircuts, and legal limitations. Recoveries may come from borrower payments, assigned refunds, collateral rights, or other valid sources and should be modelled by product and cohort. See Parts 1, 2b, and 5.

Repayment Velocity. An Explicit Liquidity Feature comparing actual principal repayment with contractually scheduled repayment over a governed window. It may indicate acceleration, deterioration, cure, or restructuring effects and should not be interpreted without contract status. See Parts 2a and 2b.

Revolving credit. A facility under which a borrower may draw, repay, and redraw up to an approved limit, subject to policy and contract. The model distinguishes ordinary operating utilization from a stress-driven full-draw or adverse-utilization pattern. See Parts 1 and 5.

Risk-based credit-pricing model (RBCPM). The applicable framework under which the total lending rate is expressed as KESONIA plus a borrower-specific premium, with fees and charges forming part of total cost of credit. Governance must distinguish customer pricing from SPV financing terms. See Part 5.

Risk-neutral measure $\mathbb{Q}$. A measure used for market-consistent valuation under conditions supporting such a transformation or calibration. It is distinct from the physical probability used to forecast actual defaults and is not required for every customer or SPV pricing calculation. See Part 5 and its Appendix A.

S

Significant increase in credit risk (SICR). The IFRS 9 assessment that credit risk has increased significantly since initial recognition, subject to the reporting entity's approved policy and evidence. A model signal may inform SICR but does not bypass accounting governance. See Parts 3 and 4.

SHAP. SHapley Additive exPlanations, a family of methods allocating a model output difference among input features under a chosen background and conditionality assumption. SHAP supports model analysis but does not automatically provide a causal or legally sufficient customer reason. See Parts 2b and 4.

Special-purpose vehicle (SPV). A ring-fenced legal and financing entity established to acquire eligible receivables, issue or receive funding, maintain controlled accounts, and distribute cash through a waterfall. It is not the insurer, originator, technology platform, or customer-facing product issuer. See Parts 1, 5, and 6.

Strong heredity. An interaction-selection principle under which an interaction is included or receives ma-

terial prior support only when its constituent main effects are also represented. It reduces unstable interaction discovery and improves interpretability. See Parts 2b and 6.

Structured finance. Financing in which asset eligibility, legal isolation, subordination, reserves, covenants, servicing, and a priority of payments shape investor risk. The economic result depends on the actual asset and cash-flow model rather than the transaction label. See Part 5.

Subledger. A governed instrument-level or product-level accounting record supporting balances, accruals, payments, arrears, modifications, ECL attributes, and reconciliation to the general ledger. See Parts 4 and 6.

Sufficiency threshold. A governed boundary comparing the driver's available net operating income with essential living and operating obligations. It is a policy and affordability concept, not a claim that default becomes mathematically certain below one universal ratio. See Parts 1 and 3.

T

Tail dependence. Dependence that persists in extreme regions of a joint distribution. Lower-tail dependence is relevant when several products deteriorate together during severe driver or macroeconomic stress. See Part 2b.

Telematics. Digitally captured information about vehicle movement, diagnostics, trips, and operating context. Telematics can support safety and underwriting but requires quality, consent, purpose, retention, fairness, and security controls. See Parts 1 through 4.

Transformer. A neural architecture based principally on attention rather than recurrence. In Mwendo Pamoja it represents lower-frequency structural and contextual sequences and contributes an embedding to the fusion layer. See Part 2a.

True sale. A legal transfer intended to move receivables and associated rights from an originator to an SPV rather than create only a secured loan. Its effectiveness depends on transaction documents, conduct, perfection, recourse, insolvency analysis, and applicable law. See Parts 1, 5, and 6.

U

Unearned premium. The portion of an insurance premium associated with the remaining cover-

age period. Any cancellation refund depends on policy terms, law, timing, charges, insurer process, and valid assignment; it is not automatically liquid collateral at full face value. See Parts 1 and 5.

Unexpected loss. A loss measure beyond expected loss, commonly defined through a portfolio-loss quantile, expected shortfall, or another approved risk metric. Its definition and confidence level must be stated and must not be conflated automatically with regulatory capital. See Part 2b.

Usage-based insurance (UBI). Insurance in which permitted driving or usage information affects pricing, coverage, incentives, or risk intervention. The insurer retains underwriting and contractual authority under the applicable arrangement. See Parts 1 and 3.

Utilization. The amount drawn under a revolving facility divided by the approved limit. High utilization may be ordinary for working capital or evidence of stress depending on repayment, income, timing, and other conditions. See Parts 1, 2a, and 5.

V

Variance inflation factor (VIF). A diagnostic measuring how much the variance of a regression coefficient is inflated by linear association with other predictors. It is one diagnostic for redundancy and is interpreted with posterior correlation, conditioning, ablation, and stability evidence. See Parts 2b and 6.

W

Wallet volatility. An Explicit Liquidity Feature measuring variability in governed wallet inflows, out-

flows, or net balances over a specified window. Its value depends on transaction classification, seasonality, reversals, cash-outs, and missing-wallet coverage. See Part 2a.

Waterfall. The contractual priority in which available SPV cash pays taxes or statutory costs, servicing, senior interest and principal, mezzanine obligations, reserve replenishment, mandatory sweeps, and residual equity distributions. Trigger events can change or block lower-priority payments. See Part 5.

Whitening. A transformation that centres and rescales a multivariate block so that its training-sample covariance is approximately the identity matrix. The neural residual is whitened within training folds to improve conditioning. Whitening does not guarantee independence or future-regime orthogonality. See Parts 2b and 6.

WORM repository. Storage designed so records are written once and read many times, protecting approved evidence from ordinary alteration during the retention period. It supports assurance but still requires access control, retention policy, indexing, and reconciliation. See Part 4.

Z

Zero-emission vehicle (ZEV). A vehicle producing no tailpipe emissions during operation. The paper touches on ZEV transition deadlines, charging access, routing, asset obsolescence, and the interaction between vehicle type and credit or operating risk. See Parts 1 through 3.

Closing orientation

The glossary should not be read as a claim that every discipline is resolved by one platform. Its purpose is to preserve boundaries. The predictive model estimates risk; the policy gate authorises action; the accounting process records approved economic events; the insurer governs insurance contracts; the lender governs credit; the bank owns its prudential calculations; and the SPV applies its transaction documents and waterfall. The value of the architecture lies in connecting those responsibilities without collapsing them into one undifferentiated model.